



NetApp ONTAP 9.5 Cluster Administration



Senthilkumar Muthusamy

NetApp ONTAP Agenda

- ▶ Module 1: NetApp ONTAP 9 Clusters
- ▶ Module 2: Cluster Management
- ▶ Module 3: Network Management
- ▶ Module 4: Physical Storage Management
- ▶ Module 5: Logical Storage Management
- ▶ Module 6: Data Access
- ▶ Module 7: Data Protection
- ▶ Module 8: Storage Efficiency
- ▶ Module 9: Cluster Maintenance





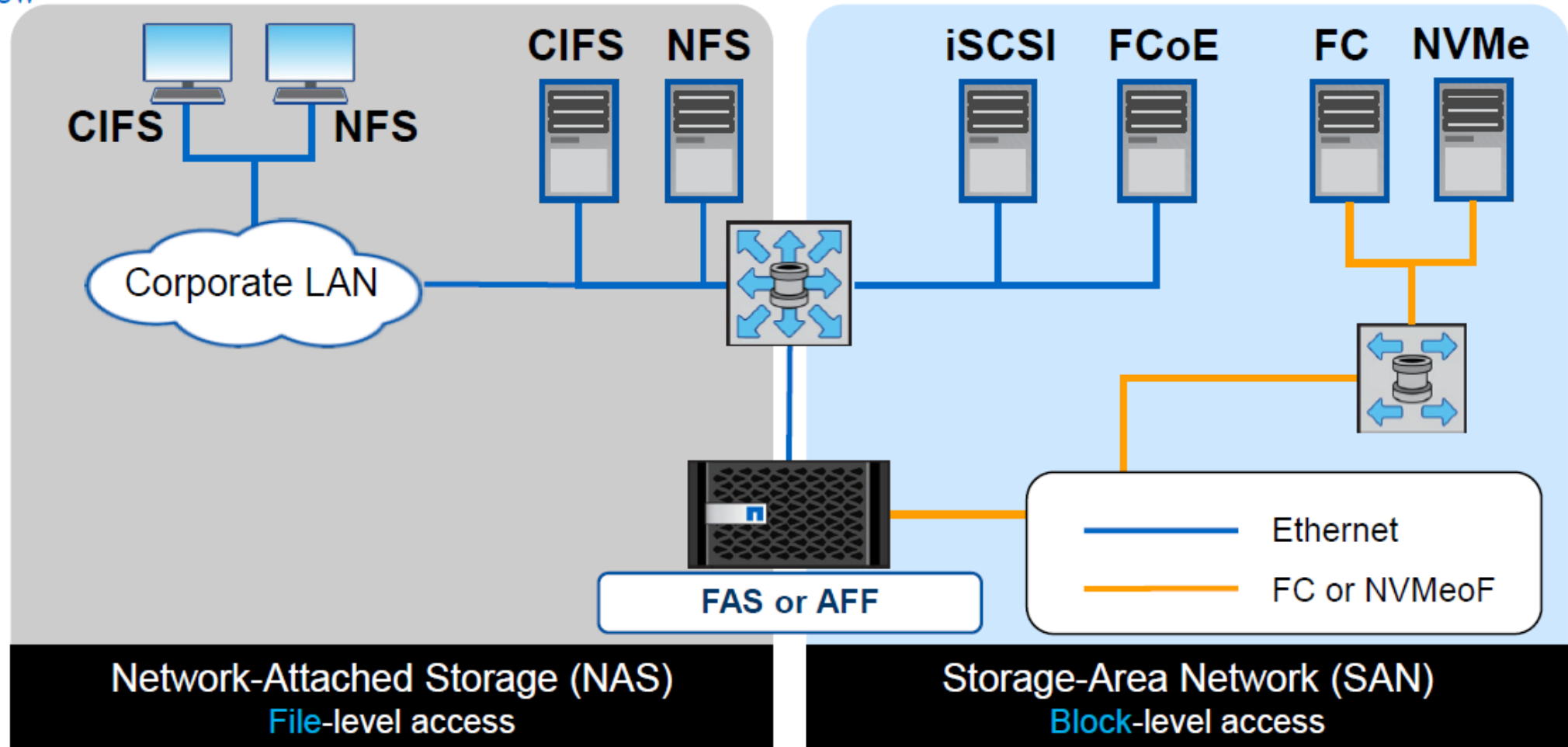
Introduction to NetApp ONTAP



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Unified Storage

Review



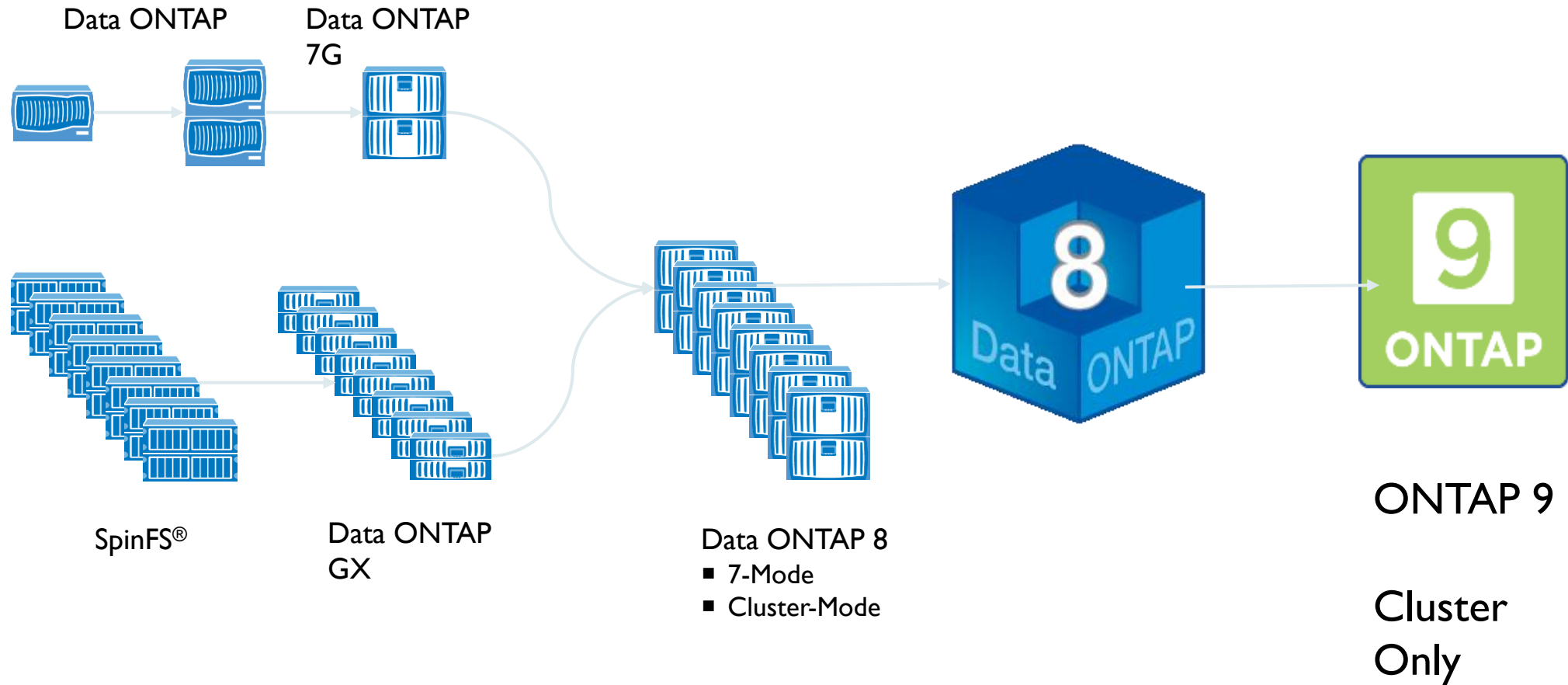
ONTAP 9.5 Supported Hardware Systems

January 2019

AFF	FAS
<div data-bbox="1057 429 1490 506">Enterprise Level</div> <div data-bbox="249 464 598 711"></div> <div data-bbox="835 515 1128 696"> - AFF A800 - AFF A700 - AFF A700s </div> <div data-bbox="1057 715 1490 792">Mid Level</div> <div data-bbox="249 778 598 901"></div> <div data-bbox="835 801 1105 843"> - AFF A300 </div> <div data-bbox="1057 935 1490 1012">Entry Level</div> <div data-bbox="249 1021 598 1115"></div> <div data-bbox="835 1021 1105 1136"> - AFF A220 - AFF A200 </div> <div data-bbox="1274 429 2423 1243" data-cs="2" data-kind="parent"><div data-bbox="1964 444 2313 686"></div><div data-bbox="1319 511 1574 554"> - FAS9000 </div><div data-bbox="1964 786 2313 882"></div><div data-bbox="1319 801 1574 843"> - FAS8200 </div><div data-bbox="1964 1043 2313 1110"></div><div data-bbox="1319 1021 1574 1136"> - FAS2700 - FAS2600 </div></div>	



A Tale of Two Products



NetApp ONTAP Deployments

As Software



ONTAP Select

In a Box



ONTAP 9

In the Cloud



ONTAP Cloud (Cloud Volumes)



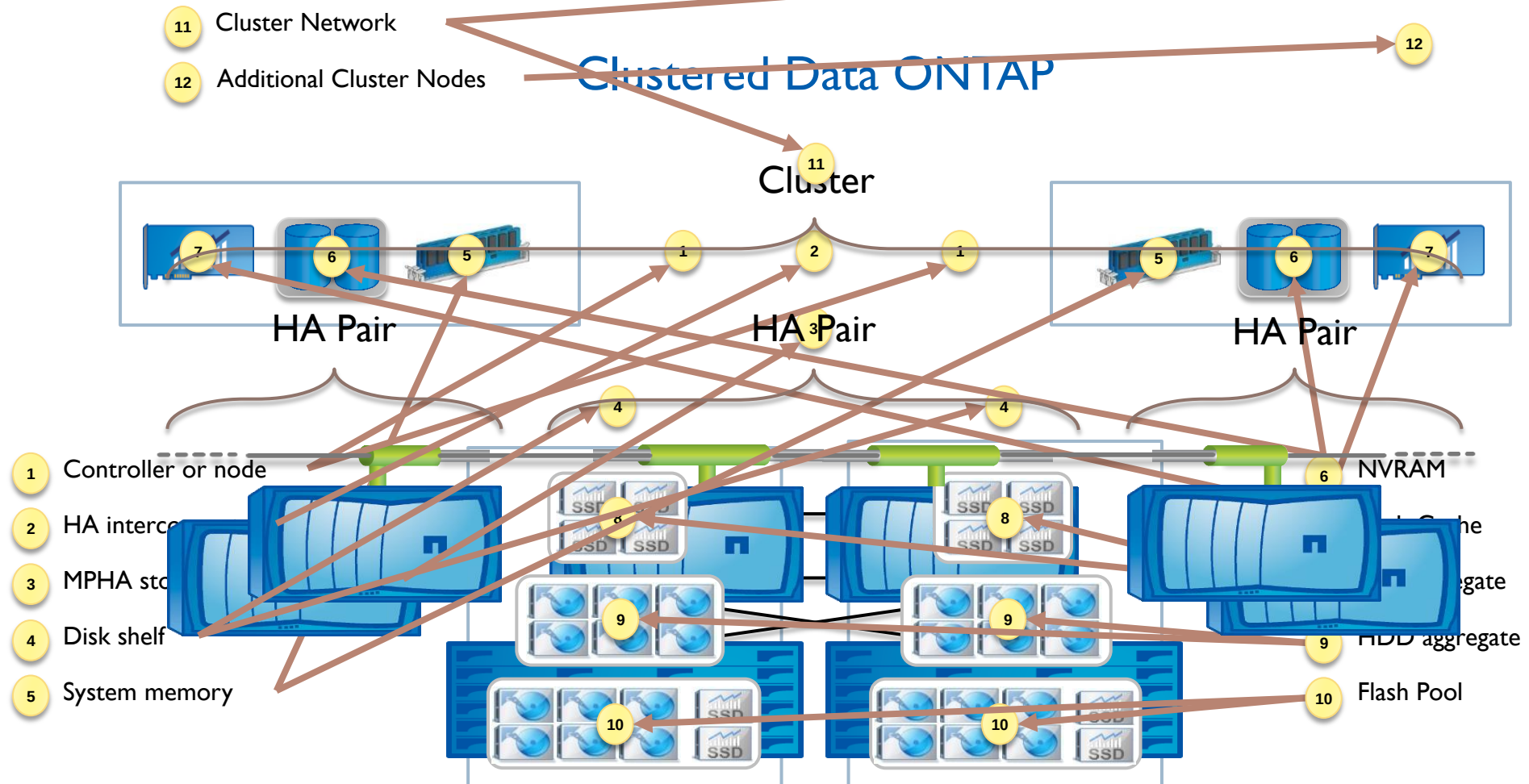
NetApp Cluster Components



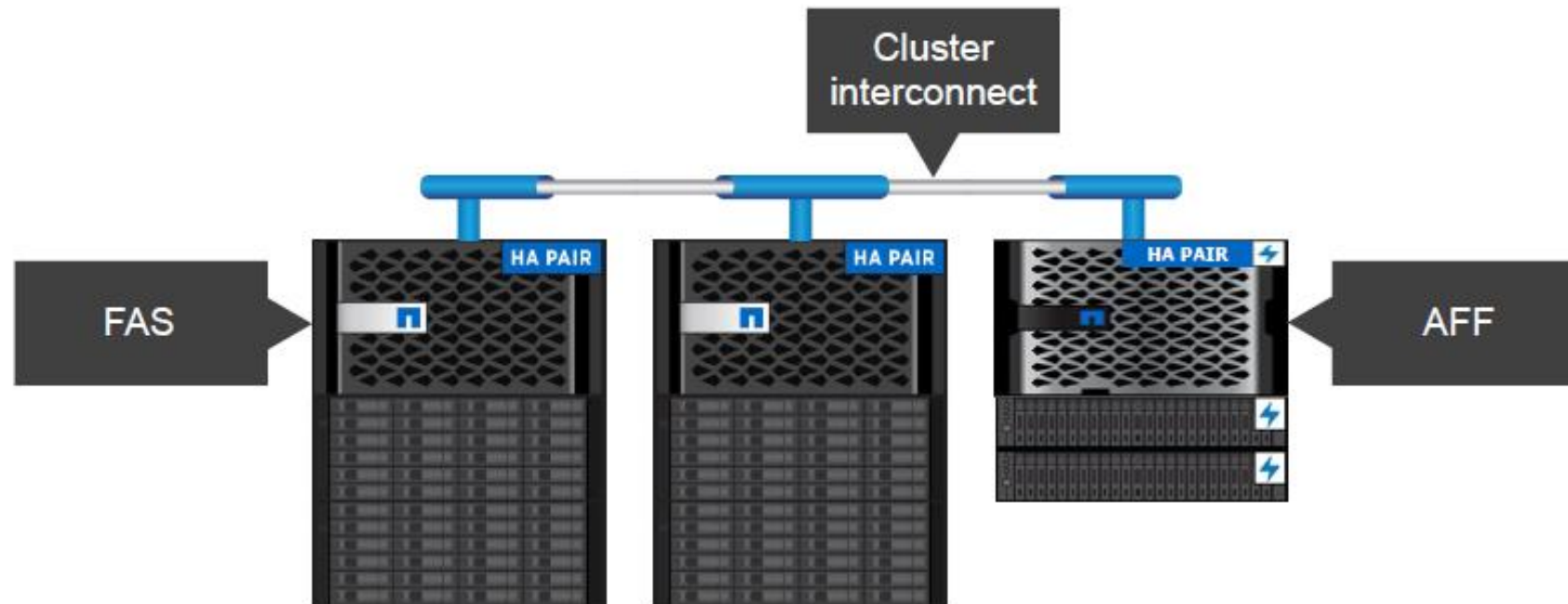
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System Architecture

- What makes up a clustered Data ONTAP storage system?



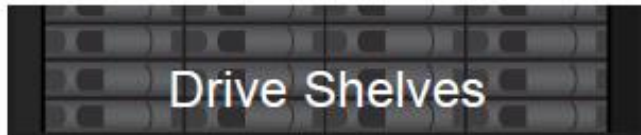
The Cluster



A cluster is one or more FAS or AFF controllers that run the ONTAP software. In ONTAP terminology, a controller is called a node. In clusters with more than one node, a cluster interconnect is required so that the nodes appear as one cluster.

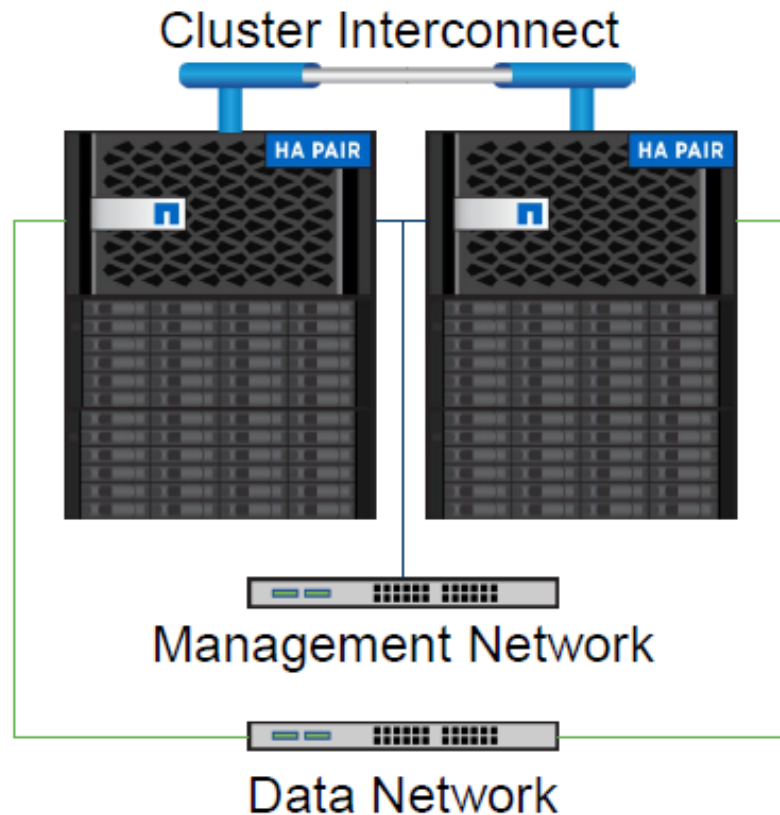
Node

Rear View



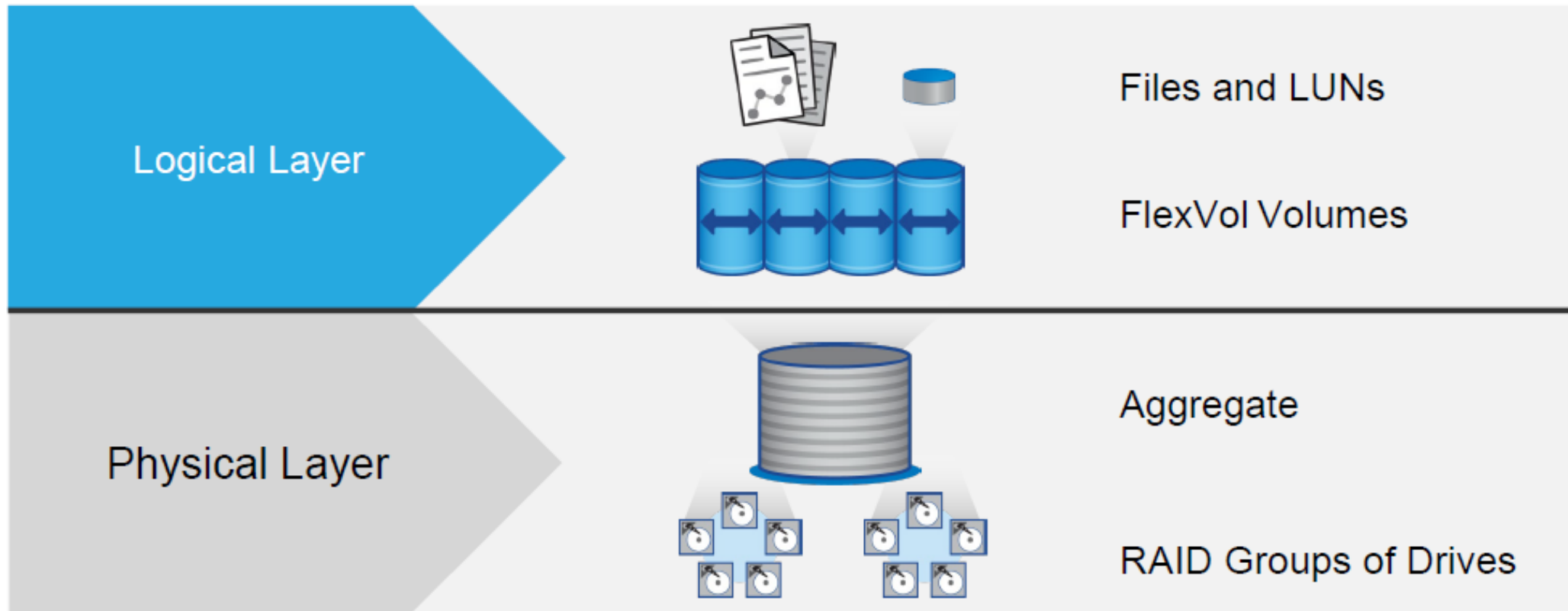
- A FAS or AFF storage controller that runs ONTAP software
- Storage and network ports
- Expansion slots
Not all entry-level systems have expansion slots.
- NVRAM or NVMEM
- Drive shelves or internal drives or both

Network



- Cluster interconnect:
 - Connection of nodes
 - Private network
- Management network:
 - Cluster administration
 - Ethernet network that can be shared with data
 - Recommended practice: dedicated management network
- Data network:
 - One or more networks for data access from clients or hosts
 - Ethernet, FC, or converged network

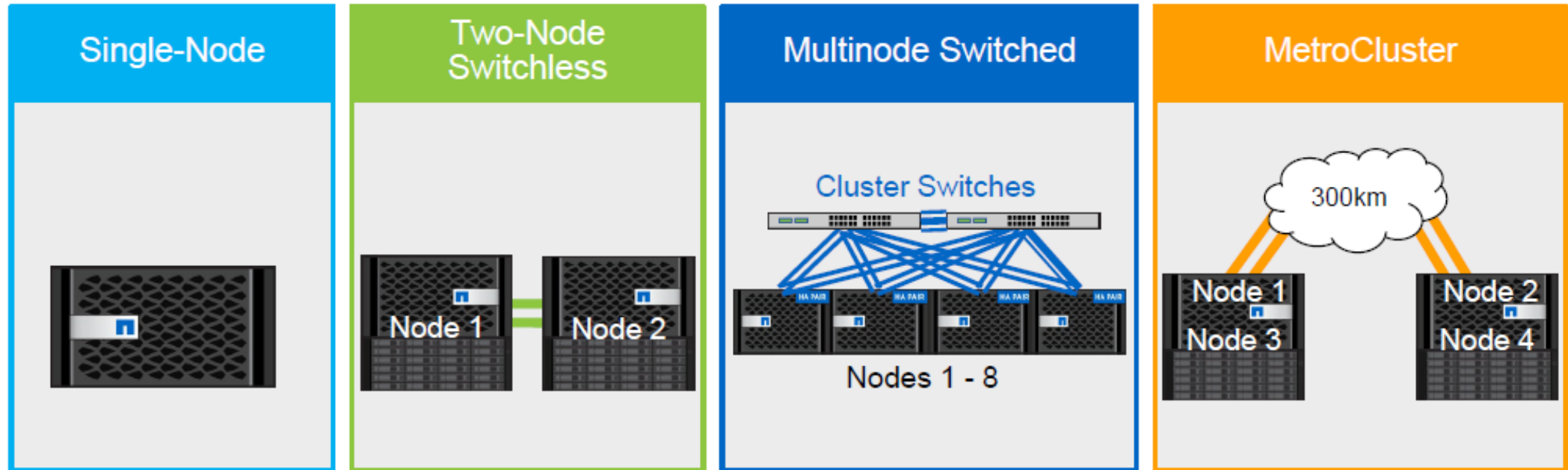
ONTAP Storage Architecture



NetApp ONTAP Cluster Setup

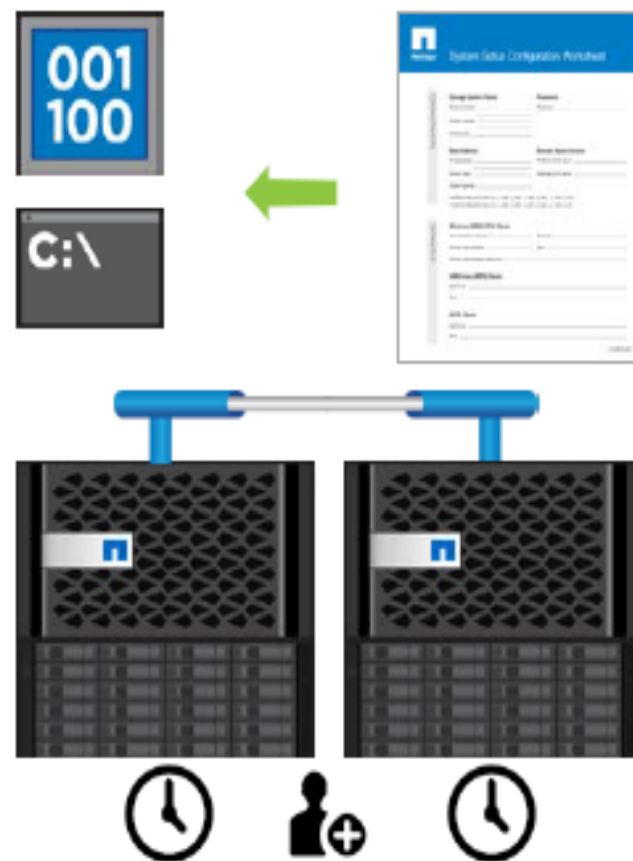
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Supported Cluster Configurations



Creating a Cluster

- **Cluster creation methods:**
 - Cluster setup wizard, using the CLI
 - Guided Cluster Setup, using OnCommand System Manager
- **The CLI method:**
 1. Create the cluster on the first node.
 2. Join the remaining nodes to the cluster.
 3. Configure the cluster time and AutoSupport.
- **The Guided Cluster Setup method:**
 1. Configure the node management interface using CLI.
 2. Connect to the node management IP address using your web browser.

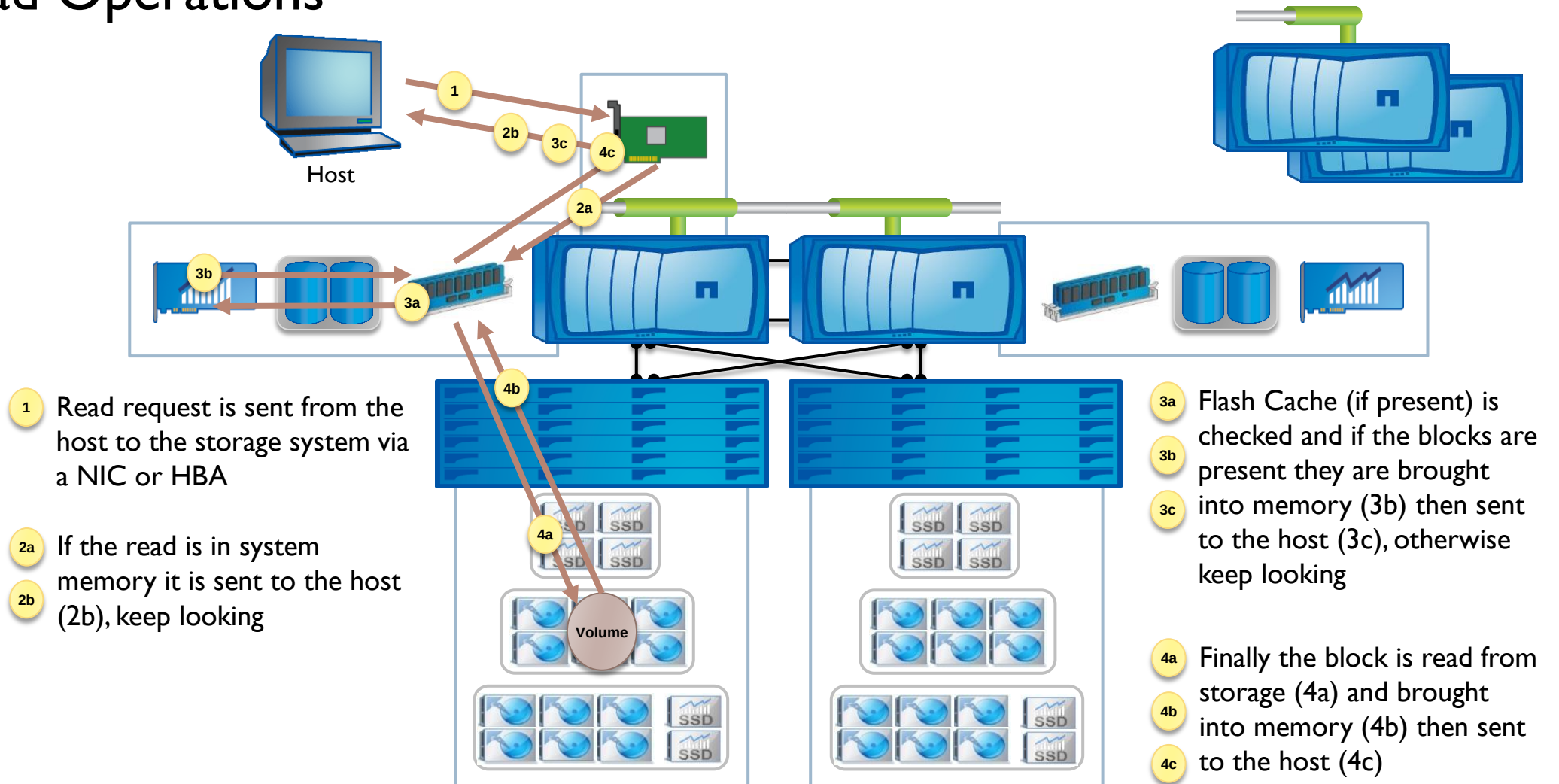


NetApp Data Flow (Read & Write)

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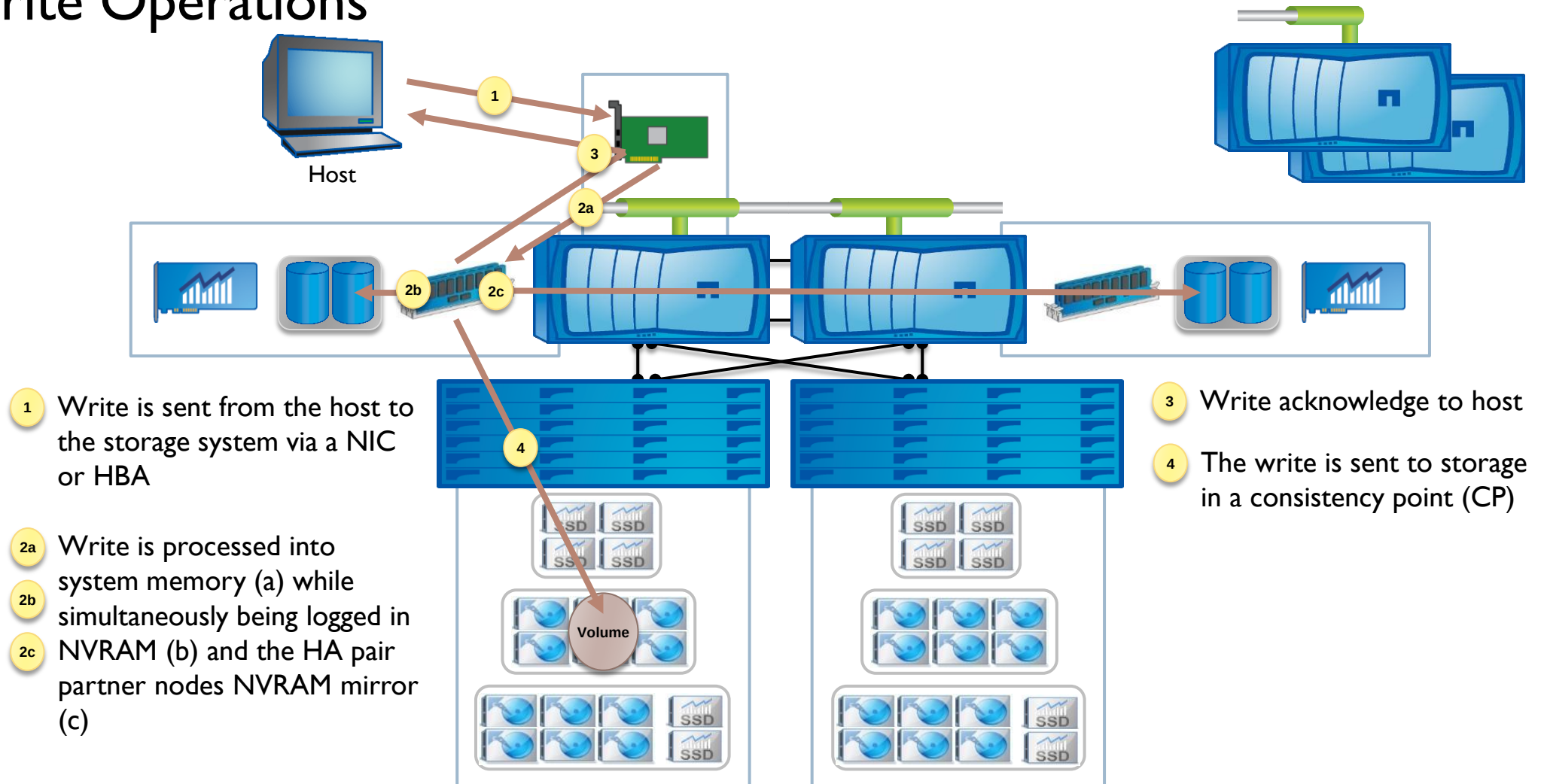
Operations within a Cluster

► Read Operations



Operations within a Cluster

► Write Operations



Components

- ▶ **Four major software components on every node:**
 - ▶ The Nblade (network “blade”)
 - ▶ The Dblade (data “blade”)
 - ▶ The SCSI-blade (SCSI SAN “blade”)
 - ▶ Management (also known as the M-host)
- ▶ **Other key software components on every node:**
 - ▶ The cluster session manager (CSM)
 - ▶ RDB units (the volume location database, or VLDB, for example)



RDB

- ▶ A **Replicated Database** (often referred as **RDB**) is a basis of ONTAP Cluster.
- ▶ This database contains cluster configuration information divided into **RDB units**.
- ▶ The RDB does not contain any user data, instead it provides a single point of management of all nodes within the cluster.



RDB is divided into 5 RDB units:

- ▶ **mgwd** (Management Gateway) – this unit contains info that **provides access to CLI** (Command Line). It also **enables management** of the cluster from any node.
- ▶ **vifmgr** – VIF (Virtual Interface(s)) Manager. This unit **stores and monitor LIF configuration**, as well as LIF failover policies. It is also responsible for automatic and manual LIF failover
- ▶ **vldb** – Volume Location Database. The VLDB tracks where the volumes and aggregates are across the cluster. It is an index of **which aggregates owns a volume**, and also, **which host owns the aggregate**.
- ▶ **bcomd** – Block Configuration and Operations Management. This unit stores **LUN map definitions**, and **igroups** (initiator groups) configuration.
- ▶ **crs** – Configuration Replication Services. This unit is related to **MetroCluster configuration**. It is responsible for being able to replicate configuration and operation data between clusters.



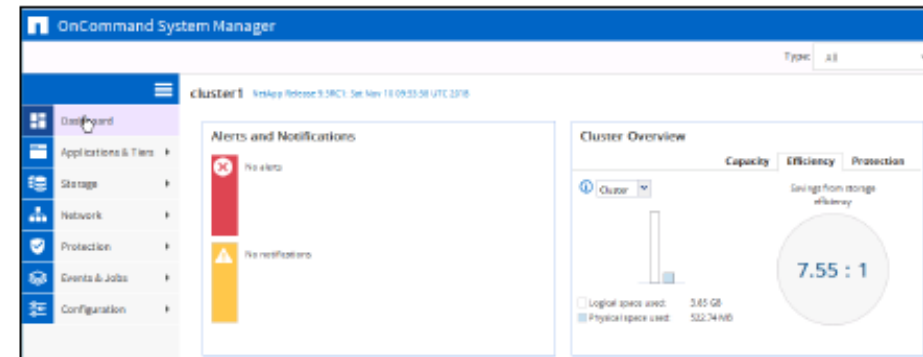


Administration Interfaces



Managing Resources in a Cluster

- OnCommand System Manager:
 - Visual representation of the available resources
 - Wizard-based resource creation
 - Best practice configurations
 - Limited advanced operations
- The CLI:
 - Manual or scripted commands
 - Manual resource creation that might require many steps
 - Ability to focus and switch quickly among specific objects
- Automation Tools:
 - OnCommand Workflow Automation (WFA)
 - API/ZAPI
 - Ansible



```
login as: admin
Using keyboard-interactive authentication.
Password: *****
cluster1::> cluster show
Node                               Health  Eligibility
-----
cluster1-01                        true    true
cluster1-02                        true    true
```

Three Available Command Shells

Clustershell



- The *clustershell* is the shell that is started automatically when you log in to the cluster.
- Its scope is the entire cluster.

Nodeshell



- The *nodeshell* is the shell that enables you to run a subset of 7-Mode commands from within clustered Data ONTAP.
- Its scope is the local controller.

Systemshell



- The *systemshell* is the low-level shell that is used for diagnostic and troubleshooting purposes only.
- Its scope is the local controller.



Clustershell

The default CLI, or shell, in ONTAP is called the clustershell and features the following:

- Inline help
- Online manual pages
- Command history
- Ability to reissue a command
- Keyboard shortcuts
- Queries and UNIX-style patterns
- Wildcards

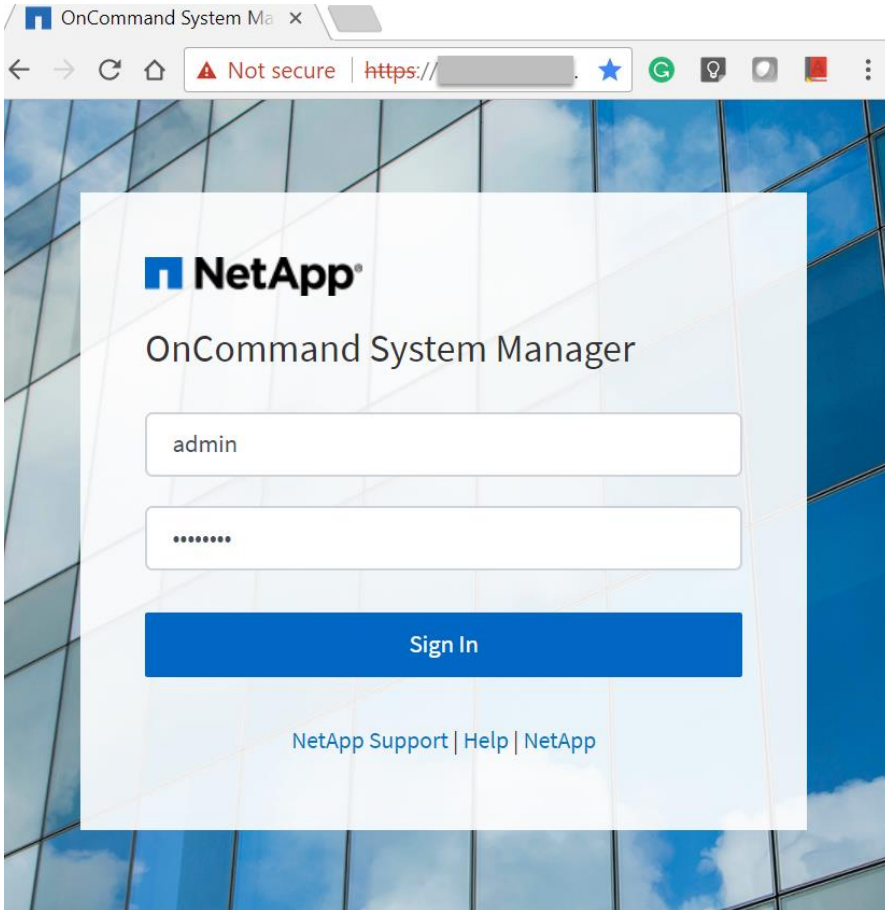
```
login as: admin
Using keyboard-interactive authentication.
Password: *****
svl-nau::> cluster show
```

Node	Health	Eligibility
-----	-----	-----
svl-nau-01	true	true
svl-nau-02	true	true

```
svl-nau::>
```

Technical overview

OnCommand System Manager 9.x



- **Simplified product launch experience:**
 - Access System Manager using cluster management LIF
 - Login using cluster administrator credentials
- HTML 5, no Java or Flash dependency
- End-to-end HTTPS communication by default
- Latest Web browsers supported

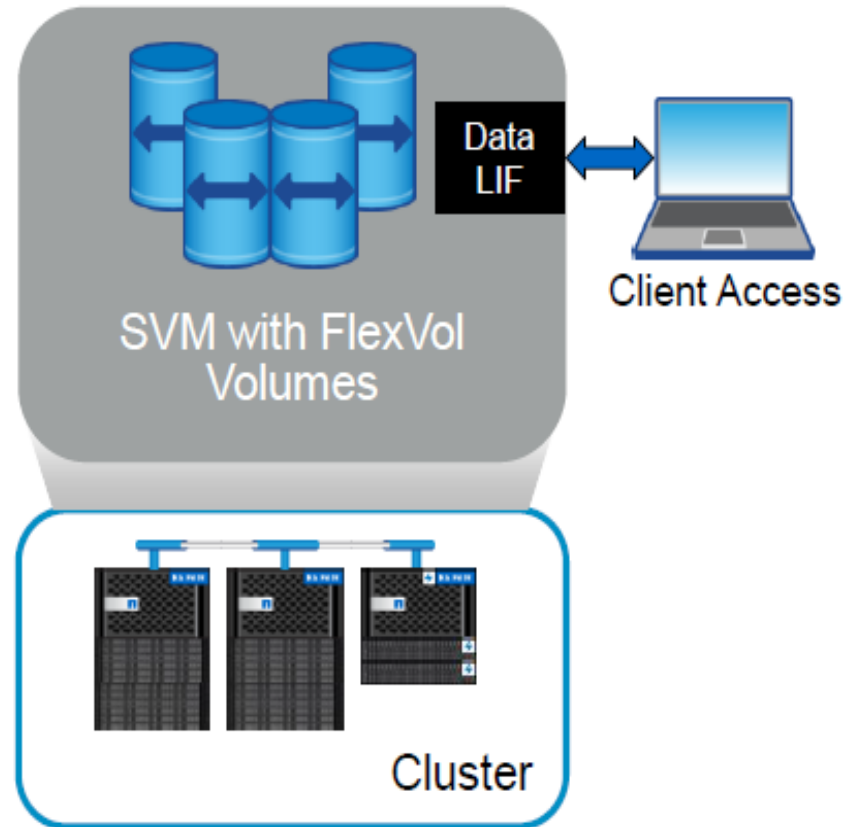
Storage Virtual Machine(SVM)

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What SVM's Are

- Storage Virtual Machines (SVMs, formerly known as Vservers) contain data volumes and one or more LIFs through which they serve data to the clients.
- SVMs securely isolate the shared virtualized data storage and network, and each SVM appears as a single dedicated server to the clients. Each SVM has a separate administrator authentication domain and can be managed independently by its SVM administrator.
- In a cluster, SVMs facilitate data access. A cluster must have at least one SVM to serve data. SVMs use the storage and network resources of the cluster.
- A cluster can have one or more SVMs with FlexVol volumes

SVM Components



- SVM:
 - Container for data volumes
 - Access of client data through a LIF
- Volume:
 - Logical data container for files or LUNs
 - One or more FlexVol volumes on an SVM or one scalable infinite volume on an SVM
- LIF:
 - Representation of the network address that is associated with a port
 - Access to client data

SVM Types

Admin SVM:

- Is created during cluster setup
- Represents the cluster
- Exists once per cluster
- Owns cluster-scoped resources

Node SVM:

- Is created during cluster setup
- Represents an individual node
- Exists once per node in the cluster
- Owns node-scoped resources

Data SVM:

- Provides client access to user data
- Includes data volumes, LIFs, and protocols and access control
- Is for multiple use cases:
 - Secure multitenancy
 - Separation of resources and workloads
 - Delegation of management

```
svl-nau::> vserver show
```

Vserver	Type	Subtype	Admin State	Operational State	Root Volume	Aggregate
svl-nau	admin	-	-	-	-	-
svl-nau-01	node	-	-	-	-	-
svl-nau-02	node	-	-	-	-	-
svm_green	data	default	running	running	svm_green_root	svl02_data_001
svm_red	data	default	running	running	svm_red_root	svl01_data_001
svm_yellow	data	default	running	running	svm_yellow_root	svl01_data_002

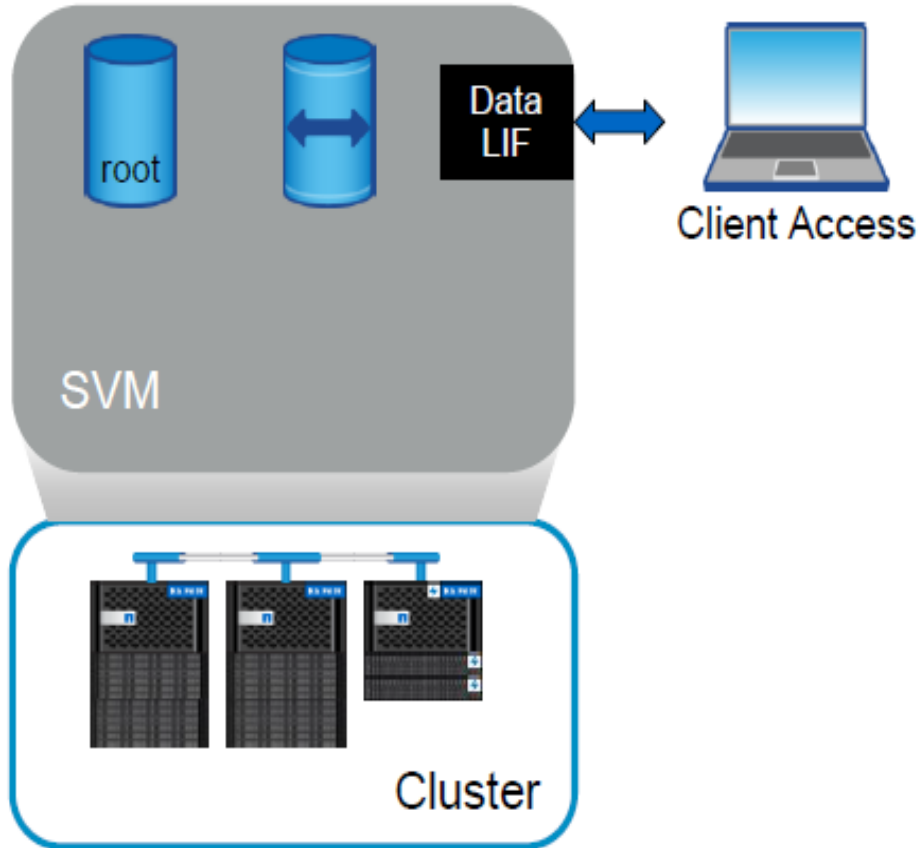
6 entries were displayed.

SVM Benefits

- Unified storage—SVMs with FlexVol volumes:
 - NAS protocols: CIFS and NFS
 - SAN protocols: iSCSI and FC (FCoE included)
- Nondisruptive operations (NDO) and nondisruptive upgrades (NDU):
 - Resource migration
 - Resource availability during hardware and software upgrades
- Scalability:
 - Adding and removing SVMs as needed
 - Modifying SVMs for data throughput and storage requirements on demand
- Secure multitenancy:
 - Partitioning of a storage system
 - Isolation of data and management
 - No data flow among SVMs in the cluster
- Delegation of management:
 - User authentication and administrator authentication
 - Access assigned by the cluster administrator



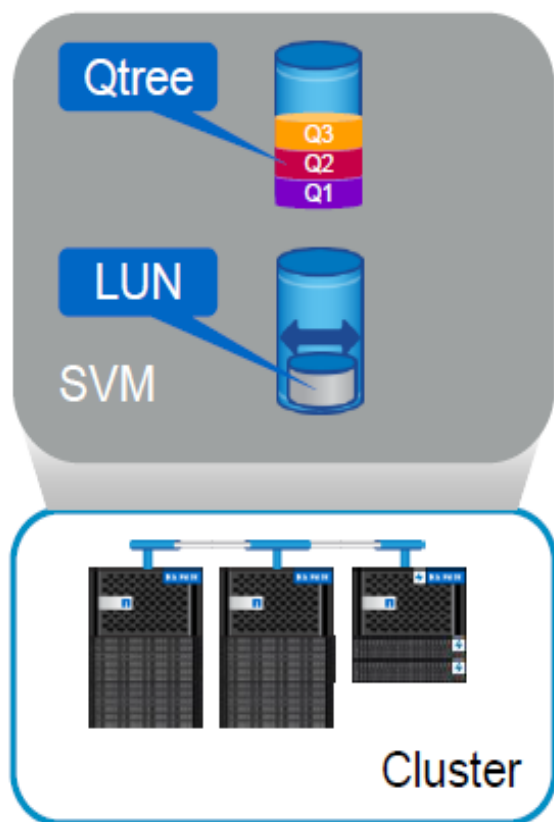
SVM Root Volume



Characteristics of root volume:

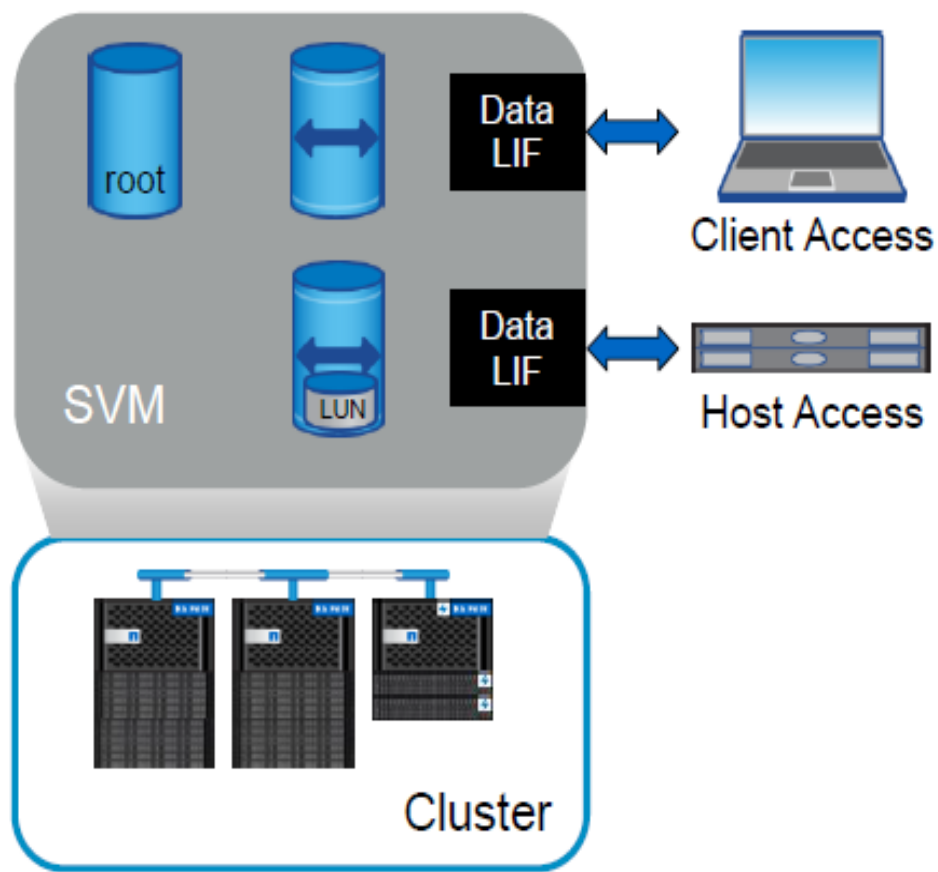
- Is created when the SVM is created
- Serves as the NAS client entry point to the namespace that an SVM provides
- Should not be used to store user data

SVM with FlexVol Volumes



- FlexVol volume:
 - Representation of the file system in a NAS environment
 - Container for LUNs in a SAN environment
- LUN: Logical unit that represents a SCSI disk
- Qtree:
 - Partitioning of FlexVol volumes into smaller segments
 - Management of quotas, security style, and CIFS opportunistic lock (oplock) settings

Data LIFs



- NAS data LIFs:
 - Multiprotocol (NFS, CIFS, or both)
 - Manually or automatically assigned IP addresses
 - Failover or migration to any node in the cluster
- SAN data LIFs:
 - Single protocol (FC or iSCSI):
 - FC LIF is assigned a worldwide port name (WWPN) when created.
 - iSCSI LIF IP addresses can be assigned manually or automatically.
 - No failover
 - Restrictions on migration



Cluster Management



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Managing Clusters

Access Control



Date and Time



Licenses



Jobs and Schedules



Administrators

- Tasks of cluster administrators:
 - Administer the entire cluster
 - Administer the cluster storage virtual machines (SVMs)
 - Create and delegate aggregates for SVM admin use
 - Can set up data SVMs and delegate SVM administration to SVM administrators
- Tasks of SVM administrators:
 - Administer only their own data SVMs
 - Can set up storage and network resources, such as volumes, protocols, logical interfaces (LIFs), and services



Admin Access

An admin account is for a pre-defined cluster administrator:

- Uses the CLI or NetApp OnCommand System Manager
- Is associated with cluster or data SVMs



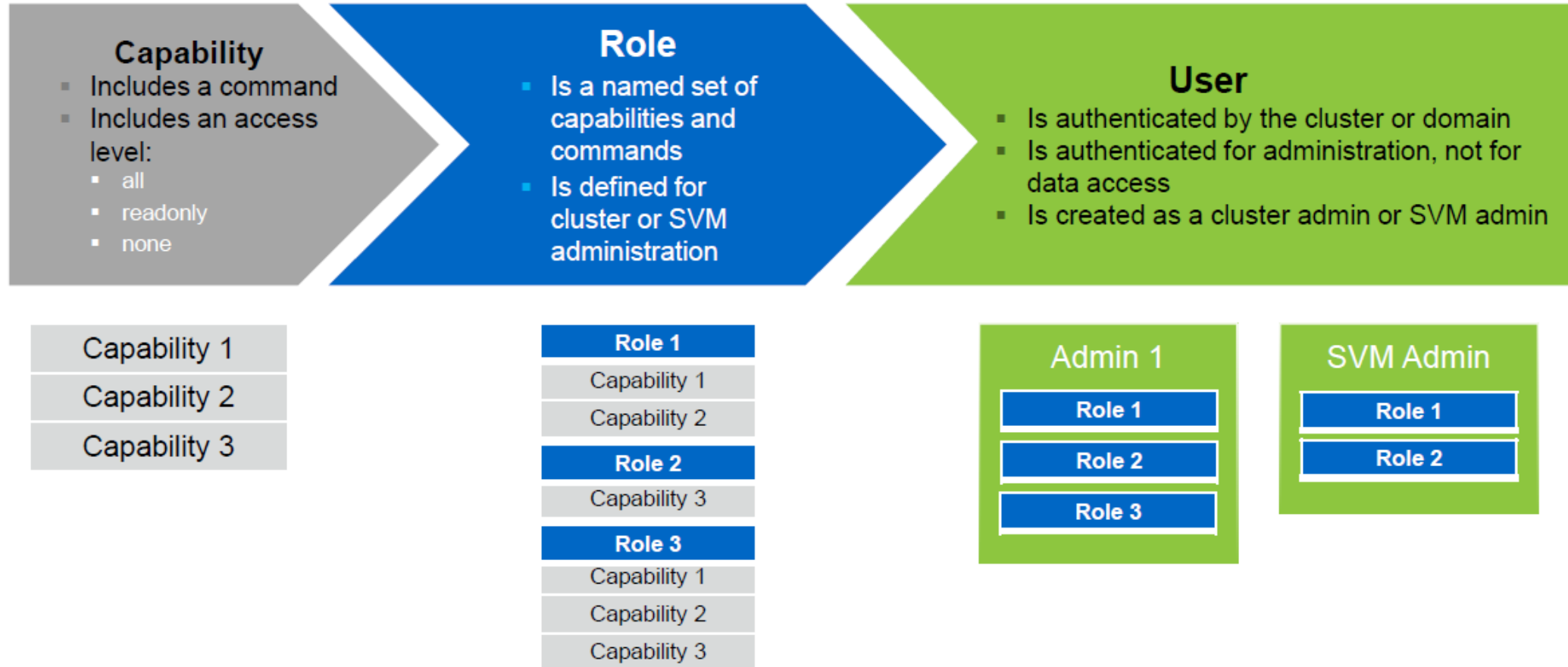
You create an admin account with role-based access control (RBAC):

```
cluster1::> security  
login
```



RBAC

RBAC users, roles, and capabilities



RBAC

Predefined roles in ONTAP software

Cluster-scoped roles (Cserver):

- admin
- backup
- readonly
- autosupport
- None

```
svl-nau::> security login role show -vserver svl-nau
```

Data SVM-scoped roles:

- vsadmin
- vsadmin-backup
- vsadmin-volume
- vsadmin-readonly
- vsadmin-protocol

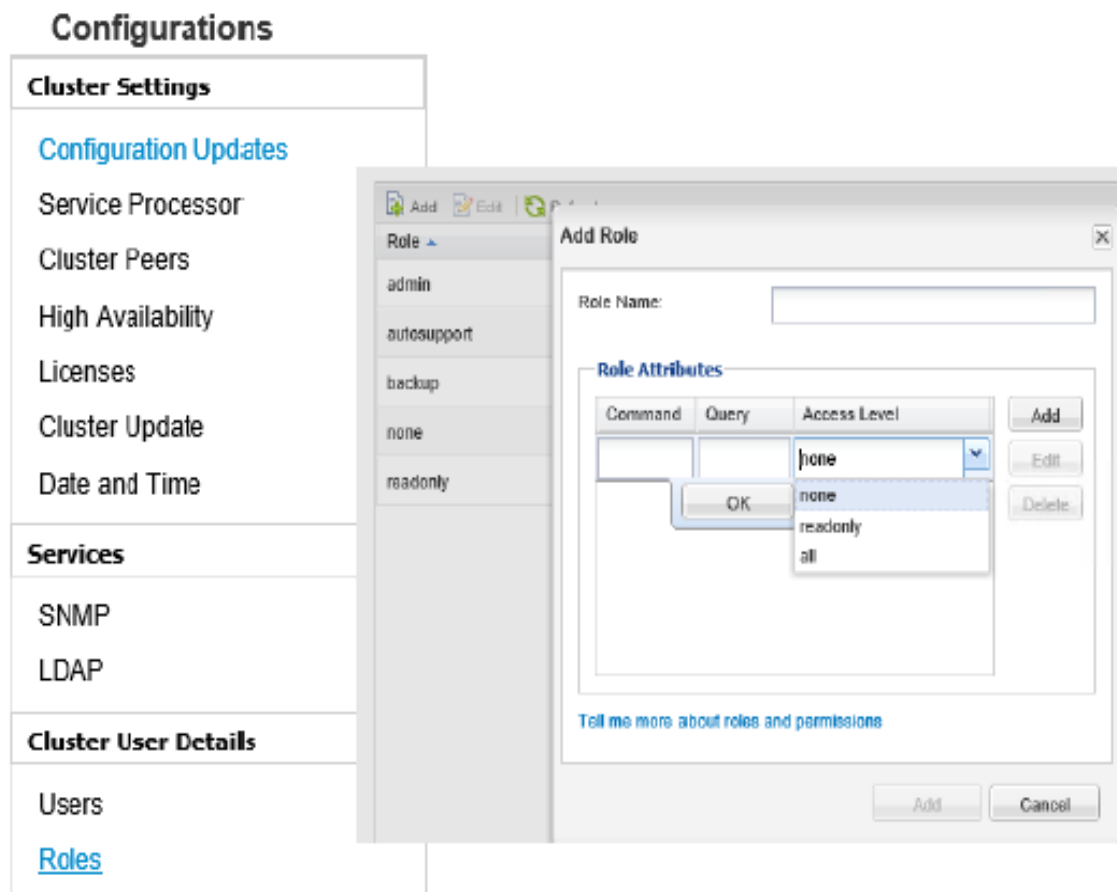
```
svl-nau::> security login role show -vserver svm_red
```



RBAC

Custom roles

- Role name
- Command directory
- Query



```
svl-nau::> security login role create...
```

```
svl-nau::> security login modify -vserver svm_red -user  
ken -role redvols
```

Date and Time

Ways to configure date and time:

- Manually: using CLI
- Automatically: using Network Time Protocol (NTP) servers

After you add an NTP server, the nodes require time to synchronize.

Configurations

Cluster Settings

Configuration Updates
Service Processor
Cluster Peers
High Availability
Licenses
Cluster Update
[Date and Time](#)

Services

SNMP
LDAP

Edit Date and Time

Date: Aug/18/2016

Time: 19:00

Time Zone: Etc/UTC

Time Servers:

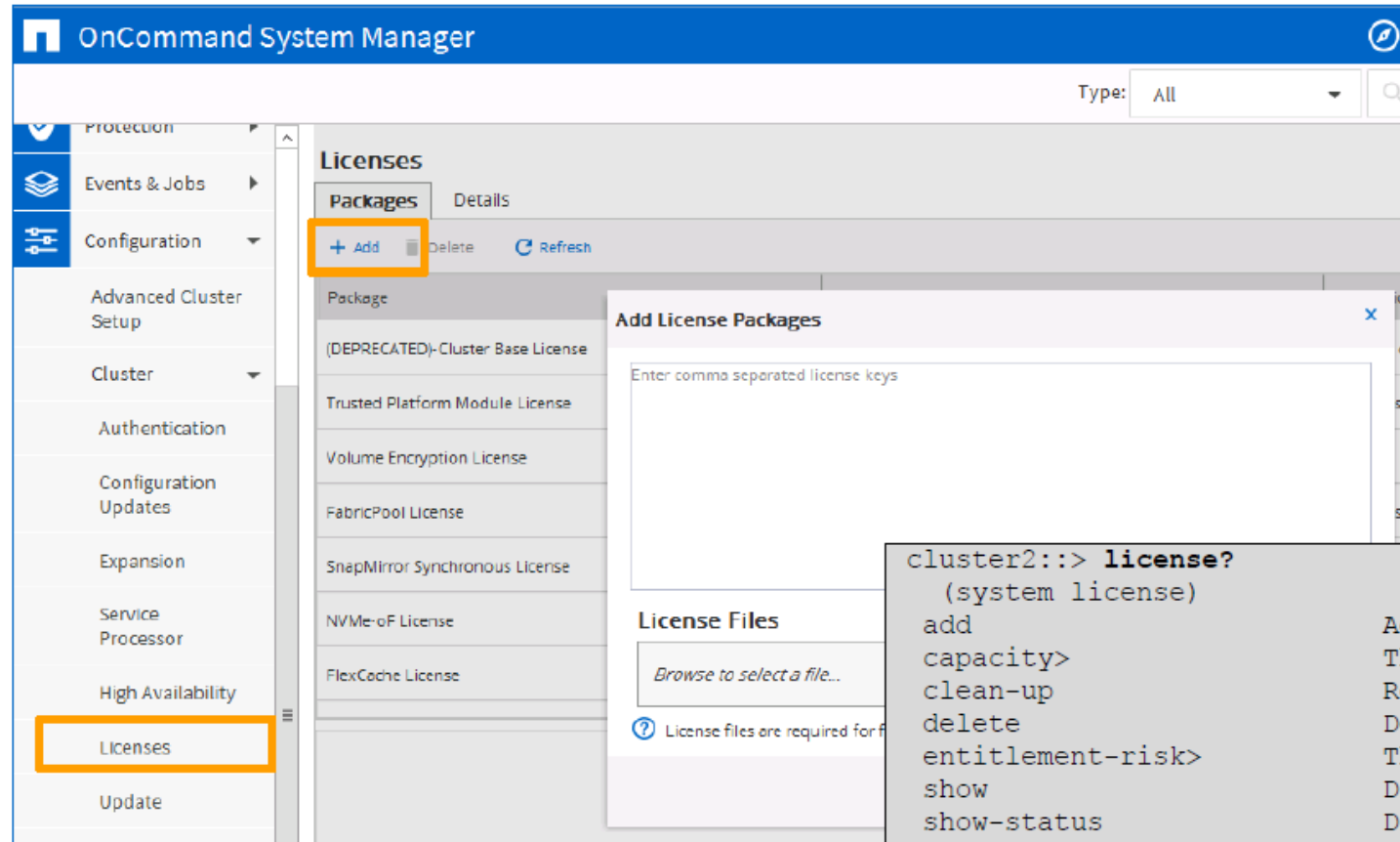
```
svl-nau::> cluster time-service ntp server create -server xx.xx.xx.xx  
svl-nau::> date
```

License Types

- Standard license
- Enterprise license
- Evaluation license
- Capacity license



License Commands



Policy-Based Storage Services

Jobs and
Schedules



Policy:

- A collection of rules that the cluster or SVM administrator creates and manages
- Predefined or created for managing data access

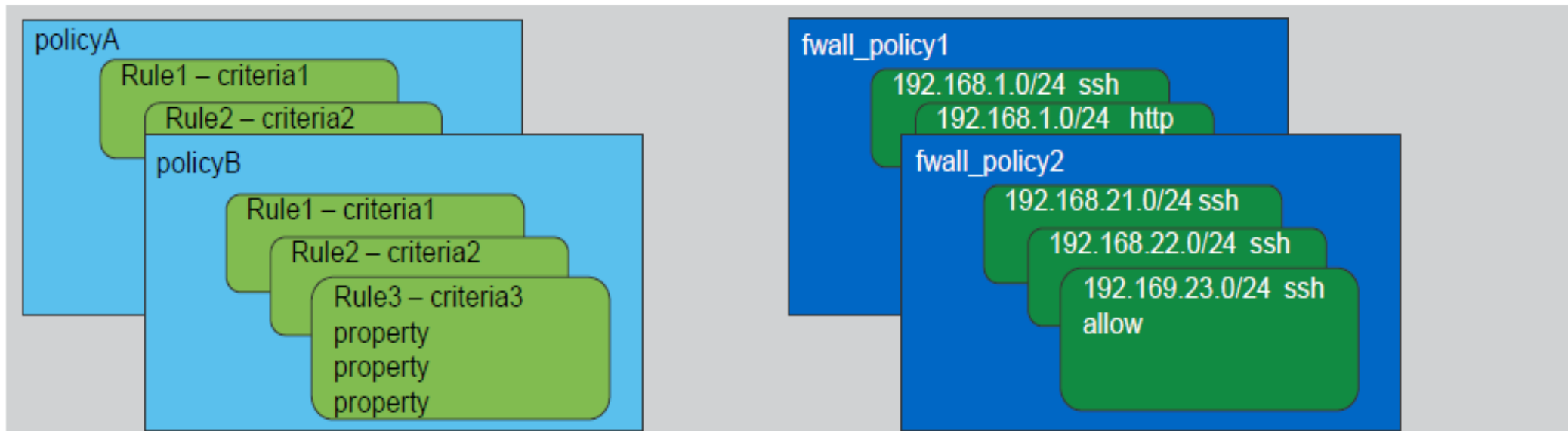
Policy examples:

- Firewall and security
- Export, quota, file, and data
- Snapshot and SnapMirror
- Quality of service (QoS)



Policy-Based Management

- Assign a policy to a service or resource.
- A rule criteria in the policy matches the service or resource.
- The matching rule properties apply to the service or resource.
- The example is a firewall to permit or deny access to a protocol for specific IP address ranges.



Jobs

- Asynchronous tasks
- Managed by the Job Manager
- Long-running operations
- In a job queue

NetApp OnCommand System Manager

Dashboard LUNs SVMs Network **Hardware and Diagnostics** Protection

Jobs

Current Jobs Job History

Refresh

Job ID	Start Time	Job Name	Node	State
1	10/20/2015 23:54:57	SnapMirror Servic...	cluster1-01	dorm
4	11/04/2015 00:00:00	Certificate Expiry...		que
5	11/03/2015 18:15:04	CLUSTER BACK...		que
6	11/03/2015 00:10:05	CLUSTER BACK		que

Jobs

System Alerts

AutoSupport

Flashpool Statistics

```
rtp-nau::> job show
```

Job ID	Name	Owning Vserver	Node	State
2	Vol Reaper	rtp-nau	-	Queued
Description: Vol Reaper Job				
6	SnapMirror Service Job	rtp-nau	rtp-nau-01	Dormant
Description: SnapMirror Service Job				

Schedules

Schedules for tasks:

- Time-based schedules, which run at specific times (similar to UNIX cron schedules)
- Interval-based schedules, which run at intervals

```
rtp-nau::> job schedule show
```

Name	Type	Description
5min	cron	@:00,:05,:10,:15,:20,:25,:30,:35
8hour	cron	@2:15,10:15,18:15
Auto Balance Aggregate Scheduler	interval	Every 1h
RepositoryBalanceMonitorJobSchedule	interval	Every 10m
daily	cron	@0:10
hourly	cron	@:05
weekly	cron	Sun@0:15

Schedules	
 Create  Edit  Delete  Refresh	
Name	Type
5min	Time based
8hour	Time based
Auto Balance Aggregate Scheduler	Interval based
daily	Time based
hourly	Time based
RepositoryBalanceMonitorJobSchedule	Interval based
weekly	Time based
Details:	
Runs at:	
Hours:	Every hour
Minutes:	0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50 and 55th minute

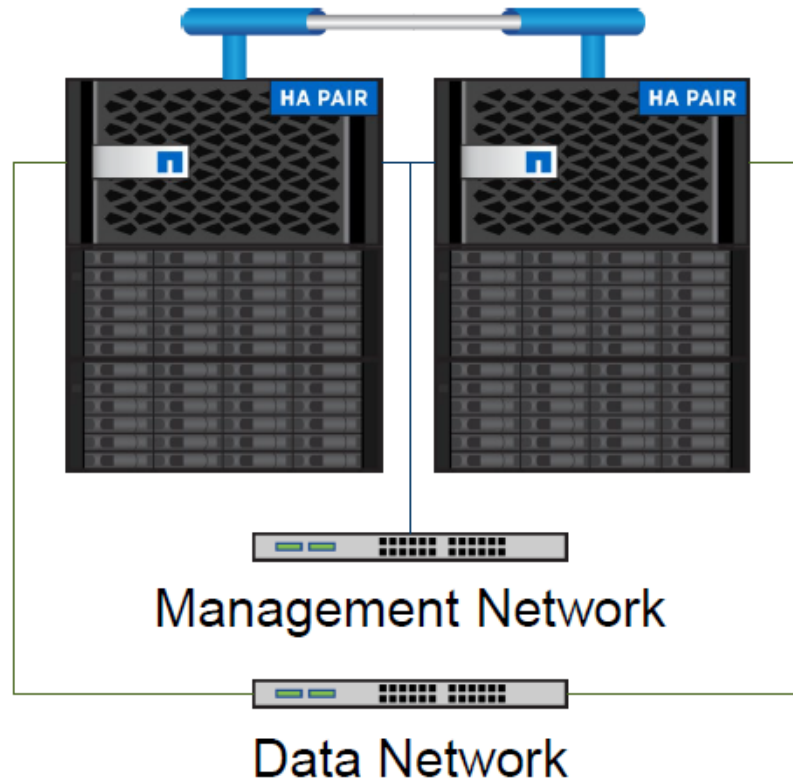


Network Management



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ONTAP Network Types



- **Cluster interconnect:**
 - Connection of nodes
 - Private network
- **Management network:**
 - Cluster administration network
 - Possibly a shared Ethernet network with data
- **Data network:**
 - One or more networks that are used for data access from clients or hosts
 - Ethernet, FC, or converged network

Port Types

- Physical port
 - Ethernet
 - FC
 - Unified Target Adapter (UTA)
 - UTA is a 10-GbE port
 - UTA2 is configured as either:
 - 10-GbE
 - or 16-Gbps FC
- Virtual port
 - Interface group (ifgrp)
 - Virtual LAN (VLAN)



Physical Port Identification

- Ethernet ports are named `e<location><letter>`:
 - e0a is the first port on the controller motherboard.
 - e3a is a port on a card in slot 3.
- FC ports are named `<location><letter>`:
 - 0a is the first port on the controller motherboard.
 - 3a is a port on a card in slot 3.
- UTA ports have both an Ethernet name and an FC name, `e<location><letter>/<location><letter>`:
 - e0a/0a is the first port on the controller motherboard.
 - e3a/3a is a port on a card in slot 3.
 - Use of show commands returns only FC label names (even in Ethernet mode).



LIFs

- An IP address or worldwide port name (WWPN) is associated with a LIF:
 - If subnets are configured (recommended), an IP address is automatically assigned when a LIF is created. Otherwise, IP addresses must be manually assigned.
 - When an FC LIF is created, WWPNs are automatically assigned.
- One node-management LIF exists per node.
- One cluster-management LIF exists per cluster.
- Cluster LIFs depend on the cluster configuration.
- Multiple data LIFs are enabled per port (client-facing for NFS, CIFS, iSCSI, and FC access).
- For intercluster peering, intercluster LIFs must be created on each node.



Creating Data LIFs

Specify the subnet name to automatically assign an IP address.

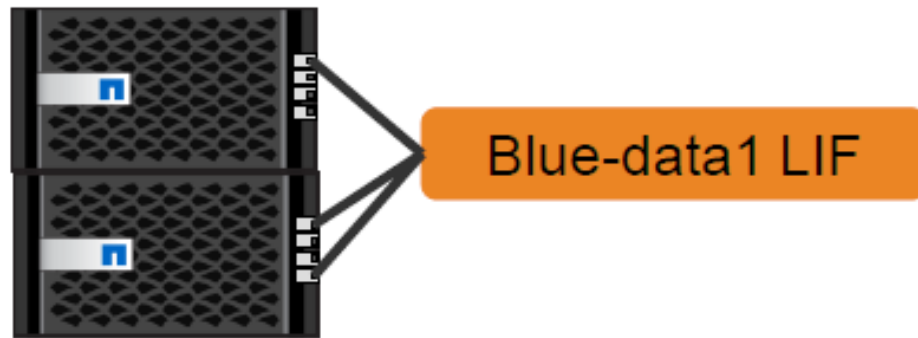
You must specify the IP address when subnets are not configured.

The screenshot shows the NetScout UI. On the left, a sidebar menu has 'Network Interfaces' highlighted. The main panel shows a list of network interfaces with columns for 'Interface Name' and status. A 'Create Network Interface' dialog is open on the right. The dialog fields are: Name: 'svm3_nfs_lif2', Interface Role: 'Serves Data', SVM: 'svm3', Protocol Access: 'CIFS' and 'NFS' checked, Management Access: 'Enable Management Access' unchecked, Assign IP Address: 'Using a subnet', Subnet: 'subnet', Port: 'e0f'. A table at the bottom shows available ports and adapters. The 'Create' button is at the bottom right.

Ports or Adapters	Hosted Interface C...	Speed
e0e	0	1000 Mbps
e0f	0	1000 Mbps
e0g	0	1000 Mbps

```
cluster1::> network interface create -vserver svm3 -lif svm3_nfs_lif2 -role data -  
data-protocol nfs -home-node cluster1-01 -home-port e0f -subnet-name  
snDefault
```

Nondisruptive LIF Features

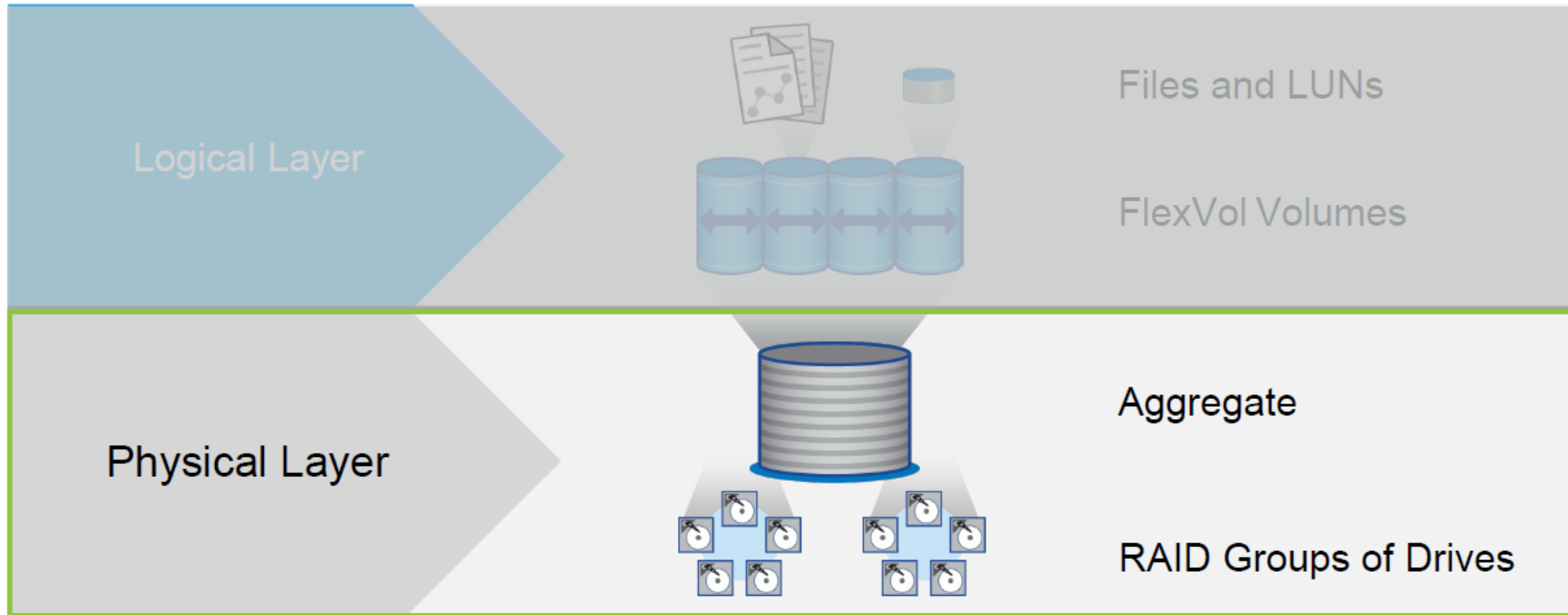


- **LIF failover:** Automatic migration that occurs because of a link failure or reboot
- **LIF migrate:** Manual move of a LIF to another port
- **LIF revert:** Manually or automatically sending a LIF back to the home (node and) port

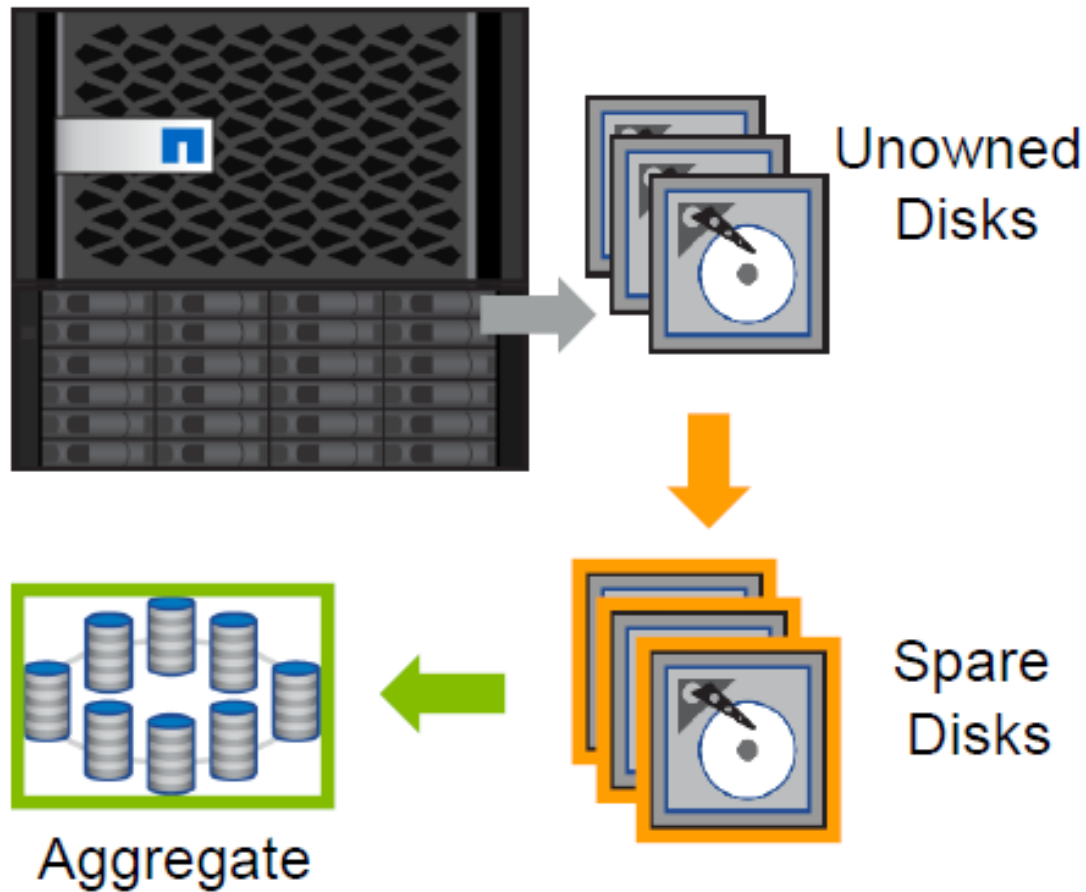
Physical Storage Management

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ONTAP Storage Architecture



Disks and Aggregates



- What happens when a disk is inserted into a system:
 - The disk is initially unassigned.
 - By default, disk ownership is assigned automatically.
 - Disk ownership can be changed.
- What happens after ownership is assigned:
 - The disk functions as a hot spare.
 - The disk can be assigned to an aggregate.

Disk Ownership

- A disk is unusable until assigned to a controller.
- Disk ownership determines which controller owns a disk:
 - Ownership is automatically assigned (default).
 - Ownership can be manually assigned or changed.
 - Software disk ownership is made persistent by writing the ownership information onto the disk.
- Disks can be unassigned.

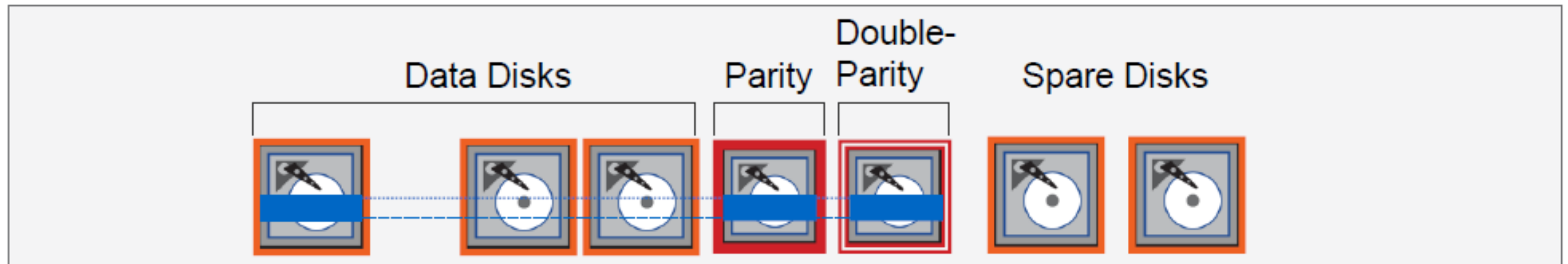
```
svl-nau::> storage disk show -container-type unassigned
```

Disk	Usable Size	Shelf	Bay	Container Type	Position	Aggregate	Owner
9.11.18	-	1	1	unassigned	present	-	-



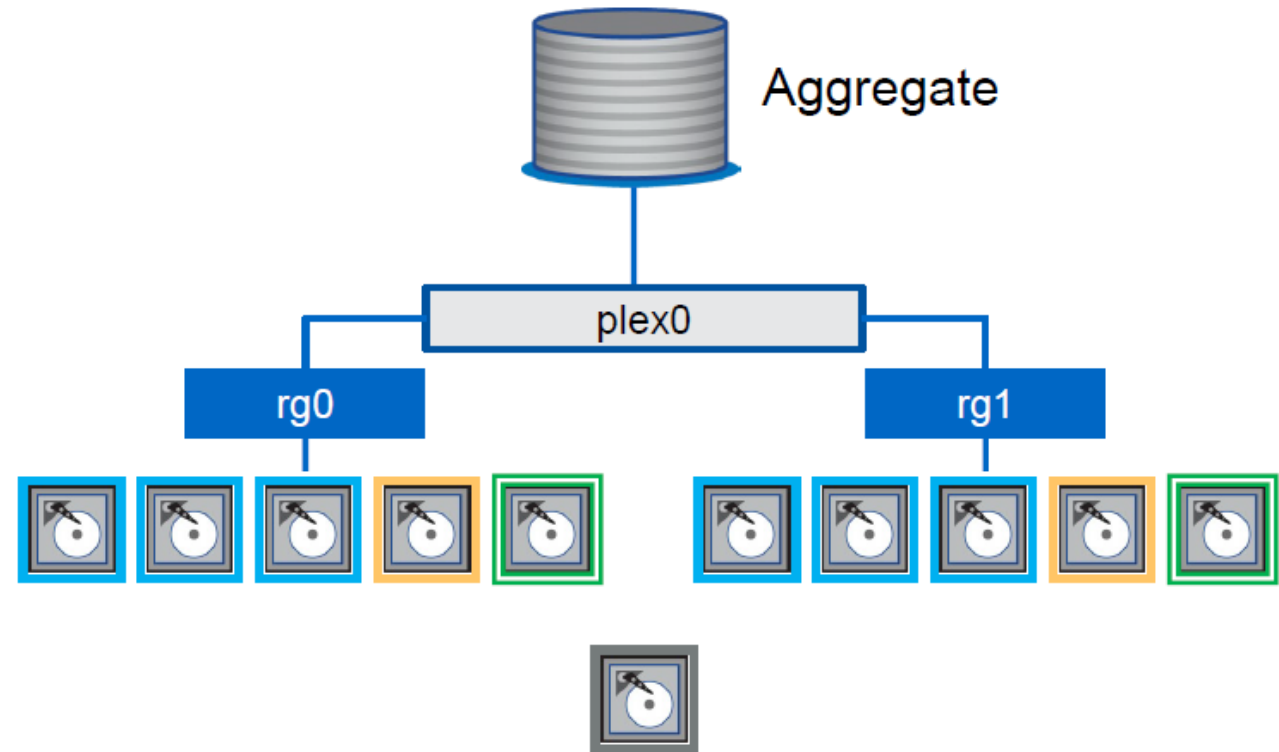
Spare Disks

- Spare disks are used to do the following:
 - Create an aggregate
 - Increase aggregate capacity
 - Replace failed disks
- Disks must be zeroed before use:
 - Disks are automatically zeroed when they are added to an aggregate.
 - NetApp recommends manually zeroing disks before use.



Physical Storage Hierarchy

- **Drive:**
Hard disk drive (HDD) or solid-state drive (SSD or NVMe)
- **RAID group:**
Drive-level protection
- **Plex:**
Logical container for RAID groups
Used by mirrored aggregates
- **Aggregate:**
Logical pool of storage



Drive Types



SATA

Same technology that is used in consumer disk drives:

- Single I/O path
- High capacity but moderate IOPS
- Low signal voltages: cable length limit of 1 meter
- SAS-to-SATA adapters that enable use in SAS shelves



SAS

Serial attached SCSI:

- Point-to-point serial protocol
- Multipath I/O
- Moderate capacity by high IOPS
- Higher signal voltage: cable lengths up to 10 meters

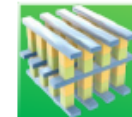


SSD

Based on flash memory chip technology that is similar to USB flash drives:

- No spinning platter
- Quick reads and writes
- Use similar to that of a SAS hard disk drive

Can also be used as an aggregate-specific cache



NVMe

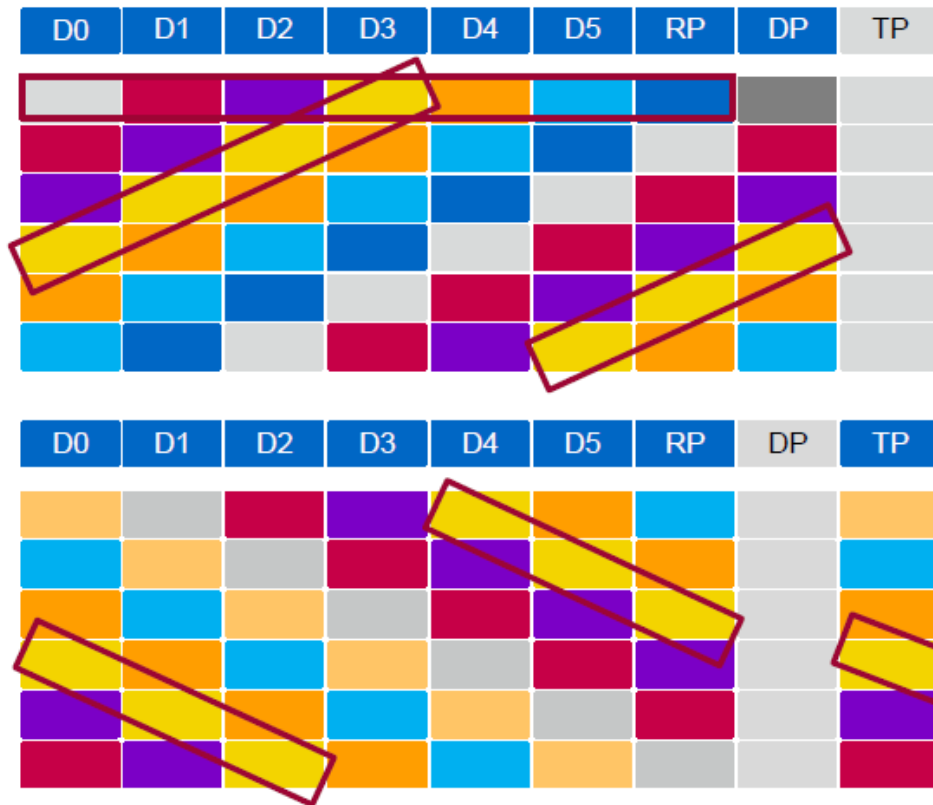
Nonvolatile memory enhanced:

- Storage-class memory (SCM) product that passes I/O directly over the Peripheral Component Interconnect Express (PCIe) bus.
- A flash module to accelerate spinning disks
- A drive to complement or substitute for SSDs



ONTAP RAID Technologies

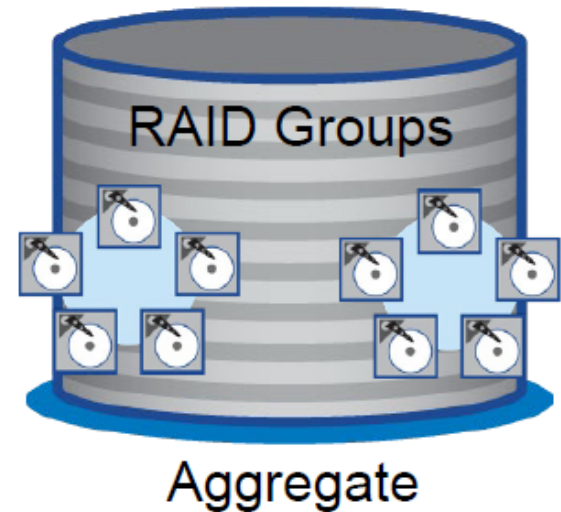
Description



- RAID 4 (row parity)
 - Adds a *row parity* drive
 - Protects against single-disk failure or media error
- RAID DP (double parity) technology
 - Adds a *diagonal parity* disk to a RAID 4 group
 - Protects against two concurrent drive failures within a RAID group
- RAID-TEC (triple erasure coding) technology
 - Adds a *triple-parity* disk to a RAID DP group
 - Protects against three concurrent drive failures

Aggregates

- Aggregates are logical containers for the drives that are managed by a node.
- Aggregates consist of one or more RAID groups.
- You can use aggregates to do the following:
 - Isolate workloads with different performance demands
 - Tier data with different access patterns
 - Segregate data for regulatory purposes
- A single node owns an aggregate, but ownership can be transferred to the partner in a high-availability (HA) pair.
 - During an HA failover, aggregate ownership is temporarily transferred to the surviving partner.



RAID Group Sizes

- Default for the following:
 - Near-line class disks (SATA or NL-SAS) of size 6TB or larger HDDs
 - Required for 10TB and larger HDDs
- Optional for other disks (SSD or SAS)
- Default RAID group sizes:
 - 21 disks for SATA or NL-SAS disks
 - 24 disks for SAS disks
- Ability to upgrade and downgrade nondisruptively between RAID types

Disk Type	Group Type	Default	Maximum
SATA	RAID4	7	7
	RAID-DP	14	20
	RAID-TEC	21	29
NL-SAS	RAID4	7	7
	RAID-DP	14	20
	RAID-TEC	21	29
SAS	RAID4	8	14
	RAID-DP	16	28
	RAID-TEC	24	29
SSD	RAID4	8	14
	RAID-DP	23	28
	RAID-TEC	24	29



Create an Aggregate

Information to provide:

- Aggregate name
- Disk type
- Owning node
- Number of disks
- RAID type

Aggregates: Create Aggregate

Enter Aggregate Details Manually Create Aggregate ☒

To create an aggregate, select a disk type then specify the number of disks.

Name:

Disk Type: [Browse](#)

Disks of 3.93 GB each from node: cluster1-02

Number of Disks: Max: 15 (excluding 1 hot spare), min: 5 for RAID-DP

RAID Configuration: RAID-DP: RAID group size of 16 disks [Change](#)

New Usable Capacity: 38.67 GB (Estimated)

☐ FabricPool [Tell me more about FabricPool](#)

☐ Cloud Tier

☐ Mirror this aggregate [Tell me more about mirrored aggregates](#)

☐ Use Flash Pool cache with this aggregate

[Cancel](#) [Submit](#)

```
cluster2::> aggr create -aggregate n2_data_02 -node cluster1-02  
-disktype fcml -diskcount 8
```

Adding Capacity to Aggregates

Provide the following information:

- Aggregate name
- Disk type
- Number of disks

You cannot shrink aggregates.

Aggregates

Add Capacity

Review the existing disks of the aggregate and select disks to add to the aggregate.

Aggregate Name: n1_data_001
Node: cluster1-01
Existing Usable Capacity: 17.58 GB
Existing Disks or Partitions:

Disk Type	Node	Disk Size	RPM	Checksum	Pool	Count	Total Capac...
FCAL	cluster1-01	3.93 GB	15000	block	Pool 0	7	27.48 GB

② Disk Type to Add:

3.93 GB disks from node: cluster1-01

Number of Disks: Max: 9 (excluding hot spares: 1)

Add Disks To: All RAID groups [Change](#)

New Usable Capacity: 24.61 GB (Estimated)

```
::> storage disk show -spare -owner cluster1-01  
::> storage aggregate add-disks -aggr n1_data_001 disks 2
```

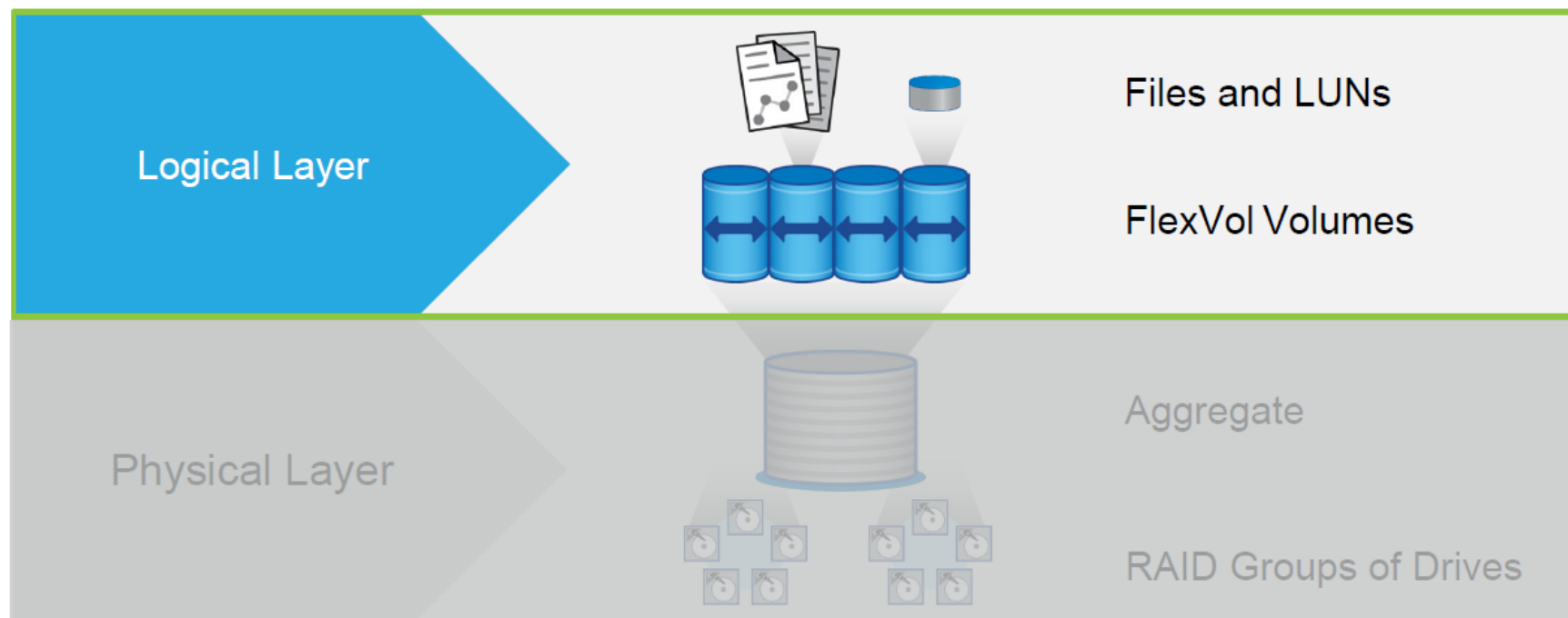


Logical Storage Management



Senthilkumar Muthusamy

ONTAP Storage Architecture

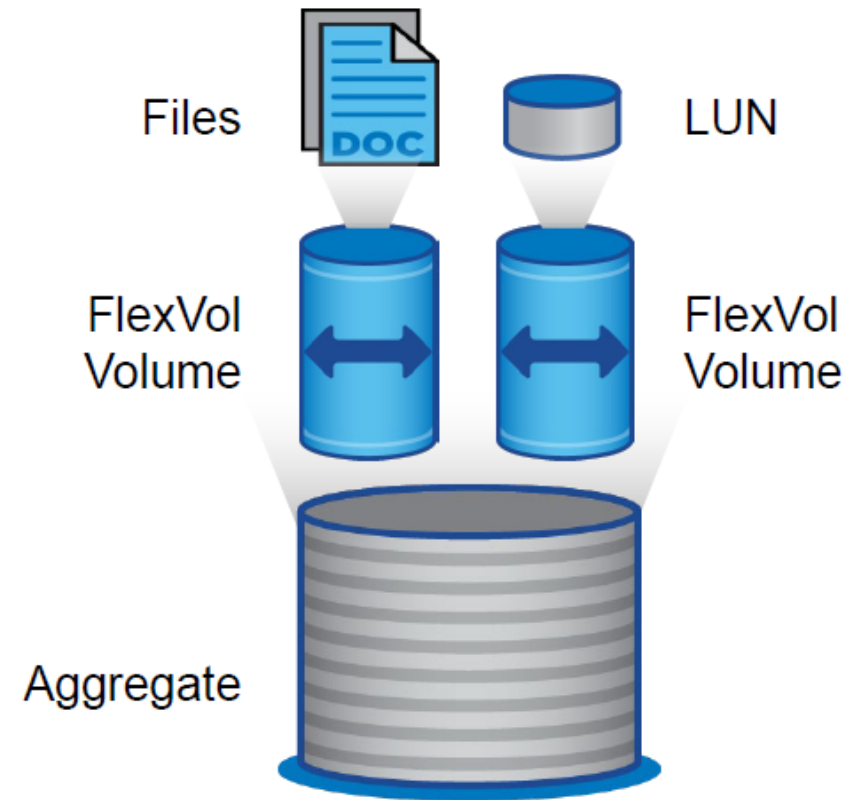


In ONTAP software, disks are grouped into RAID groups. An aggregate is a collection of physical disk space that contains one or more RAID groups. Each aggregate has a RAID configuration and a set of assigned disks. The disks, RAID groups, and aggregates make up the physical storage layer.

Within each aggregate, you can create one or more FlexVol volumes. A FlexVol volume is an allocation of disk space that is a portion of the available space in the aggregate. A FlexVol volume can contain files or LUNs. The FlexVol volumes, files, and LUNs make up the logical storage layer.

FlexVol Volumes

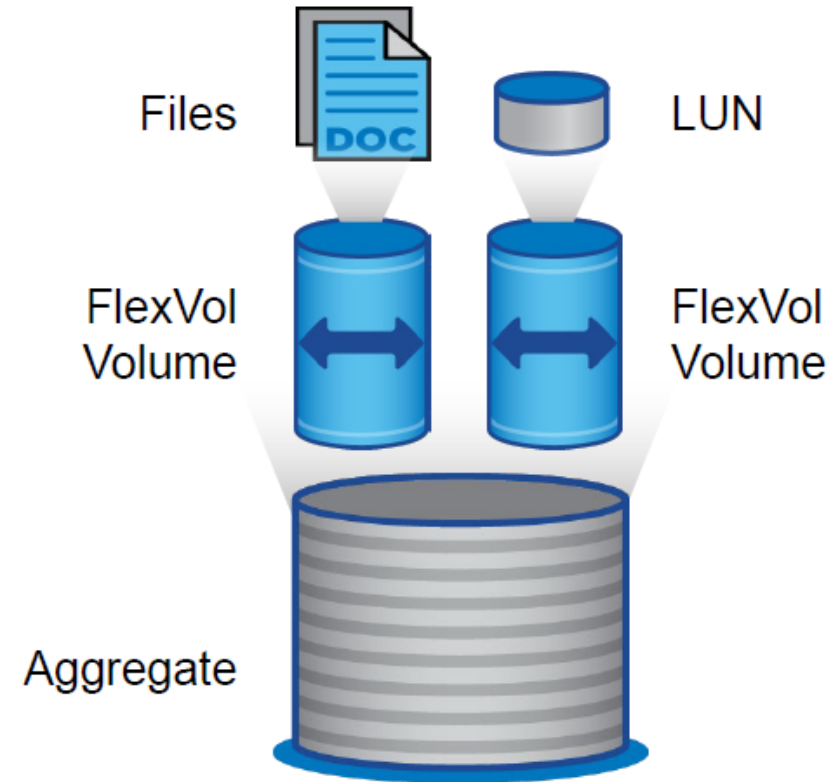
- Can contain NAS, SAN, or both types of data (mixing is not recommended).
- Are contained within an aggregate, and an aggregate can hold multiple FlexVol volumes.
- Can increase or decrease in size, as needed.



FlexVol Volumes

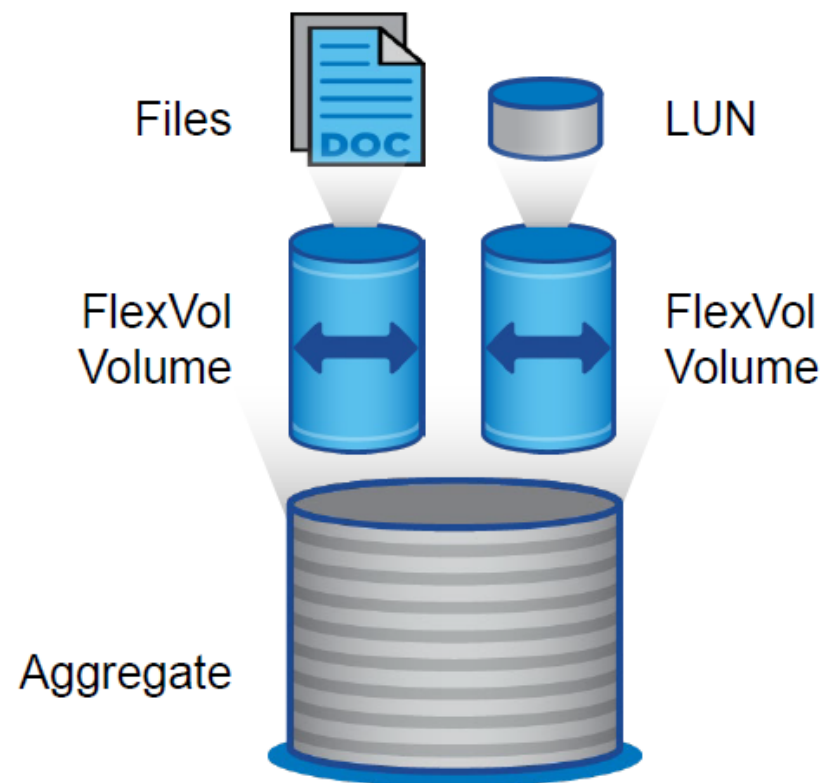
Types

- **System (or node root):**
 - Typically named vol0
 - Contains only configuration and logs
 - Cannot contain user data
 - Owned by the node SVM
- **SVM root volume:**
 - Top level of the namespace
 - Should not contain user data
- **Data:**
 - **NAS:** Contains file systems for user data
 - **SAN:** Contains LUNs



Files and LUNs

- A file refers to any data (including text files, spreadsheets, and databases) that is exported to or shared with NAS clients.
- A LUN represents a logical drive that a SCSI protocol (FC or iSCSI) addresses:
 - Block level
 - Data accessible only by a properly mapped SCSI host



Volume Properties

Actions that can be taken on volumes



Create

Edit

Resize

Delete

Clone

Move

Rehost

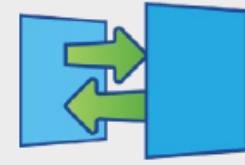
Volume options



Storage efficiency

Storage quality
of service (QoS)

Tools to protect volumes



Snapshot copies

Mirror copies**

Vaults**

Create a Flexible Volume in an SVM

Information to provide:

- Volume name
- Aggregate name
- Storage type
- Capacity

Create Volume

General | Storage Efficiency | Quality of Service | Protection

Name:

FabricPool: ☐ The SVM does not contain any FabricPool-enabled aggregates. [Configure a Cloud tier. Learn more.](#)

Aggregate:

Storage Type

☒ NAS (Used for CIFS or NFS access)

☐ Data Protection (Used as destination volume)

Size

Total Size:

Snapshot Reserve (%):

Data Space: 972.8 MB

Snapshot Space: 51.2 MB

Space Reserve

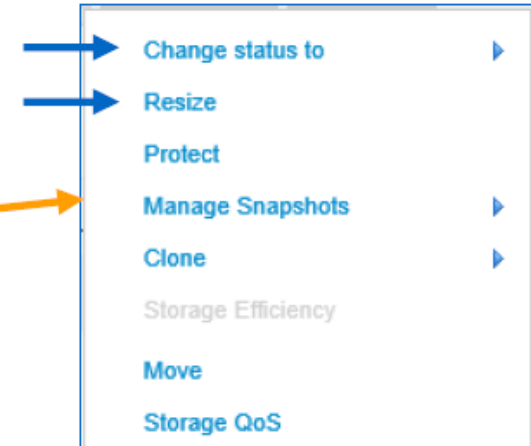
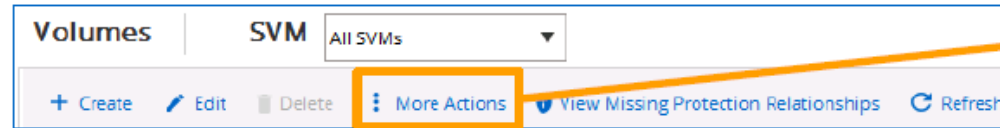
Space Reserve (optional):

[Tell me more about space reservation](#)

Discussed
later

```
::> volume create -vserver svm4 -name svm4_vol_002 -aggr  
cluster201_fc1_001 -junction-path /vol_002 -size 2gb
```

Management of FlexVol Volumes



Create

```
::> volume create -vserver svm4 -name svm4_vol1 -aggr cluster201_fcsl_001 -size 200gb
```

Resize

```
::> vol modify -vserver svm4 -name svm4_vol1 -size +10gb
```

Offline and online

```
::> vol offline -vserver svm4 -name svm4_vol1  
::> vol online -vserver svm4 -name svm4_vol1
```

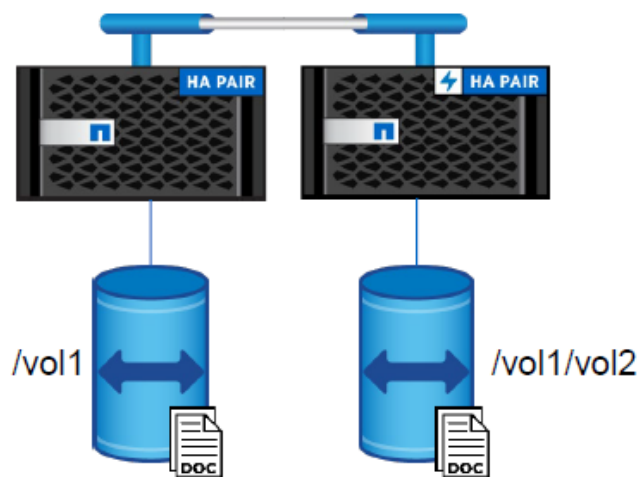
Destroy

```
::> vol delete -vserver svm4 -name svm4_vol1
```

Must be
offline

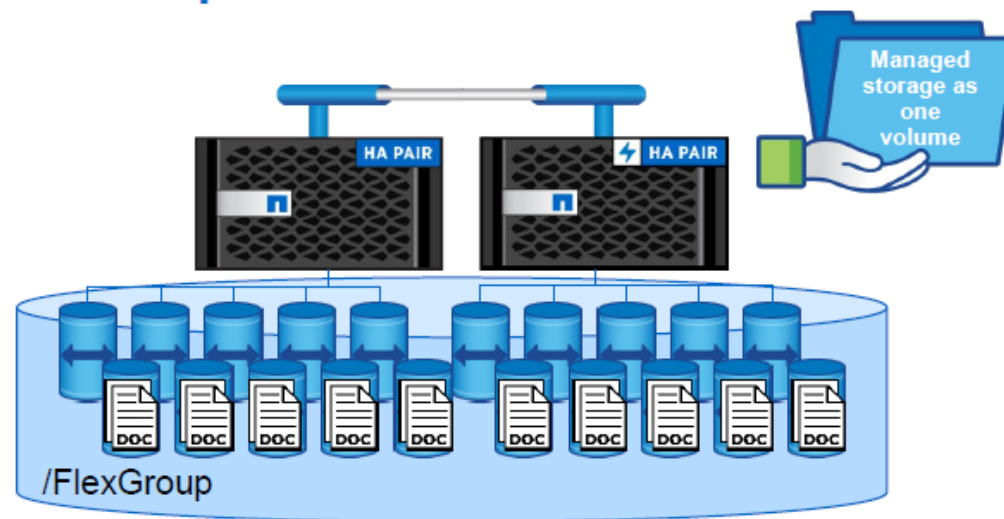
FlexVol Volumes Versus FlexGroup Volumes

How they differ at a high level



FlexVol volumes

- Owned by one node
- Span one aggregate
- Isolate reads and writes to one node and aggregate
- Are limited to storing 100TB (system-dependent)
- Are within one namespace, but with limits



FlexGroup volumes

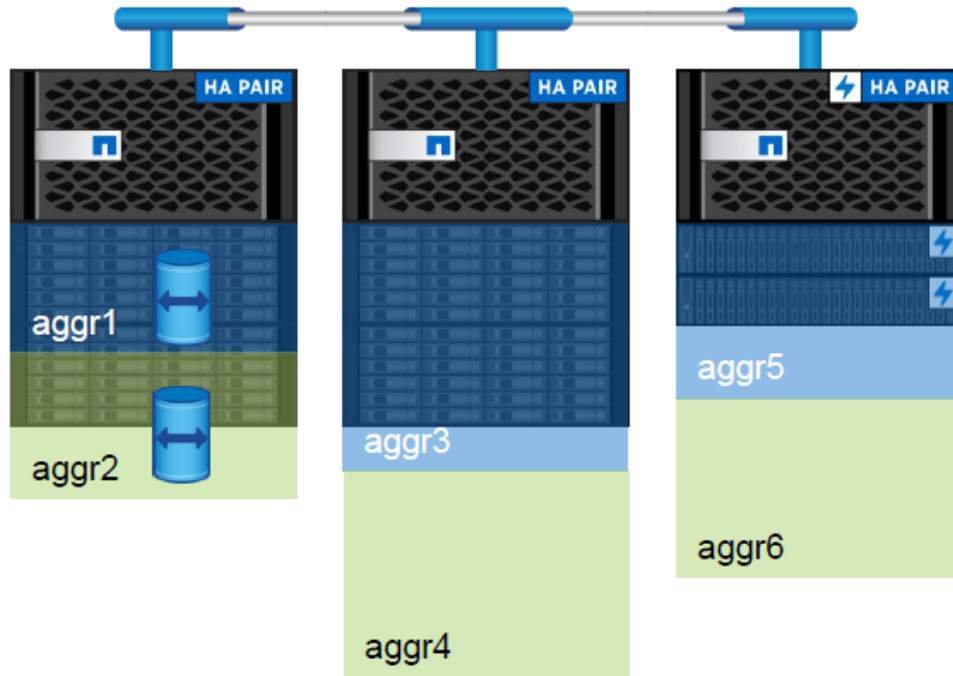
- A shared pool of FlexVol volumes
- Component volumes span multiple aggregates
- Balance reads and writes across all nodes and aggregates
- Can store up to 20PB (200 FlexVol volumes)
- Are within one namespace, almost without limits



Vol move



Volume Move



- **Rules:**
 - Move only within the SVM.
 - Move to any aggregate to which the SVM has permission.
 - Move is nondisruptive to the client.
- **Use cases:**
 - **Capacity:** Move a volume to an aggregate with more space.
 - **Performance:** Move to an aggregate with different performance characteristics.
 - **Servicing:** Move to newly added nodes or from nodes that are being retired.

How a Volume Move Works

1. A volume is created on the destination aggregate.
2. A Snapshot copy of the source volume is created.
3. The Snapshot copy is replicated to the destination volume.
4. When replication is complete, client access is temporarily blocked.
5. A final replication is performed to reach consistency.
6. Cutover is initiated.
7. Clients access the destination volume, and the source volume is cleaned up.



The volume move Command

Volumes | SVM: svm4

+ Create | Edit | Delete | More Actions | View Missing Pro

Status	Name
+ ✓	svm4_vol002
+ ✓	svm4_root
+ ✓	svm4_vol1_Cl

More Actions menu:

- Change status to
- Resize
- Protect
- Manage Snapshots
- Clone
- Storage Efficiency
- Move**
- Storage QoS
- Provision Storage for VMware
- Change Tiering Policy

Move Volume

Source Volume

Name: svm4_vol002
Committed Size: 2 GB
Aggregate: cluster2_fcald_001
Storage Type: FCAL

Destination Aggregate

Name	Available Space	Total Space	Storage Type	FabricPool
cluster2_fcald_002	27.12 GB	28.13 GB	FCAL	No

```
cluster2::> vol move start -vserver svm4 -vol svm4_vol_002 -destination-  
aggr cluster2_fcald_002  
cluster2::> vol move trigger-cutover -vserver svm4 -vol svm4_vol_002
```

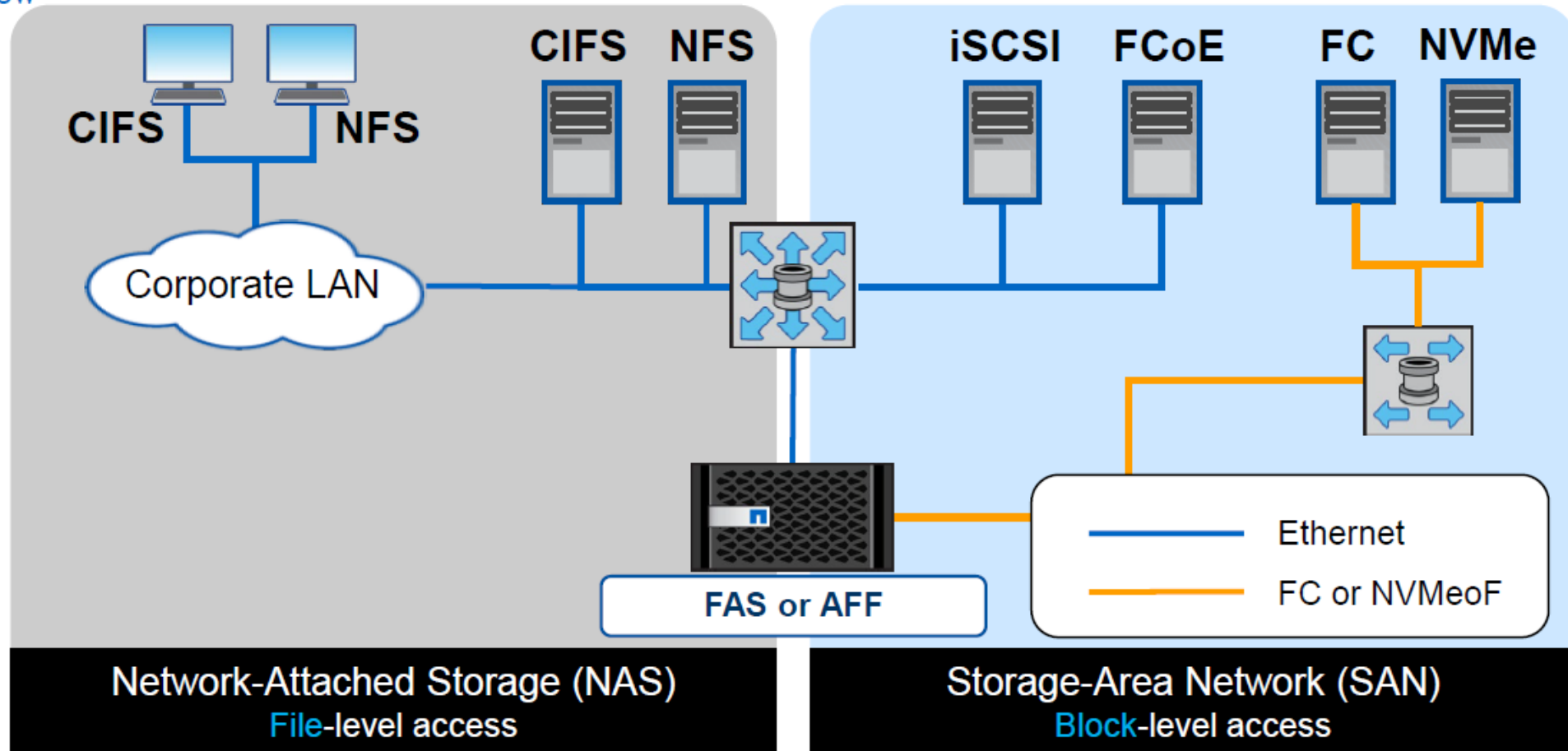


Data Access (NAS – SAN)

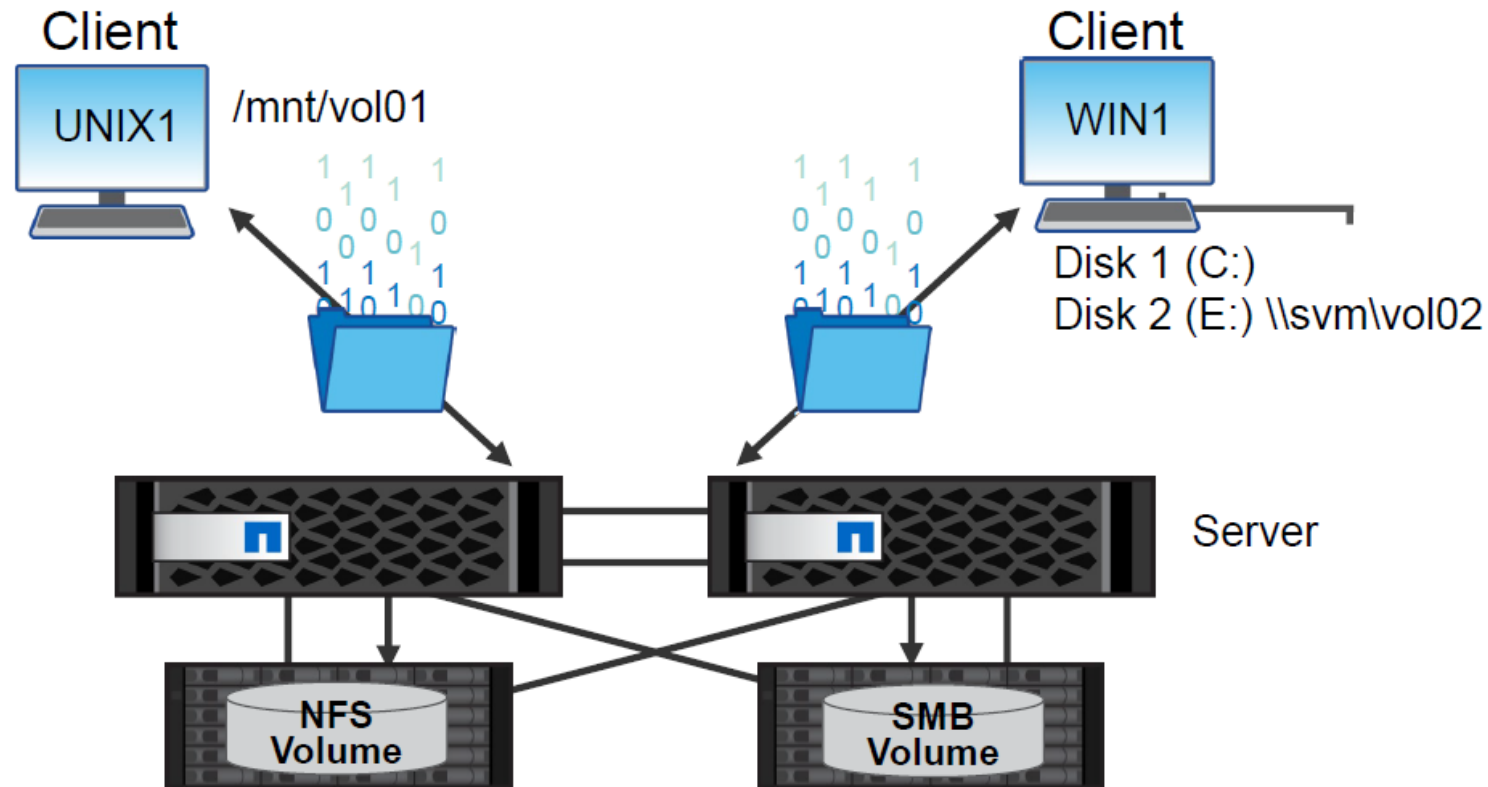
Senthilkumar Muthusamy

Unified Storage

Review



The NAS File System

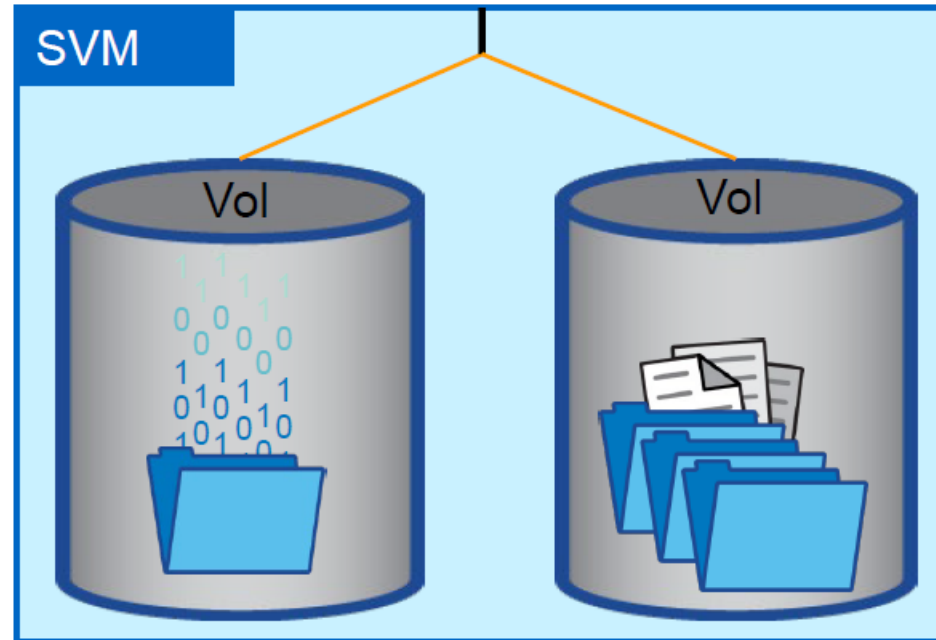


NAS is a distributed file system that enables users to access resources, such as volumes, on a remote storage system as if the resources were on a local computer system.

NAS provides services through a client-server relationship. Storage systems that make file systems and other resources available for remote access are called *servers*. The server is set up with a network address and provides file-based data storage to other computers, called *clients*, that use the server resources.

Storage System Resources

- FlexVol Volume
 - Data containers to manage data in a storage virtual machine (SVM)
 - Exportable by mounting to namespace junction
- Qtree
 - Volume partition created on storage system
 - Exportable by mounting to namespace junction
- Directory
 - Volume partition created on the NAS client
 - Not exportable

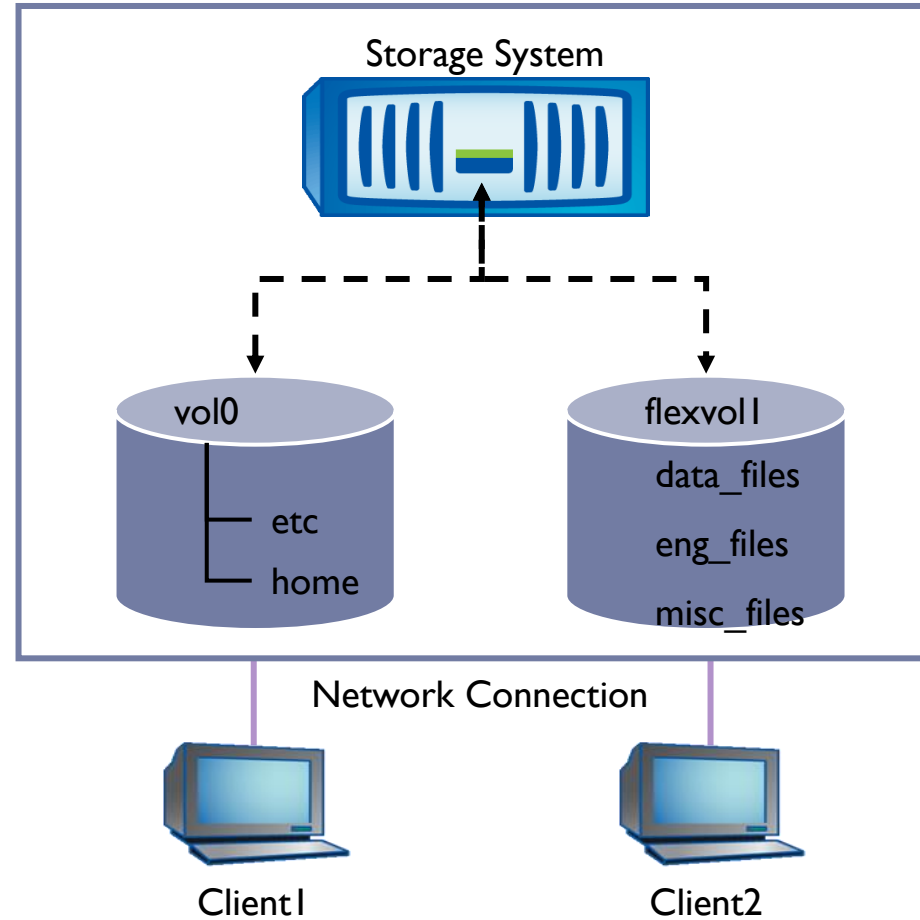


NFS Overview

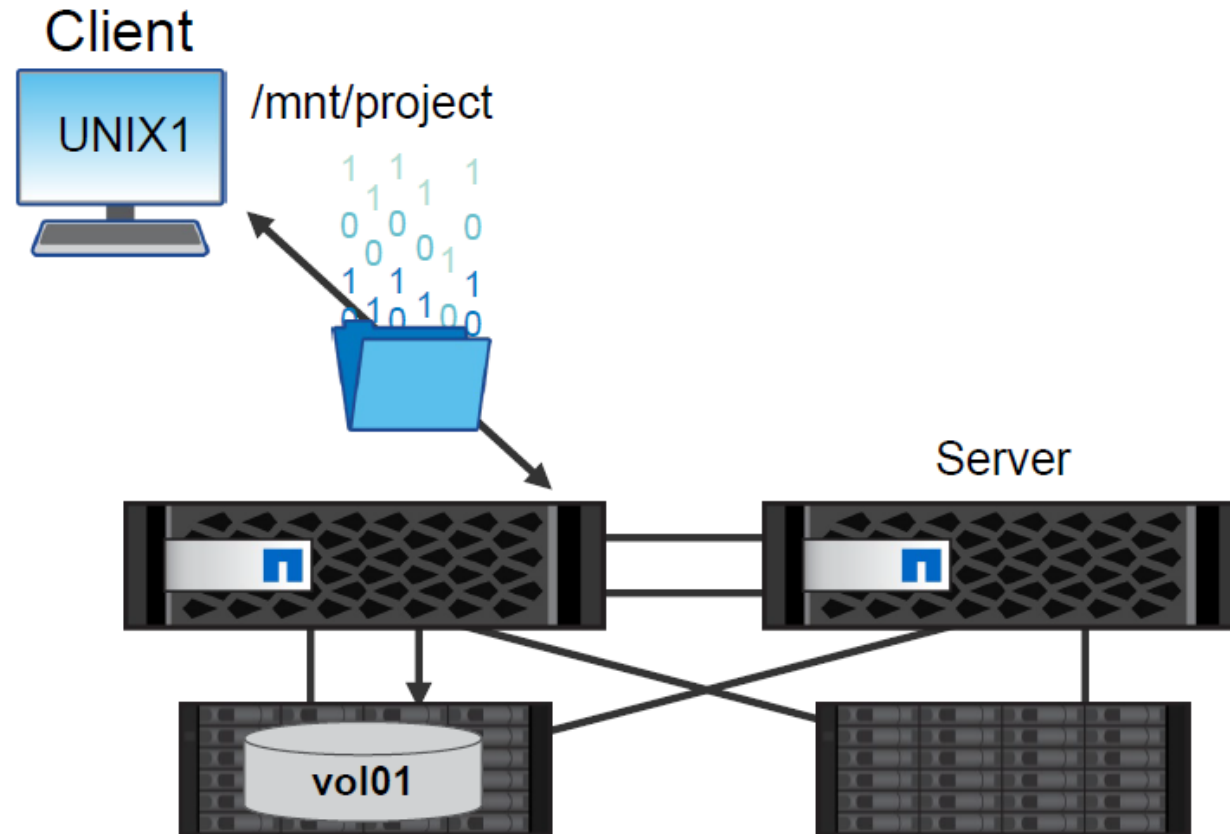
- ▶ NFS enables **network file systems** (clients) to share files and directories that are stored and administered centrally from a storage system.
- ▶ These platforms usually support NFS:
 - ▶ Sun Microsystems Solaris[™]
 - ▶ Linux[®]
 - ▶ HP-UX
 - ▶ And more



Exported Resources Overview



NFS



- `vol01` is **exported** to UNIX1 with read/write permission.
- UNIX1 **mounts** `vol01` to `/mnt/project` with read/write permission.

NFSv3 Implementation Steps

1. Verify or add the NFS protocol license.
2. Enable NFS functionality on the SVM.
3. Create or identify the necessary resources.
4. Export the available resources.
5. Configure NFS authentication.
6. Authorize the user.
7. Mount the exported resources.



SVM Create Wizard: NFS

SVM basic details

SVMs

+ Create Edit

Create

Storage Virtual Machine (SVM) Setup

1 Enter SVM basic details

SVM Details

? Specify a unique name and the IPspace

SVM Name: svm4

? IPspace: [dropdown]

? Data Protocols: ☐ CIFS ☒ NFS ☐ FC/FCoE **Protocols**

? Default Language: C.UTF-8 [c.utf_8]

The language of the SVM specifies the default language encoding setting for the SVM and its volumes. Using a setting that incorporates UTF-8 character encoding is recommended.

? Security Style: UNIX

Root Aggregate: n1_data_003 **SVM Root Aggregate**

DNS Configuration

Specify the DNS domain and name servers. DNS details are required to configure the CIFS protocol.

? Search Domains: DEMO.NETAPP.COM

SVM Create Wizard: NFS

Configure the NFS protocol

Select an IP address
from the subnet?

Network
Port

(Optional) Network
Information Service
(NIS) information

Storage Virtual Machine (SVM) Setup

1

2

3

Enter SVM basic details

Configure CIFS/NFS protocol

Enter SVM administrator details

Configure NFS protocol

? To enable CIFS, specify the data interfaces and the CIFS server details. If you are configuring NFS, specify NIS details.
To enable access to the NFS ports, add rules to the default export policy or create a new policy for the SVM.

▲

Data LIF Configuration

☐ Retain the NFS data LIFs configuration for CIFS clients.

Data Interface details for NFS

Assign IP Address: ▼

Subnet: subnet [Change](#)
auto-assign mode

? Port:

▲

NIS Configuration (Optional)

Configure NIS domain on the SVM to authorize NFS users.

Domain Names:

IP Addresses:

? Database Type: ☒ group ☒ passwd ☒ netgroup

▼

Create a volume to export.

Provision a volume for NFS storage.

Export Name:

Size: GB ▼

Permission: [Change](#)



SVM Create Wizard: NFS

SVM administrator details

Storage Virtual Machine (SVM) Setup

1

Enter SVM basic details

2

Configure CIFS/NFS protocol

3

Enter SVM administrator details

SVM Administration (optional)

?

Specify the following details to enable host side applications such as SnapDrive and SnapManager

?

To enable the SVM administrator to create volumes, you must assign aggregates to the SVM by using Edit SVM dialog

Administrator Details

Username:

vsadmin

Password:

.....

Confirm Password:

.....

Create an SVM administrator

Management Interface (LIF) Configuration for SVM

☐ Create a new LIF for SVM management

For CIFS and NFS protocols, data LIFs have management access enabled by default; therefore, create a new management LIF only if required. However, for iSCSI and FC/FCoE protocols, create a dedicated management LIF because data LIFs cannot be used for SVM management.

Assign IP Address:

Select

?

Port:

Browse....



Mounts

Use the UNIX `mount` command on the client to mount an exported NFS resource from the storage system.

```
unix1# mkdir /mnt/project1
```

```
unix1# mount <SVM LIF IP>:/project/pro1 /mnt/project1
```

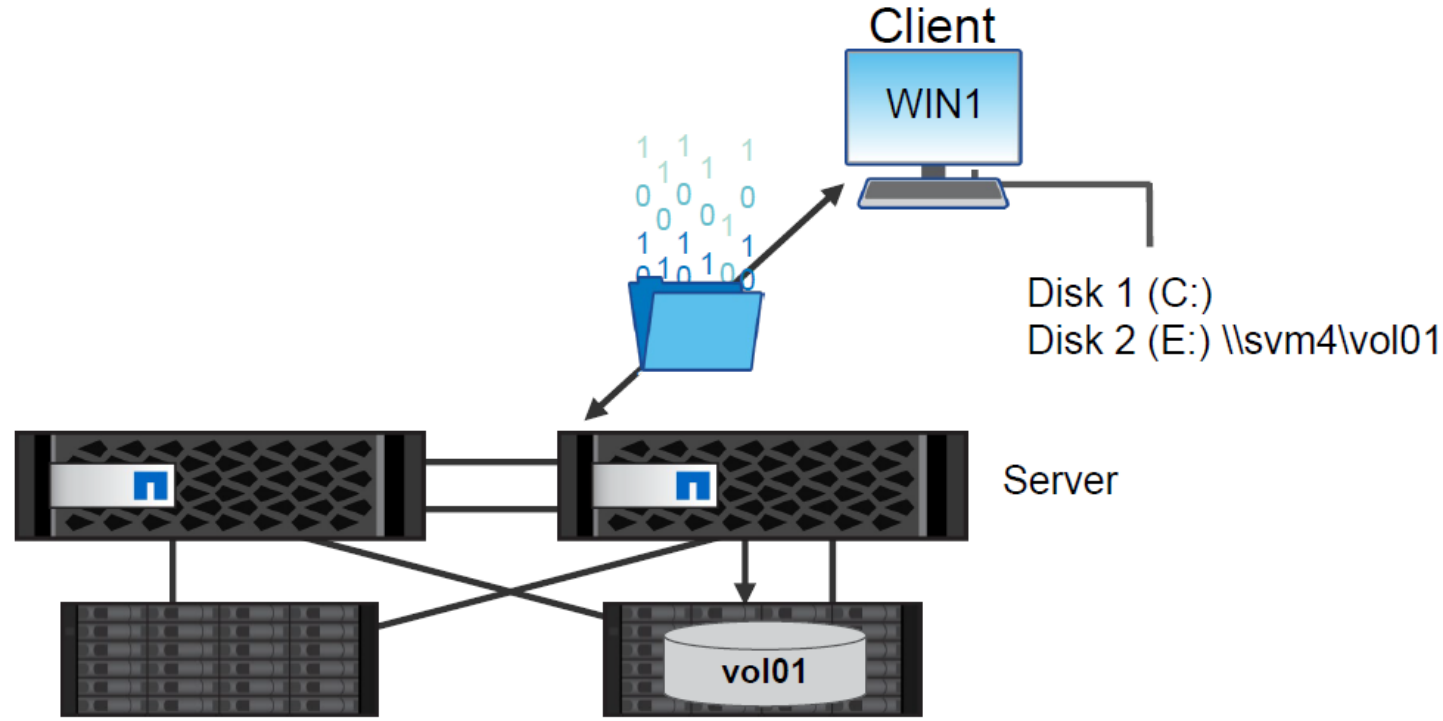




CIFS



SMB



A user or application can send network requests to read and write to files on remote computers. Messages travel from the network interface card (NIC) of the user's computer, through the Ethernet switch, to the NIC of the remote computer.

SMB provides access to files and directories that are stored on the remote computer, through sharing resources. The rules of network protocols such as IPv4 and IPv6 control the network read and write process, which is also called network I/O.

User Authentication

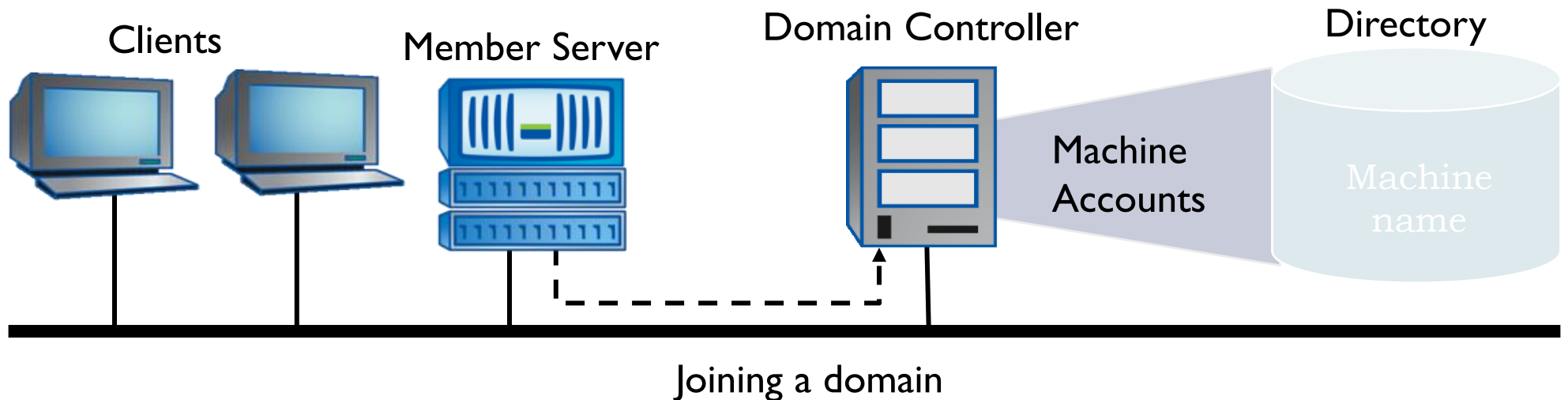
- ▶ In a CIFS environment, the storage system authenticates users in one of four ways:
 - ▶ Active Directory® authentication
 - ▶ Microsoft Windows NT® 4.0 domain authentication
 - ▶ Windows workgroup authentication
 - ▶ Authentication for non-Windows workgroups
- ▶ This module focuses on only Active Directory authentication.



Storage System Joins a Domain

When a storage system joins a domain:

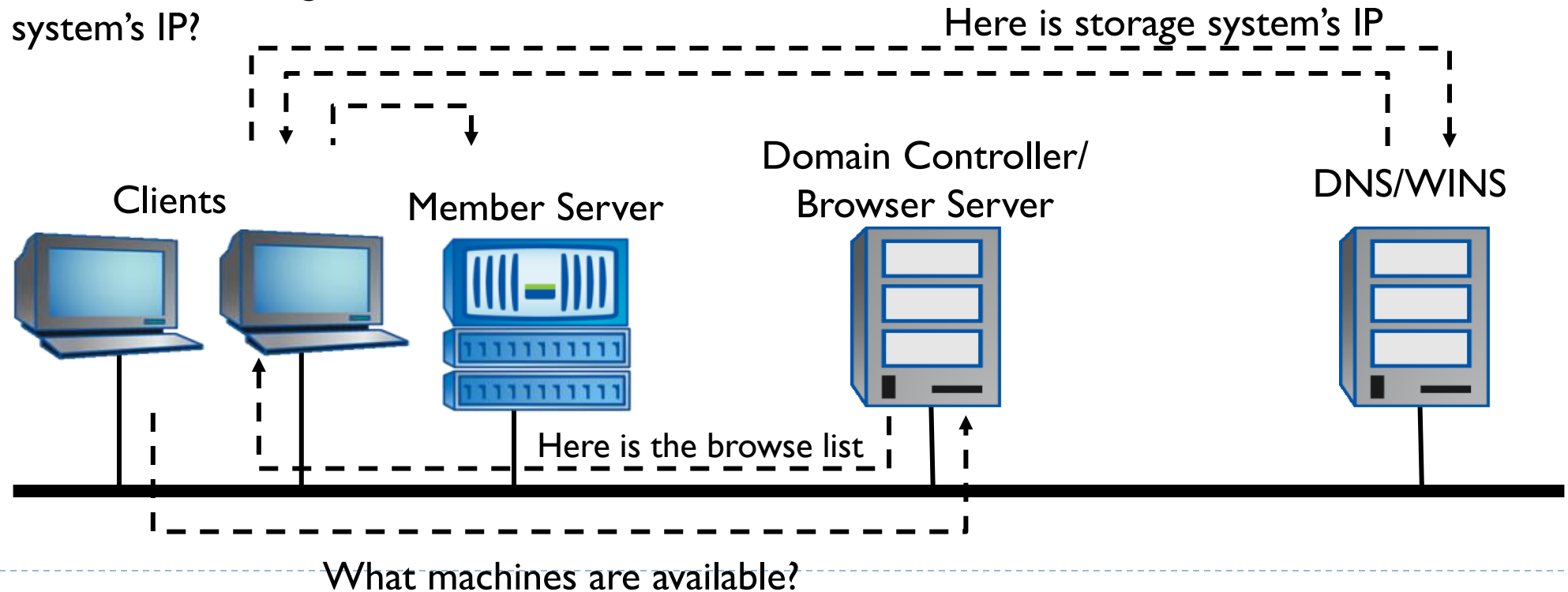
- ▶ The domain controller adds the storage system to a domain database.
- ▶ The storage system becomes a member server.



Domain Name to IP Resolution

When a client accesses a storage system's resource:

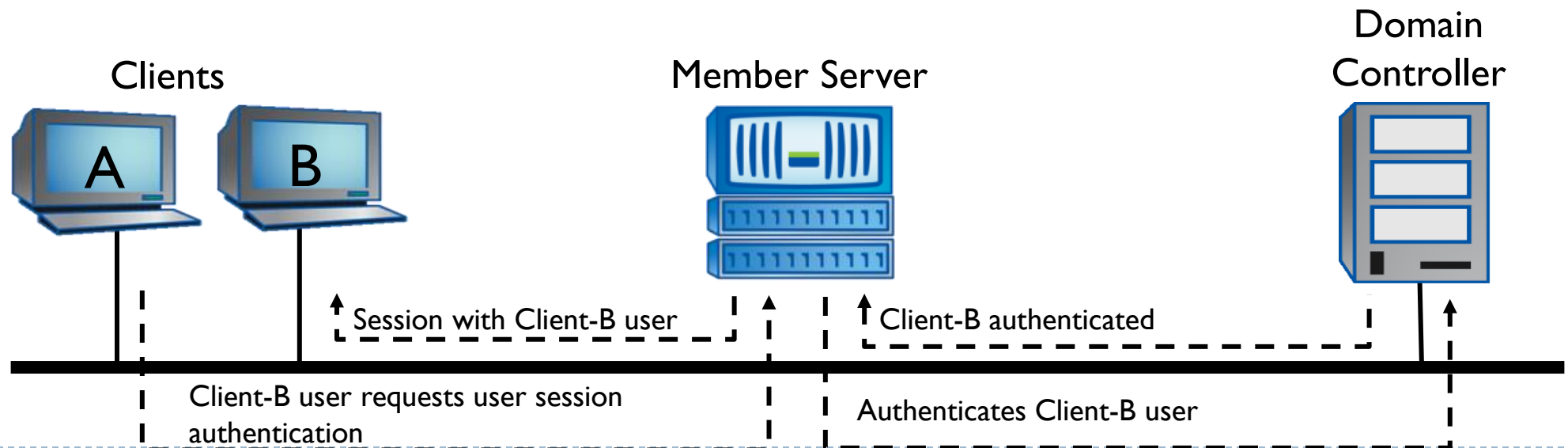
- ▶ Client requests the browse list from the domain controller.
- ▶ Domain controller contacts the DNS/WINS server for the IP address.
- ▶ Client communicates with storage system.



User Authentication

User authentication on a storage system in a domain:

- ▶ Domain users are created on domain controller.
- ▶ User session authentication occurs at the domain controller.
- ▶ Authenticated users must be authorized to access a share and resources.



SMB Implementation Steps

1. Verify or add the CIFS protocol license.
2. Enable SMB functionality on the SVM.
3. Create or identify the necessary resources.
4. Share the available resources.
5. Configure SMB authentication.
6. Authorize the user.
7. Map the shared resources.



SVM Create Wizard: CIFS

SVM basic details

SVMs

+ Create Edit

Dashboard
Applications & Tiers
Storage
Nodes
Aggregates & Disks
Cloud Tiers
Aggregates
Storage Pools
Disks
Flash Pool Statistics
SVMs
Volumes

Create

Storage Virtual Machine (SVM) Setup

1 Enter SVM basic details

SVM Details

Specify a unique name and the

SVM Name: svm_CIFS

IPspace: Default

Data Protocols: ☒ CIFS ☐ NFS ☐ FC/FCoE

Default Language: C.UTF-8 [c.utf_8]

The language of the SVM specifies the default language encoding setting for the SVM and its volumes. Using a setting that incorporates UTF-8 character encoding is recommended.

Security Style: NTFS

Root Aggregate: n1_data_003

IPspace

Protocols

SVM Root Aggregate

DNS Configuration

Specify the DNS domain and name servers. DNS details are required to configure the CIFS protocol.

Search Domains: DEMO.NETAPP.COM

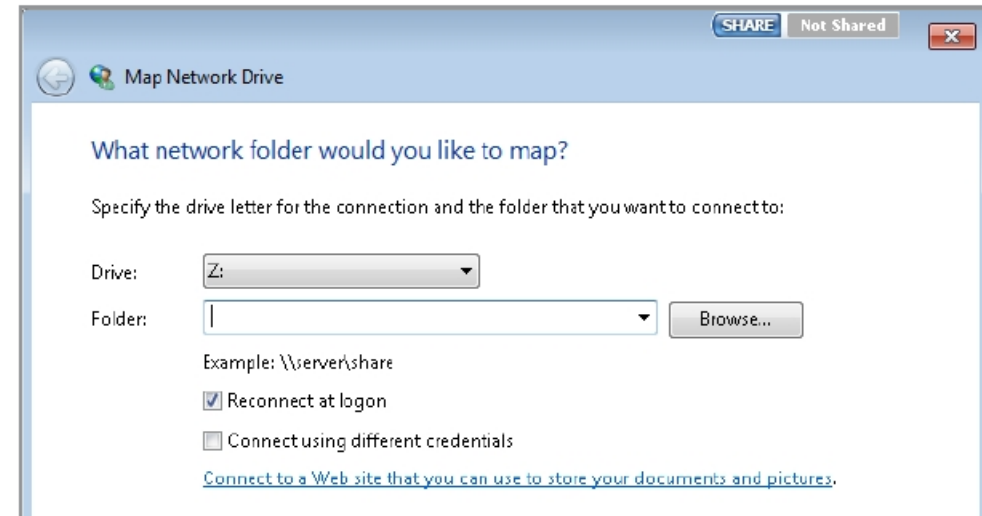
Mapping a Share to a Client

- CLI:

- `C:\> net view \\svm4`
- `C:\> net use e: \\svm4\DOCS /user:marketing\jdoe`

- UI:

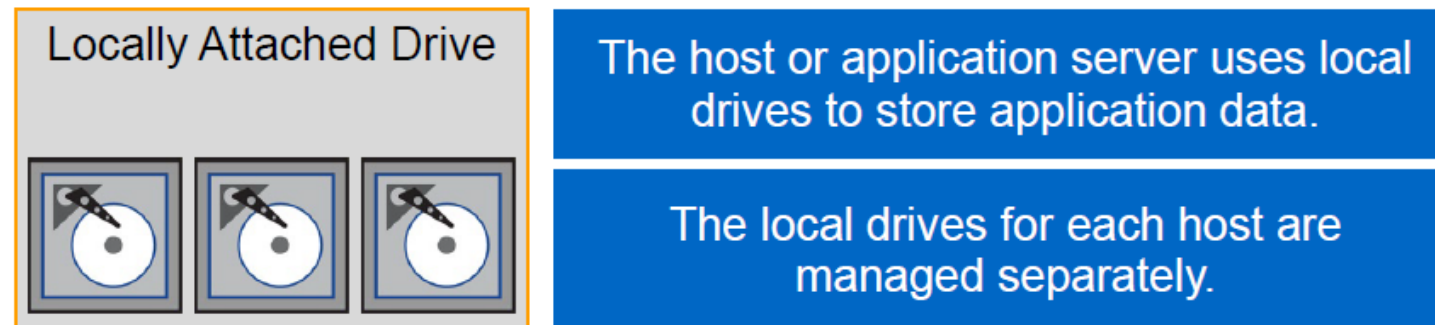
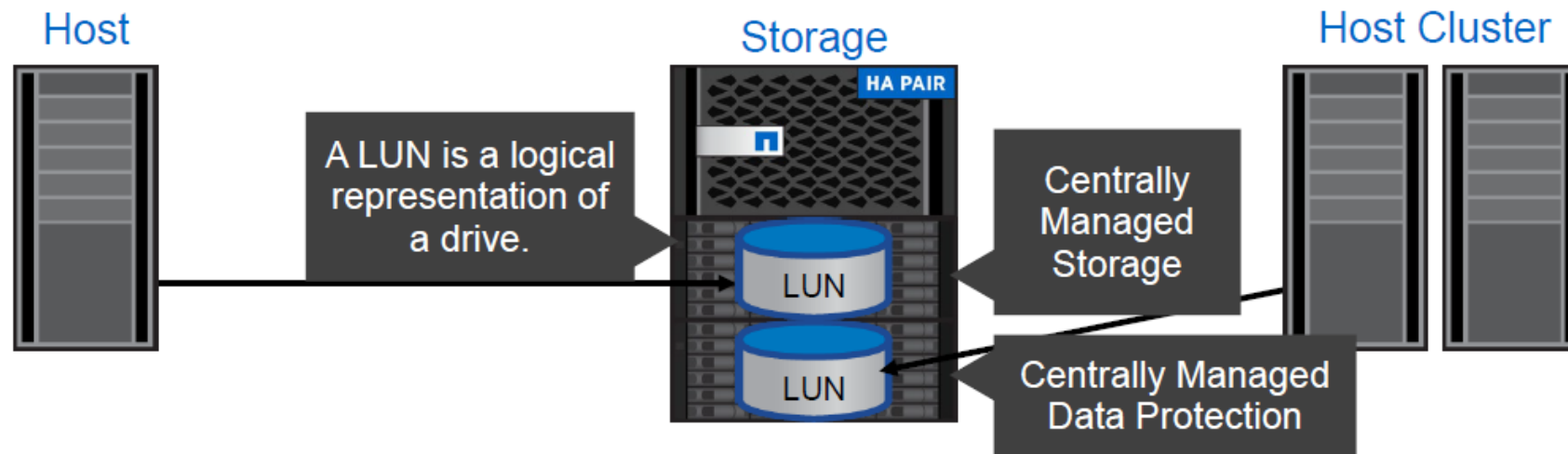
- Use the Run dialog box.
- Map a drive.





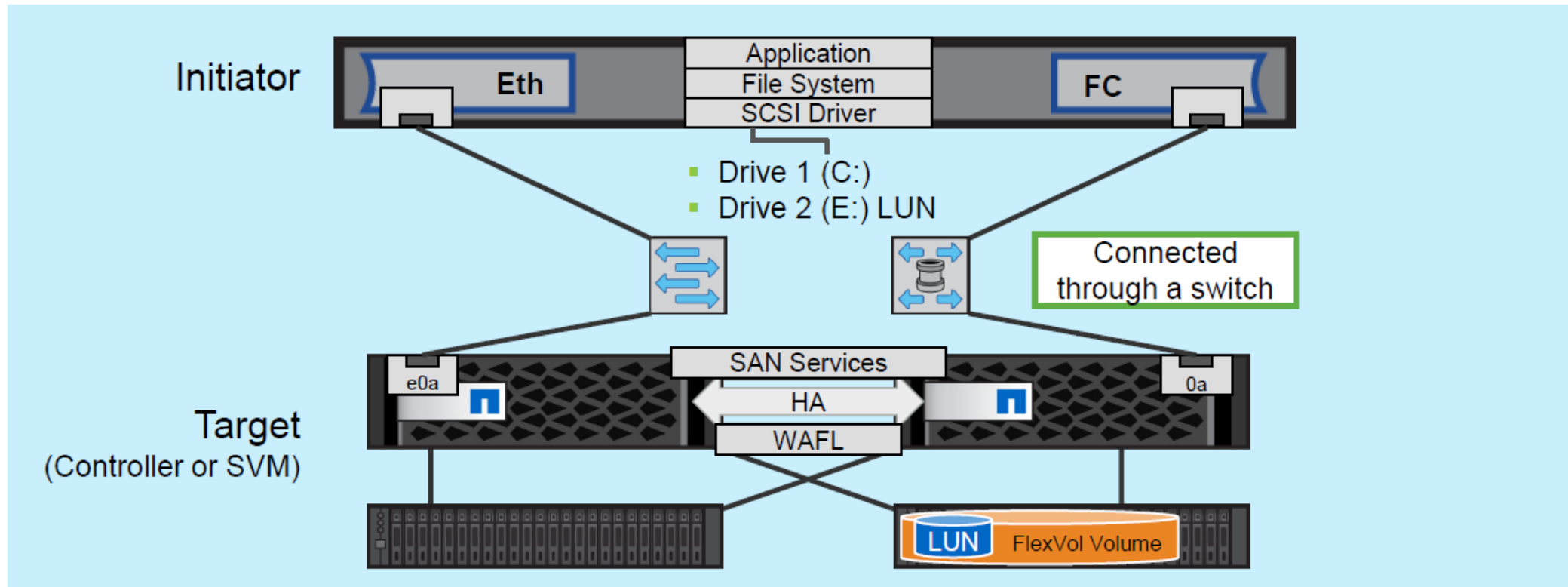
SAN Access

SAN

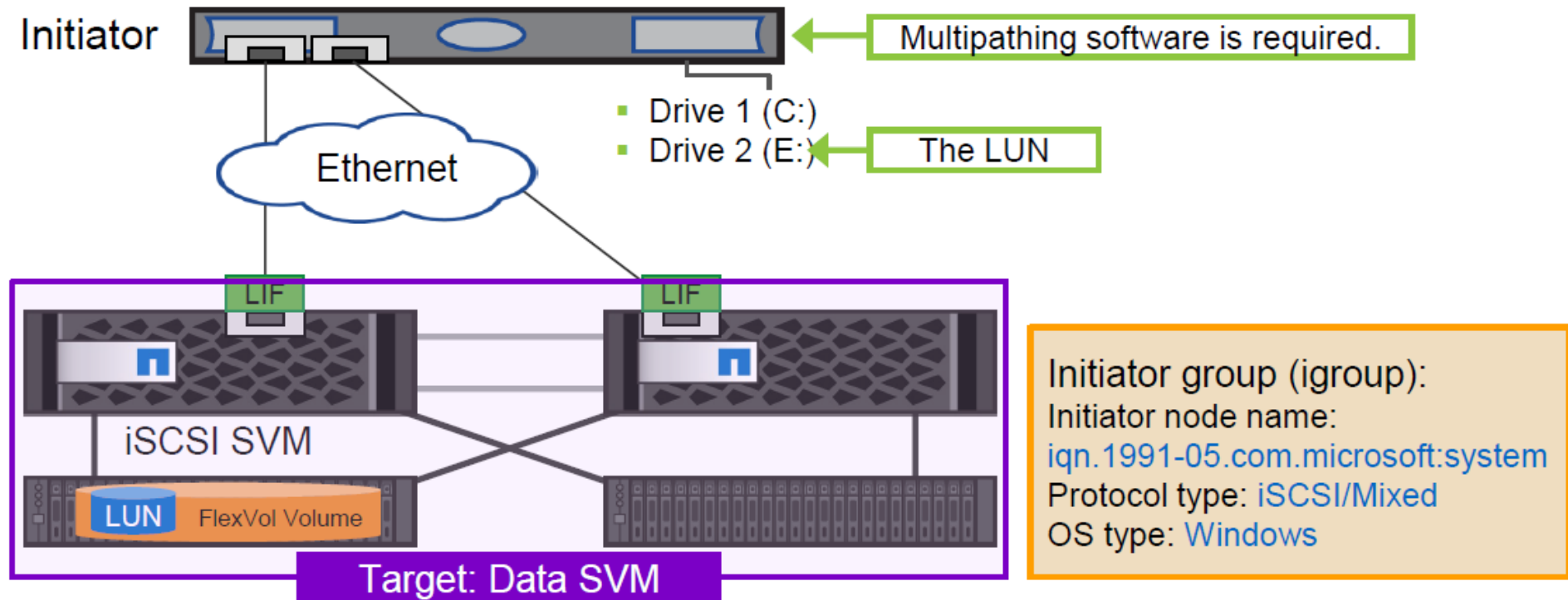


Connecting Initiator to Target

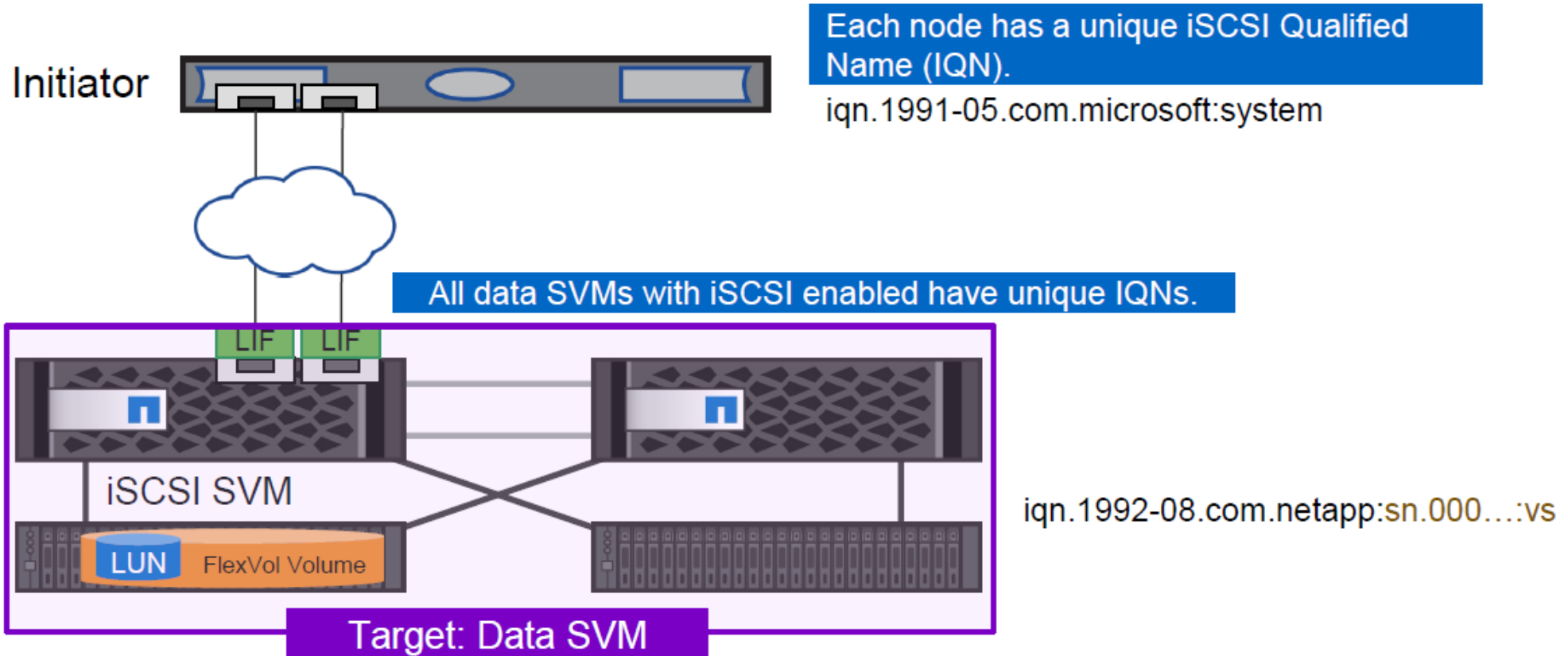
How can you connect an initiator to a target?



iSCSI Architecture



iSCSI Node Names



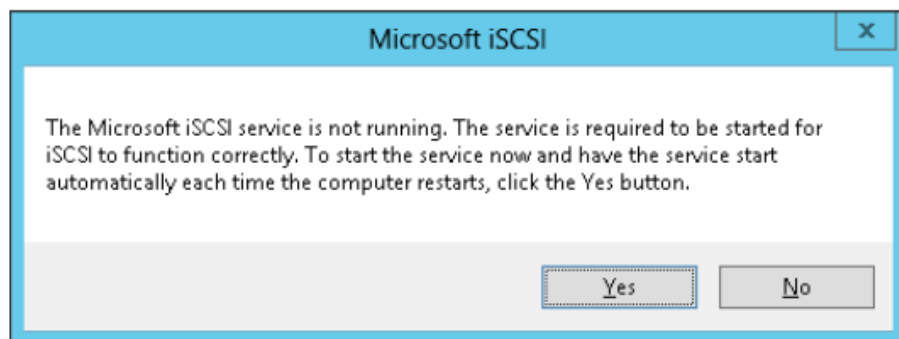
iSCSI Implementation Steps

1. Verify or add the iSCSI protocol license.
2. Enable iSCSI functionality on the SVM.
3. Create or identify the necessary resources.
4. Map the LUN to the appropriate igroup.
5. Locate the LUN on the host computer and prepare the drive.



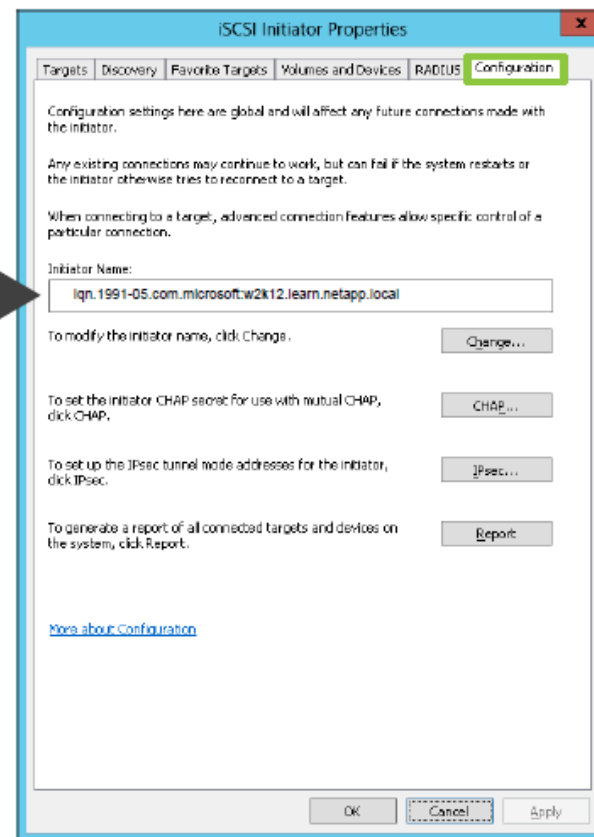
Windows iSCSI Implementation

Identify the iSCSI node name



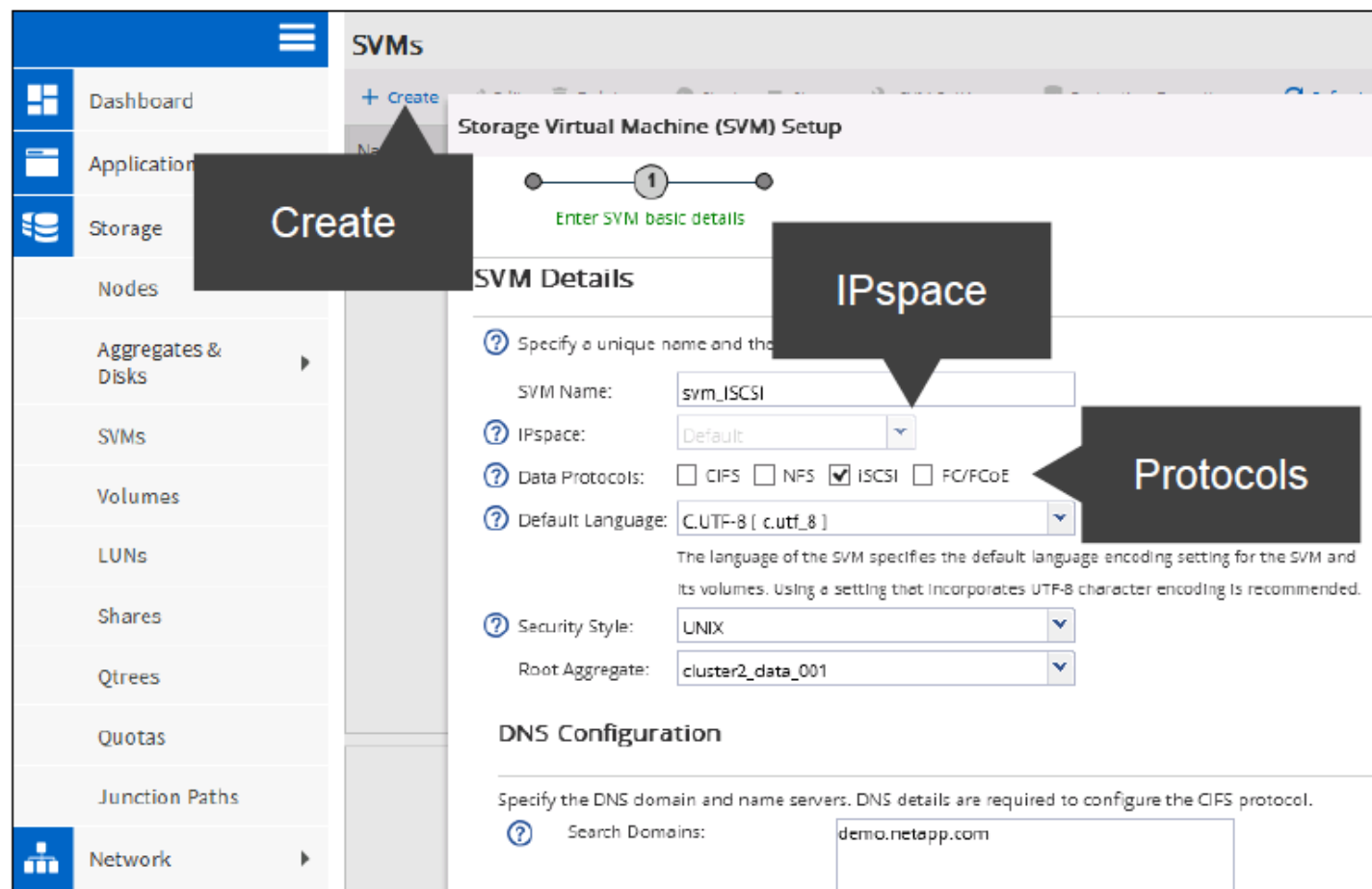
The prompt might appear the first time that you start the iSCSI initiator.

iSCSI Initiator Name



SVM Setup Wizard: iSCSI

SVM basic details



SVMs

+ Create

Storage Virtual Machine (SVM) Setup

1 Enter SVM basic details

SVM Details

Specify a unique name and the IPspace for the SVM.

SVM Name:

IPspace:

Data Protocols: ☐ CIFS ☐ NFS ☒ iSCSI ☐ FC/FCoE

Default Language:

The language of the SVM specifies the default language encoding setting for the SVM and its volumes. Using a setting that incorporates UTF-8 character encoding is recommended.

Security Style:

Root Aggregate:

DNS Configuration

Specify the DNS domain and name servers. DNS details are required to configure the CIFS protocol.

Search Domains:

SVM Setup Wizard: iSCSI

Configure iSCSI protocol



Create at least one LIF per node, per network, for each SVM that serves data with the iSCSI protocol.

Select an IP address from the subnet?

Adapter Type

LIF Configuration

Storage Virtual Machine (SVM) Setup

1 Enter SVM basic details 2 **Configure iSCSI protocol** 3 Enter SVM administrator details

Configure iSCSI protocol

Configure LIFs to access data

Data Interface (LIF) Configuration

Target Alias: **Target Alias (Optional)**

LIFs Per Node: (Minimum: 1, Maximum: 6)

Assign IP Address:

Subnet: svi_subnet auto-assign mode

Adapter Type:

☒ Review or modify LIF configuration (Advanced Settings)

Number of portsets: (Minimum: 0, Maximum: 0)

Provision a LUN for iSCSI storage (Optional):

LUN Size: GB

LUN OS Type:

Host Initiator: **Host OS**

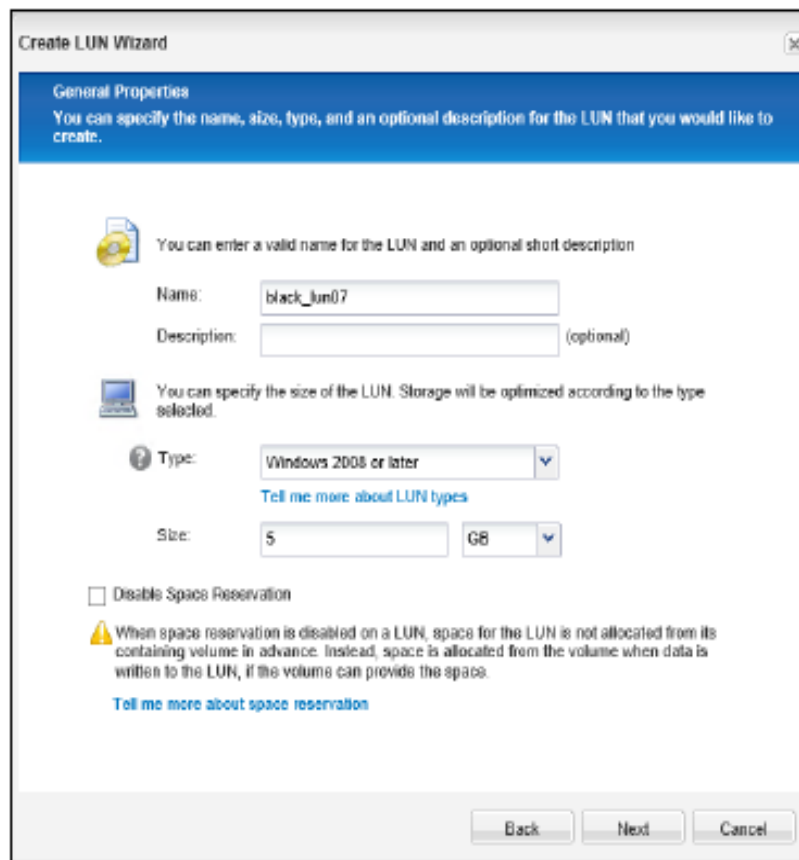
Host Initiator IQN

Double-click row to edit

Node Name	Interface Name	Home P...	Portset
svi-nau-01	svi-nau-01_iscsi_lif_1	e0c	
svi-nau-02	svi-nau-02_iscsi_lif_1	e0c	



Creating a LUN



Create LUN Wizard

General Properties
You can specify the name, size, type, and an optional description for the LUN that you would like to create.

You can enter a valid name for the LUN and an optional short description

Name:

Description: (optional)

You can specify the size of the LUN. Storage will be optimized according to the type selected.

Type: [Tell me more about LUN types](#)

Size:

☐ Disable Space Reservation

 When space reservation is disabled on a LUN, space for the LUN is not allocated from its containing volume in advance. Instead, space is allocated from the volume when data is written to the LUN, if the volume can provide the space.
[Tell me more about space reservation](#)

Back Next Cancel

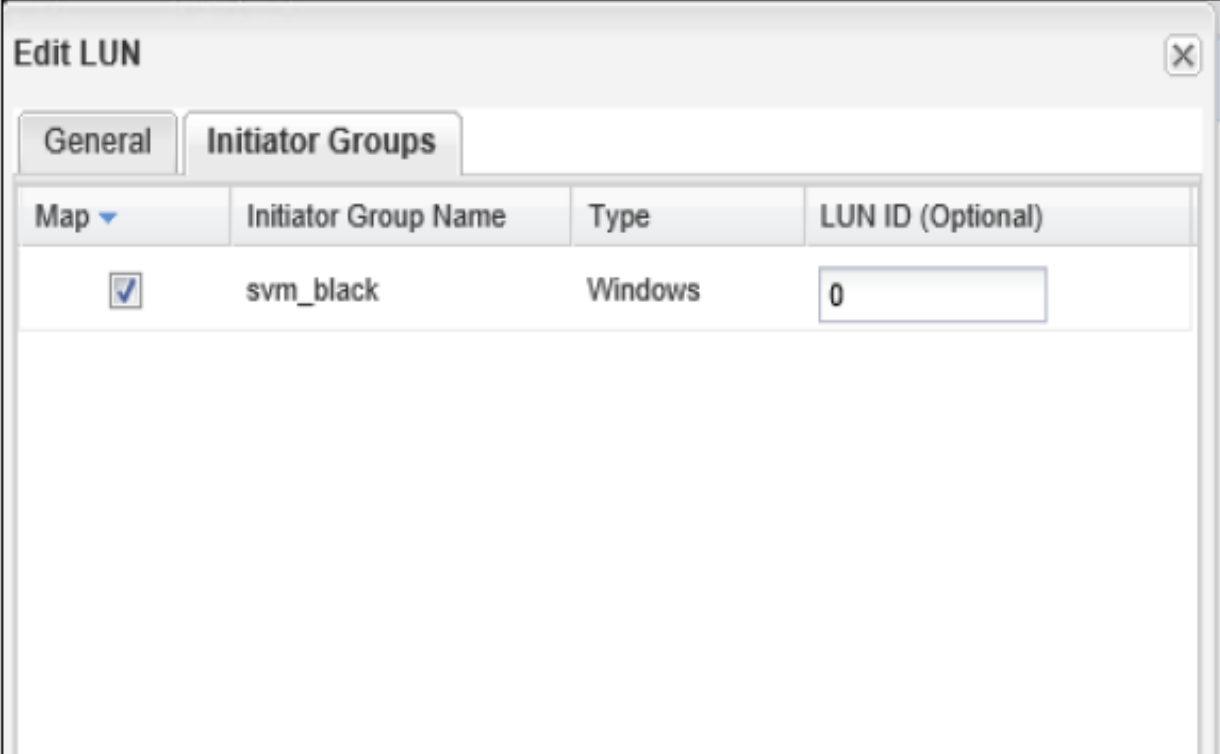
```
rtp-nau::> lun create -vserver svm_black -volume black_lun_vol07  
-lun black_lun07 -size 5GB -ostype windows_2008
```

Creating an igroup

The image displays two screenshots of the 'Create Initiator Group' GUI. The left screenshot shows the 'General' tab with the following fields: Name (ig_black_win), Operating System (Windows), and Type (iSCSI selected). The right screenshot shows the 'Initiators' tab with a list of initiators including iqn.1991-05.com.microsoft:w2k12.learn.netapp.local.

```
rtp-nau::> lun igroup create -vserver svm_black -igroup ig_black_win  
-protocol iscsi -ostype windows  
-initiator iqn.1991-05.com.microsoft:w2k12.learn.netapp.local
```

Mapping a LUN



The screenshot shows a window titled "Edit LUN" with a close button in the top right corner. It has two tabs: "General" and "Initiator Groups", with the latter being selected. Below the tabs is a table with four columns: "Map" (containing a checked checkbox), "Initiator Group Name" (containing "svm_black"), "Type" (containing "Windows"), and "LUN ID (Optional)" (containing "0").

Map	Initiator Group Name	Type	LUN ID (Optional)
☒	svm_black	Windows	0

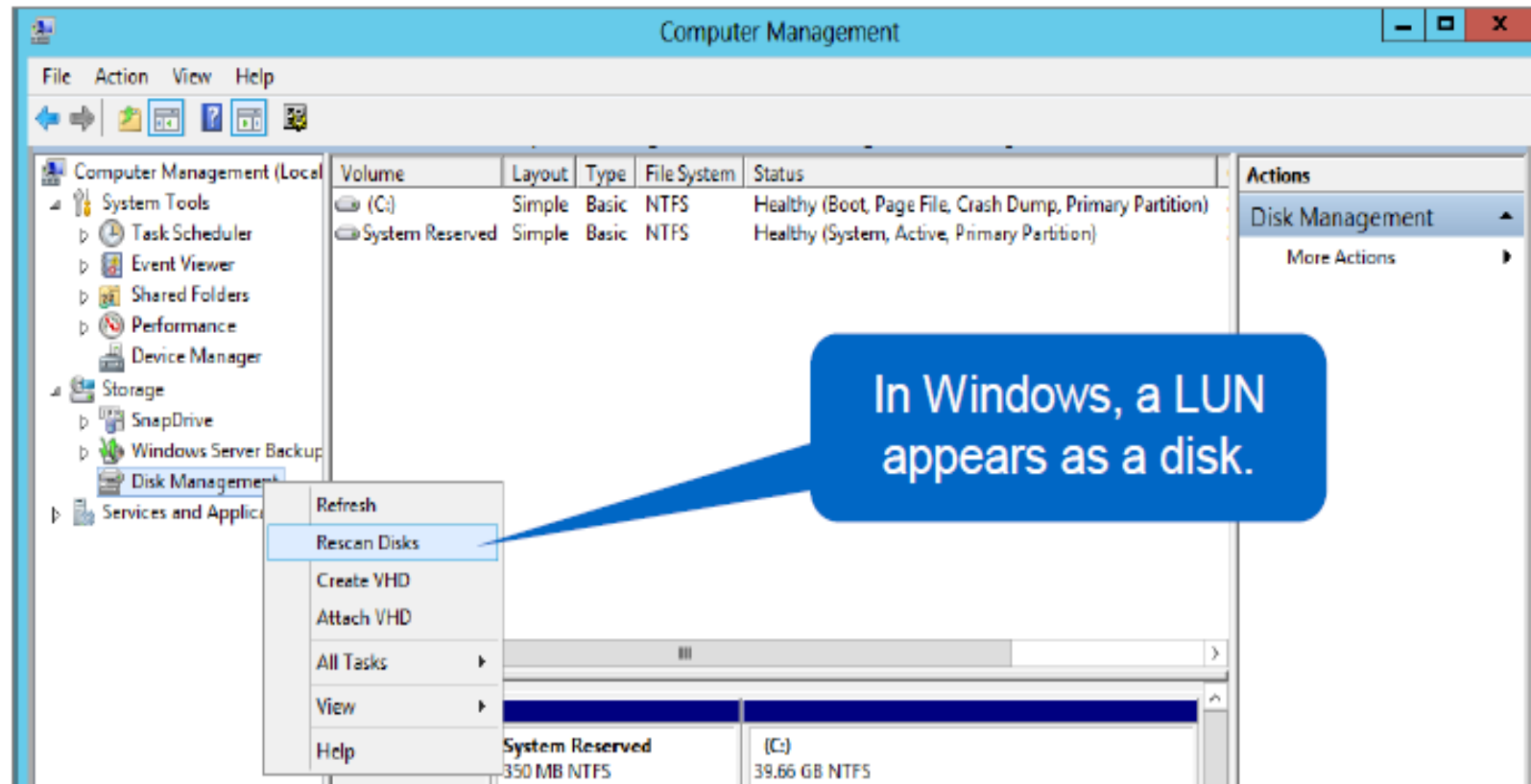
```
rtp-nau::> lun map -vserver svm1 -volume black_lun_vol07 -lun black_lun07  
              -igroup ig_black_win
```



Windows LUN Implementation

Discover LUN

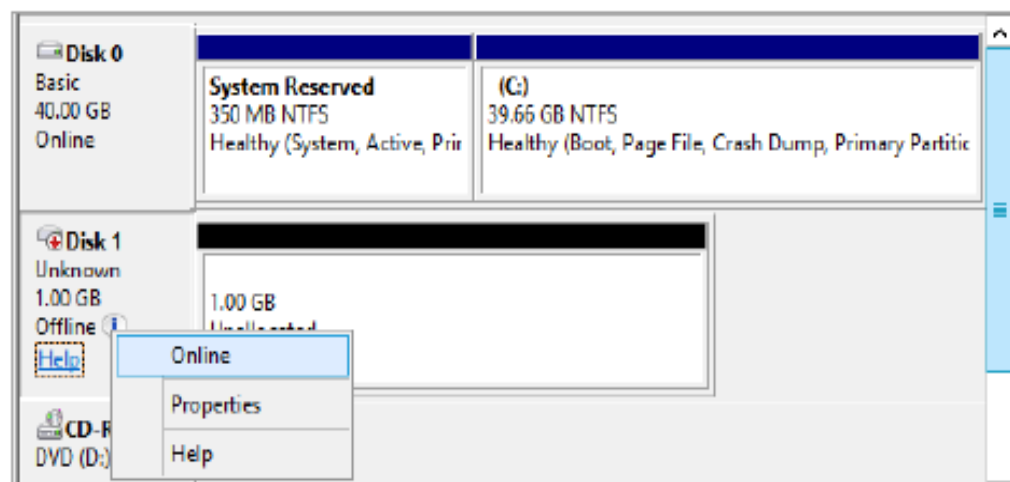
To configure an NTFS volume LUN, first discover the LUN by selecting **Disk Management > Rescan Disks**.



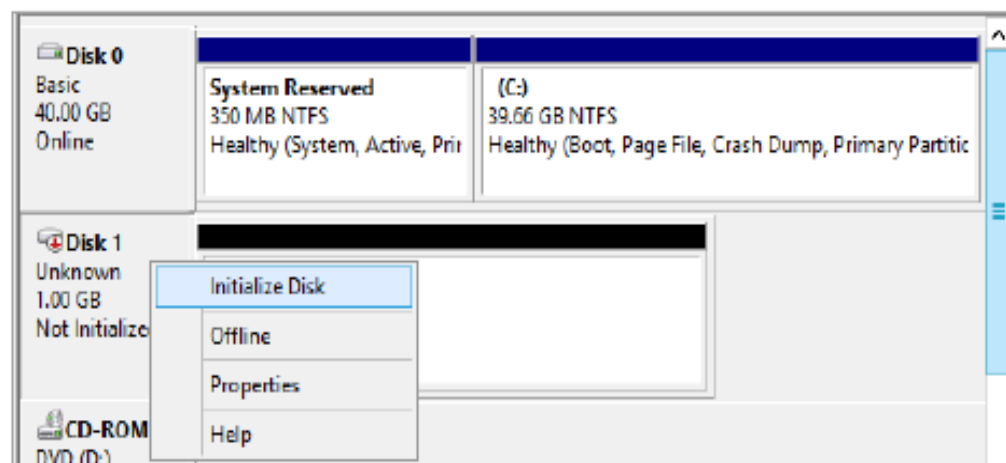
Windows LUN Implementation

Initialize disk

1. Right-click the disk, and then select **Initialize Disk**.

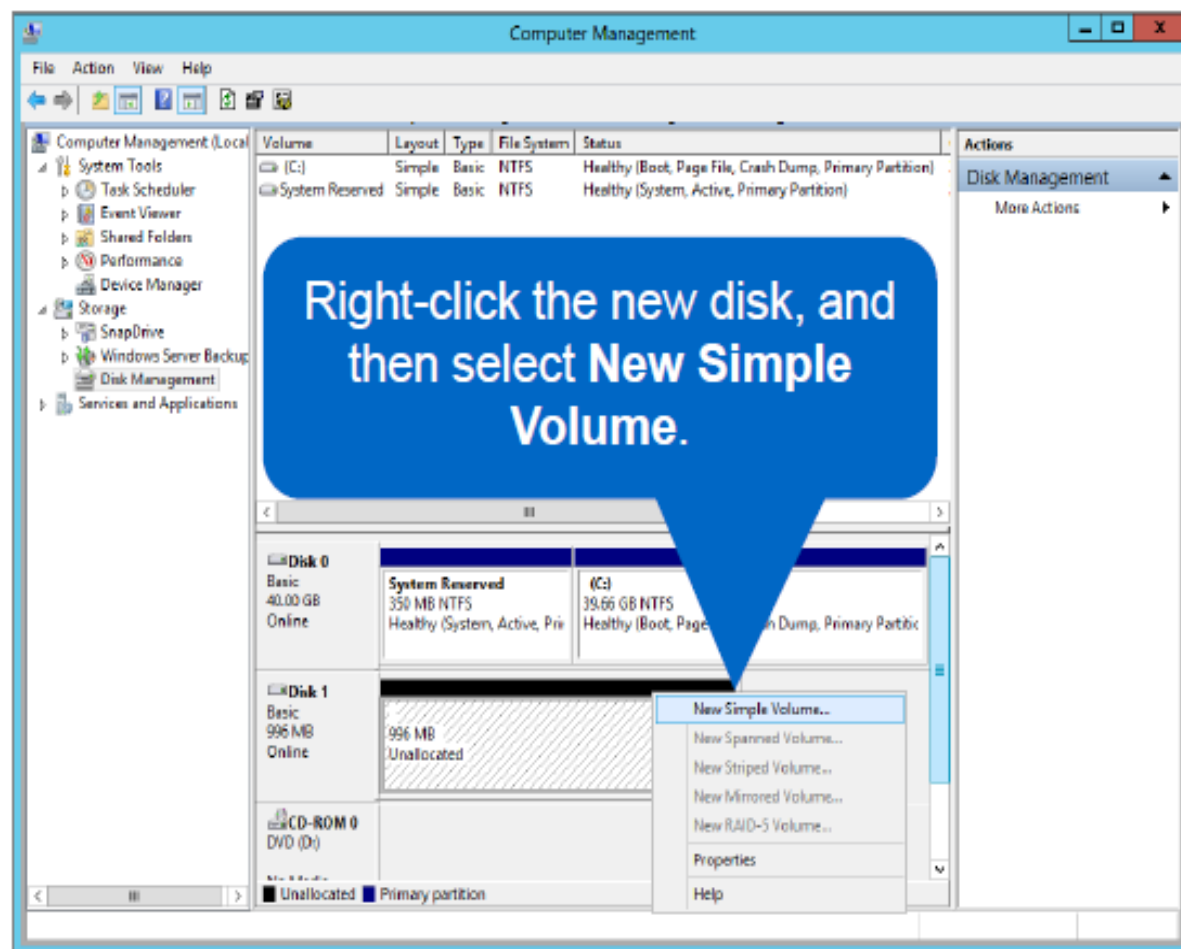


2. Right-click the new disk, and then select **Online**.

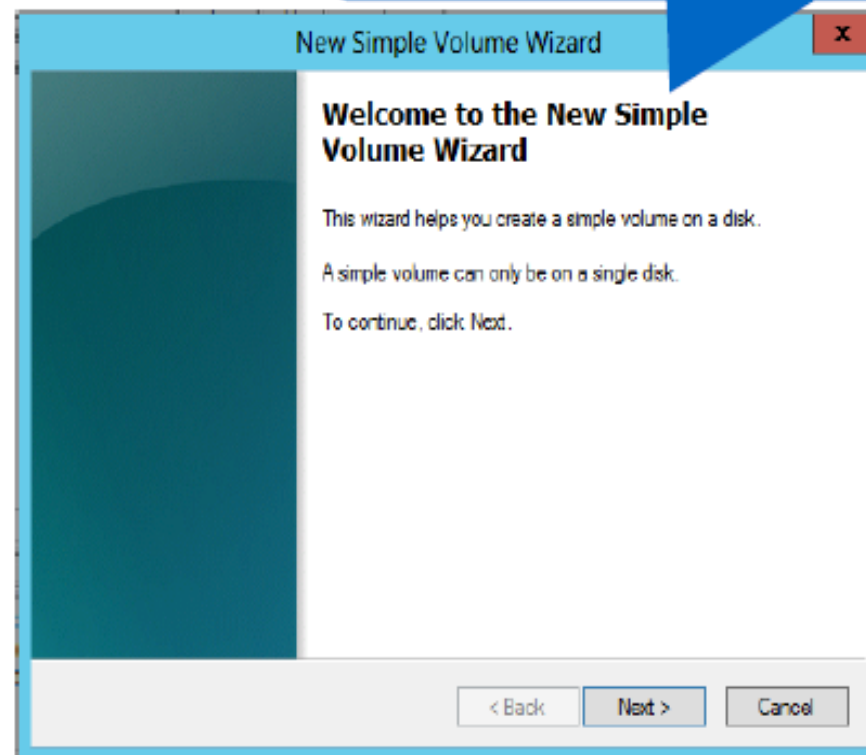


Windows LUN Implementation

Provisioning a volume (1 of 3)

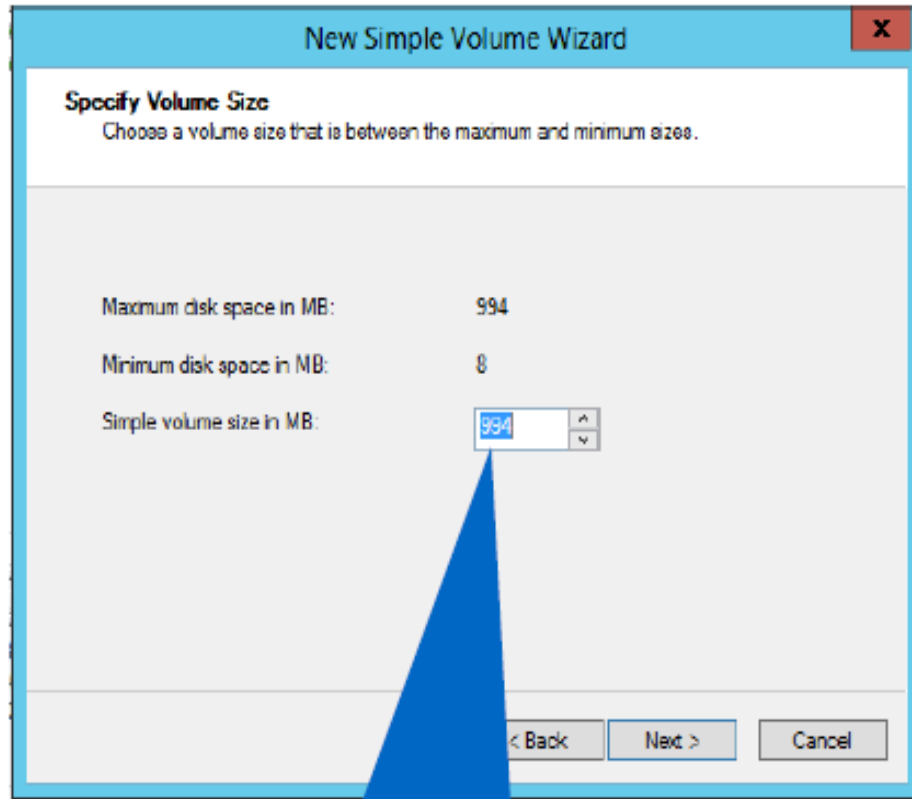


The New Simple Volume Wizard launches.



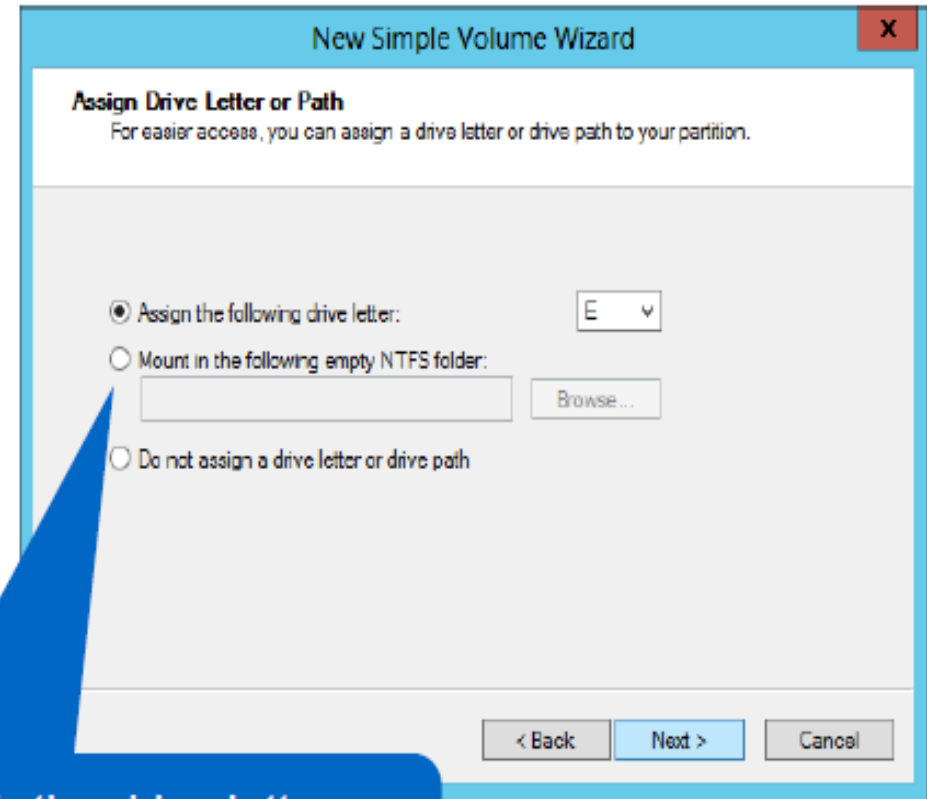
Windows LUN Implementation

Provisioning a volume (2 of 3)



The screenshot shows the 'Specify Volume Size' step of the 'New Simple Volume Wizard'. The window title is 'New Simple Volume Wizard'. The instruction says 'Choose a volume size that is between the maximum and minimum sizes.' Below this, there are three rows of information: 'Maximum disk space in MB:' with the value '994', 'Minimum disk space in MB:' with the value '8', and 'Simple volume size in MB:' with a text box containing '994' and up/down arrow buttons. At the bottom are '< Back', 'Next >', and 'Cancel' buttons.

Specify the volume size.

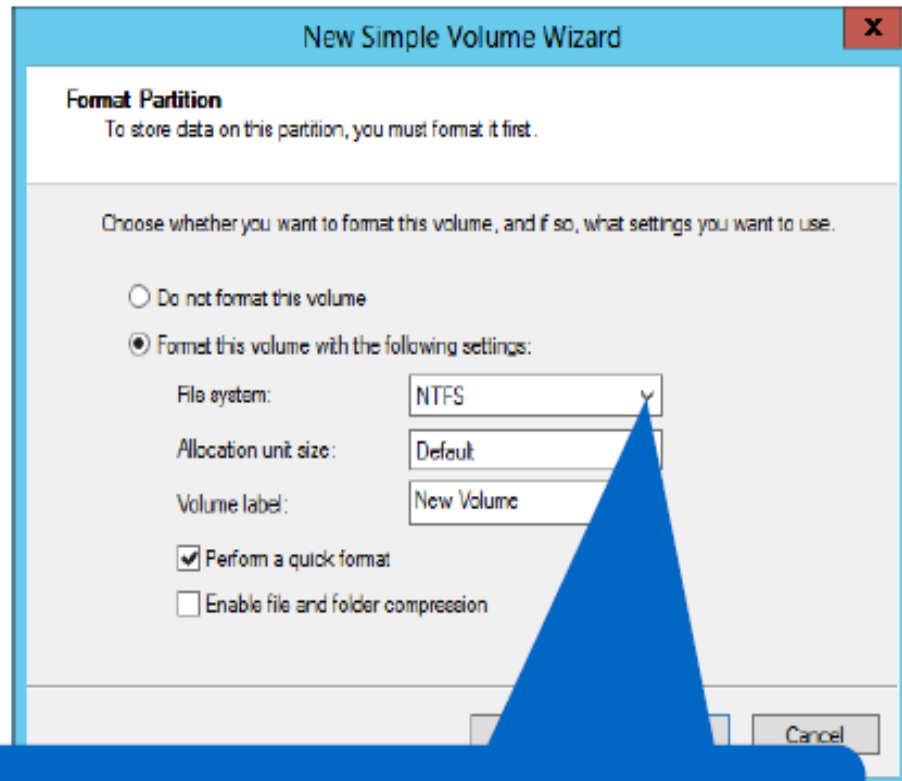


The screenshot shows the 'Assign Drive Letter or Path' step of the 'New Simple Volume Wizard'. The window title is 'New Simple Volume Wizard'. The instruction says 'For easier access, you can assign a drive letter or drive path to your partition.' There are three radio button options: 'Assign the following drive letter:' (selected) with a dropdown menu showing 'E', 'Mount in the following empty NTFS folder:' with a text box and a 'Browse...' button, and 'Do not assign a drive letter or drive path'. At the bottom are '< Back', 'Next >', and 'Cancel' buttons.

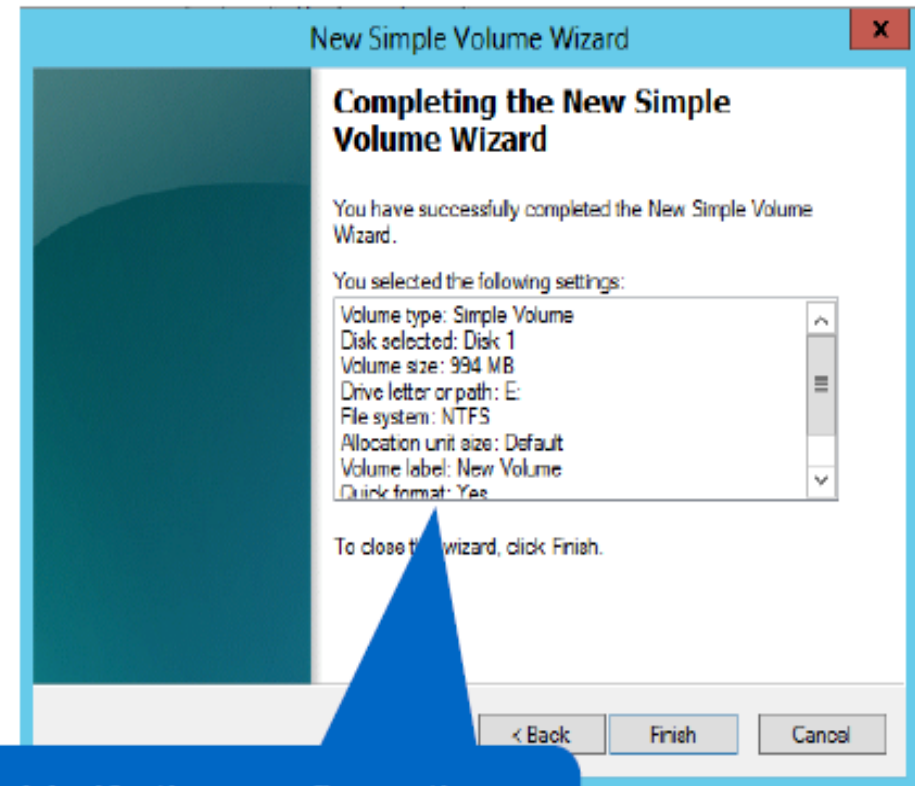
Specify the drive letter or mount point.

Windows LUN Implementation

Provisioning a volume (3 of 3)



Specify the file system format, allocation unit size, and a volume label.



Verify the configuration, and then click **Finish**.

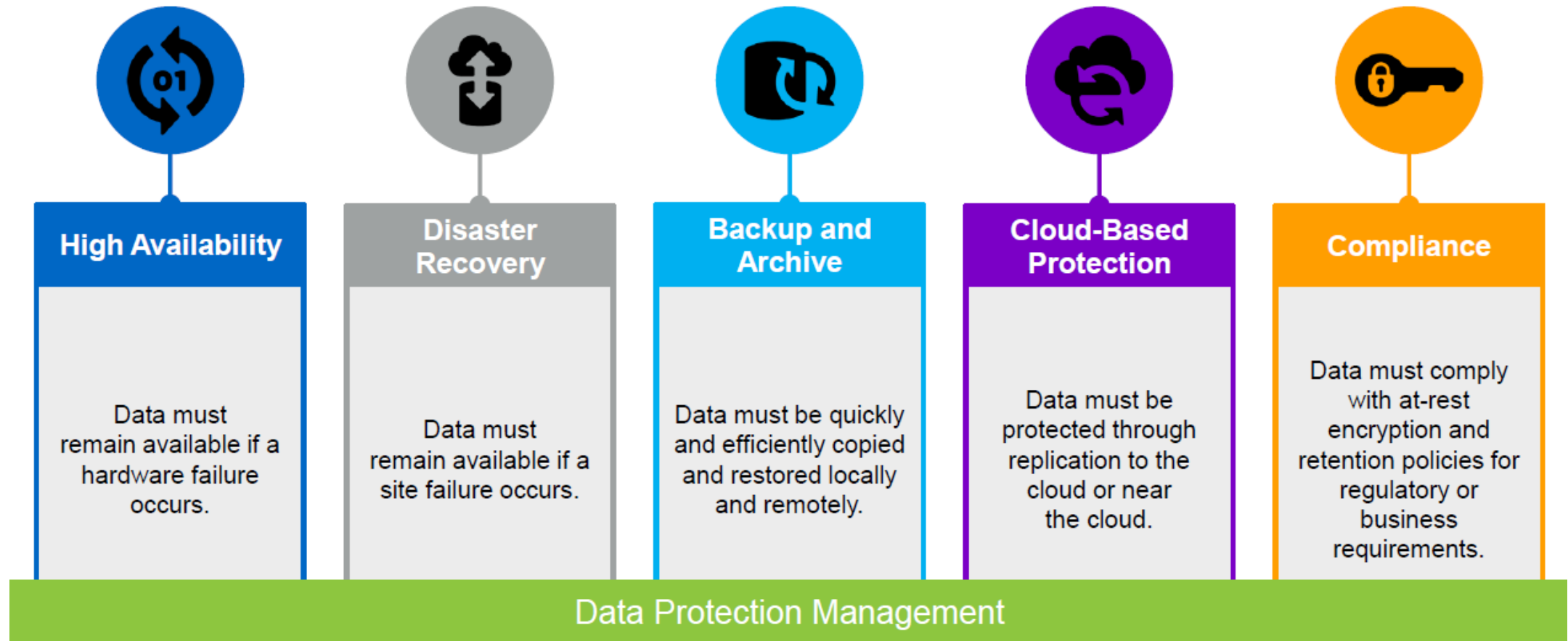


NetApp ONTAP Data Protection

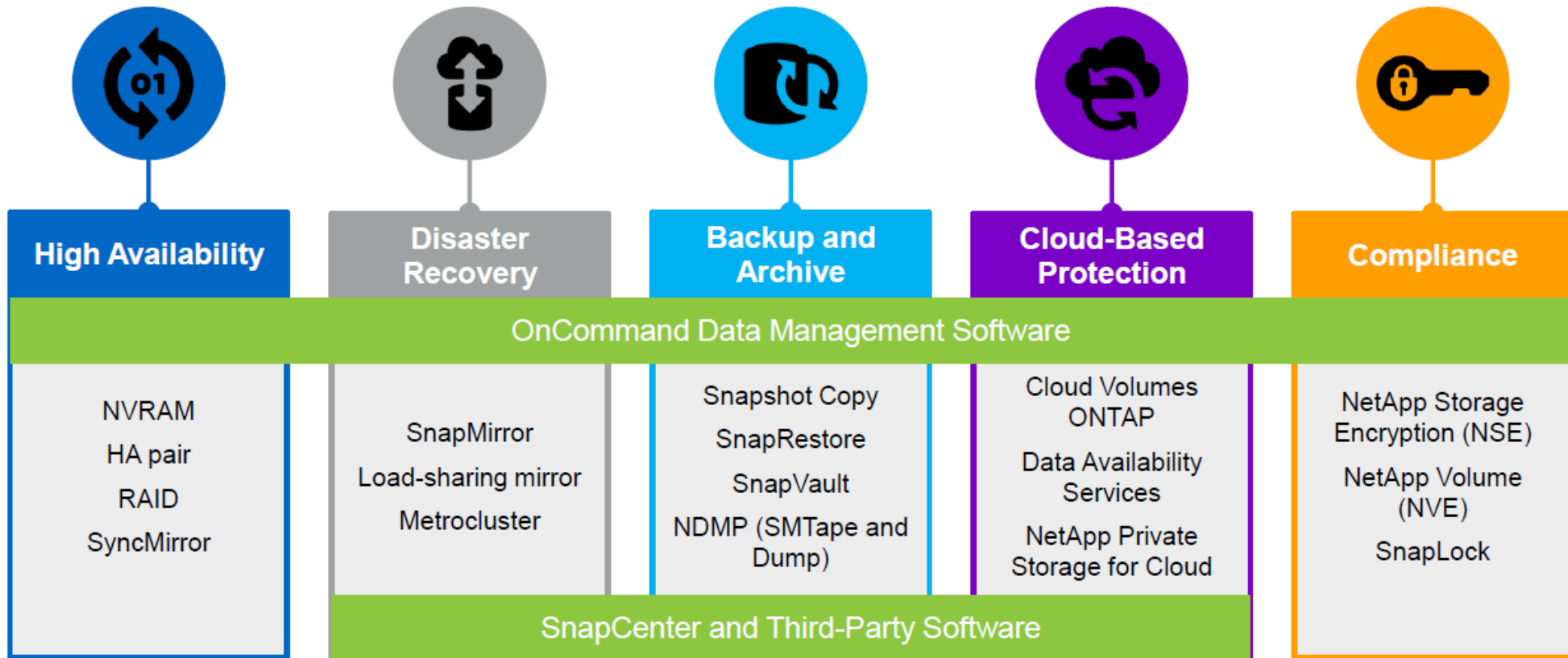


Senthilkumar Muthusamy

Data Protection Challenges



ONTAP Data Protection Solutions





Snapshot Management



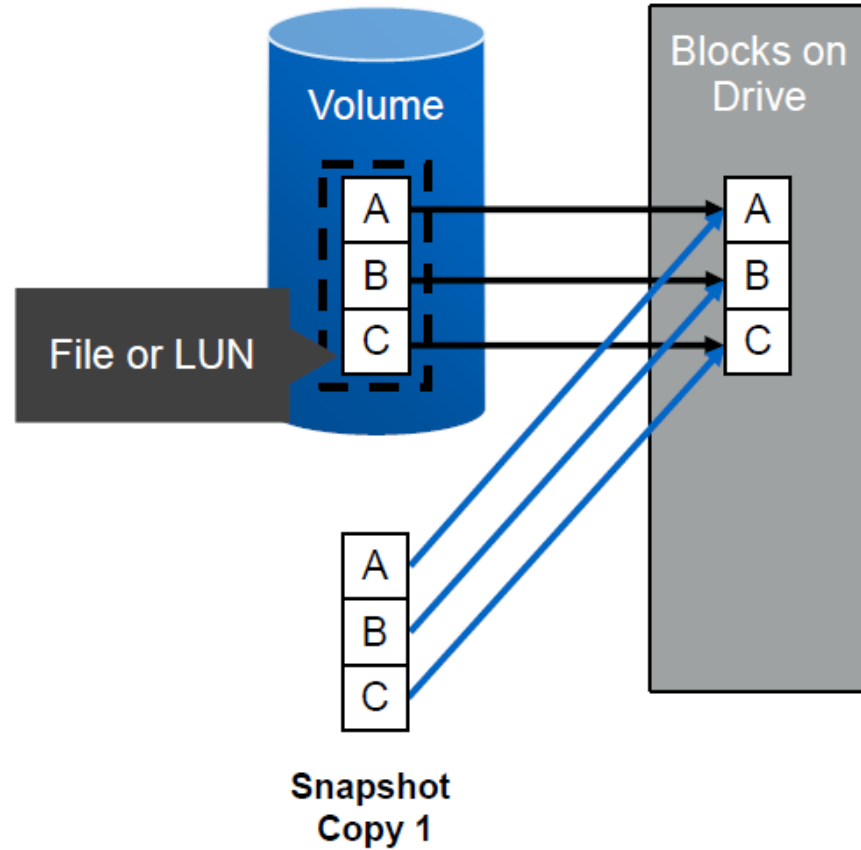
Snapshot Copies

- A Snapshot copy is a read-only, point-in-time image of a FlexVol volume.
- The image consumes minimal storage space and incurs negligible performance overhead because it records only changes to files since the most recent Snapshot copy was made.
- Snapshot copies owe their efficiency to the NetApp WAFL file system, which uses metadata to point to blocks on disk and writes to a new block rather than overwrite existing blocks.
- Instead of moving old blocks to a pool of space for Snapshot copies, old blocks remain in place. Only the pointers move from the active file system to the Snapshot copies.



Snapshot Copy Technology

Create Snapshot copy 1

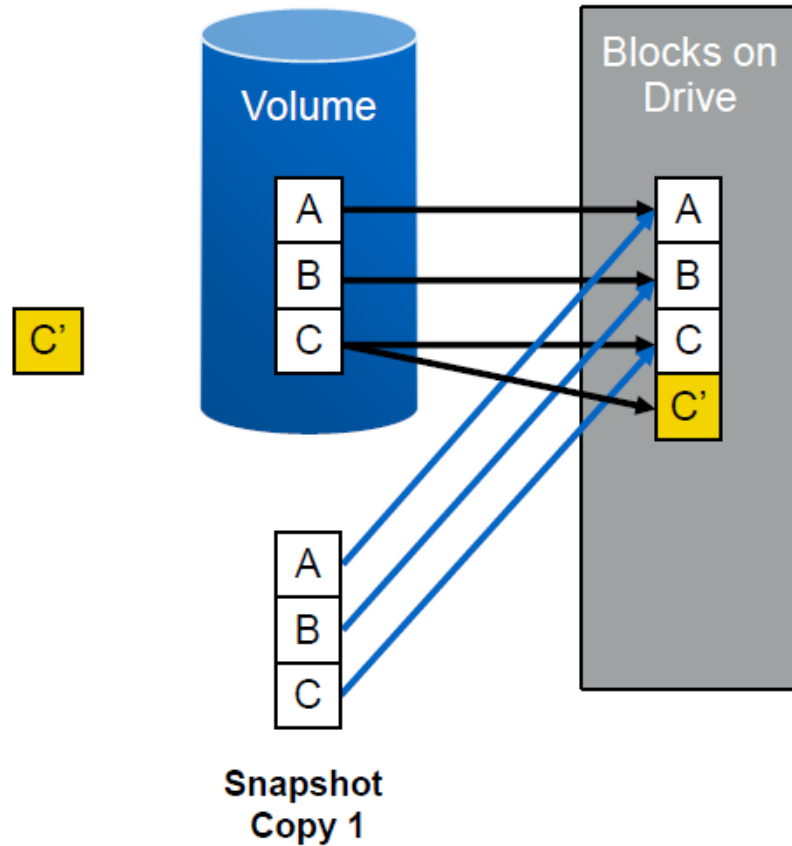


1. Create Snapshot copy 1:

- Pointers are copied.
- No data is moved.

Snapshot Copy Technology

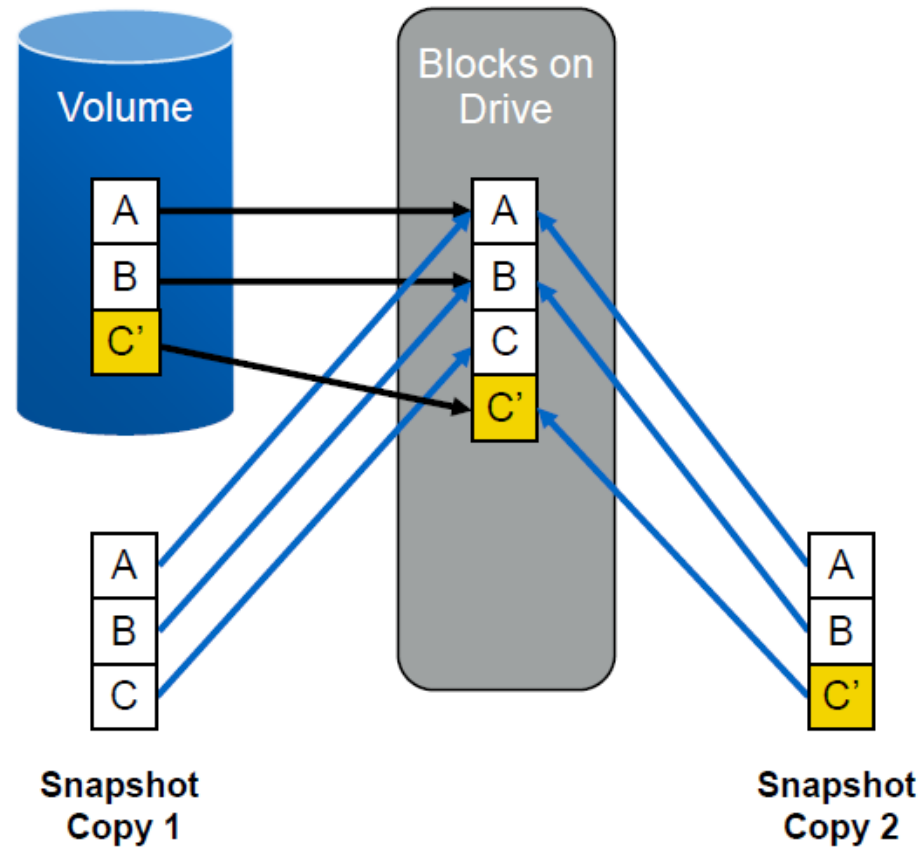
Continue writing data



1. Create Snapshot copy 1.
2. Continue writing data:
 - Data is written to a new location on the disk.
 - Pointers are updated.

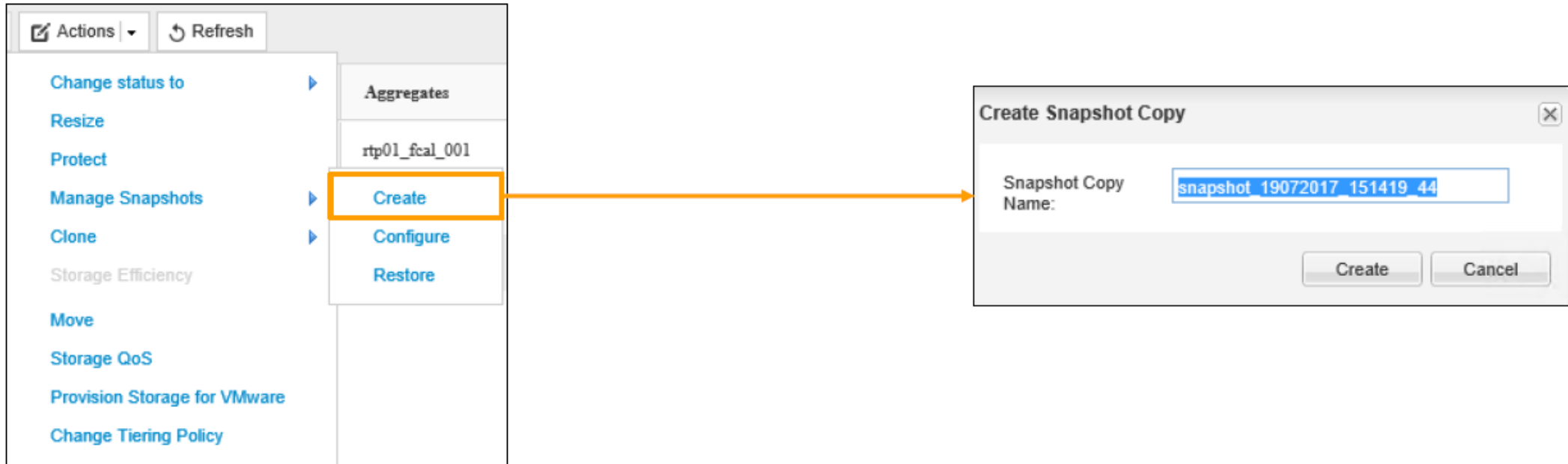
Snapshot Copy Technology

Create Snapshot copy 2



1. Create Snapshot copy 1.
2. Continue writing data.
3. Create Snapshot copy 2:
 - Pointers are copied.
 - No data is moved.
 - Block C consumes Snapshot space because the active file system no longer references Block C.

Create a Snapshot Copy



```
cluster1::> snapshot create -vserver svm4 -volume svm4_vol002
```



Naming Conventions for Snapshot Copies

- A Snapshot copy name can have a prefix or schedule name, timestamp, comment, and label:

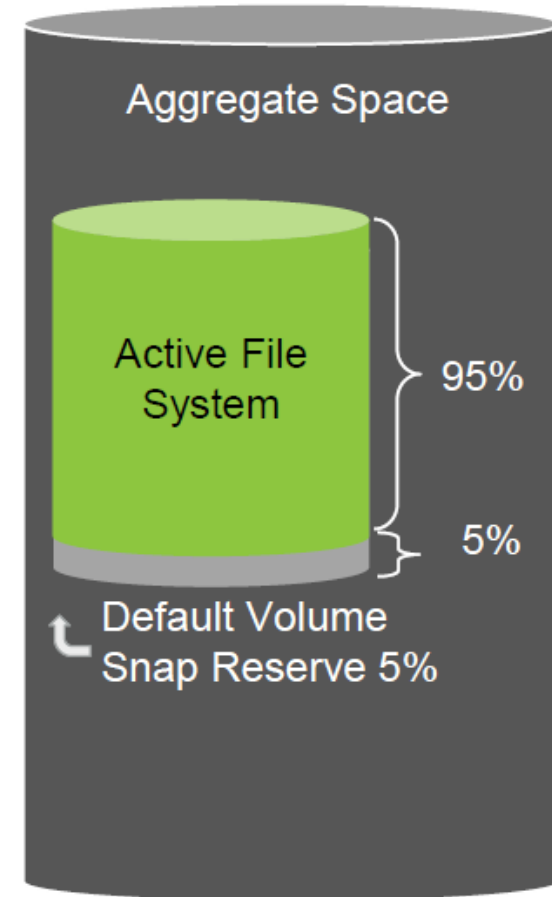
vserver	volume	snapshot
-----	-----	-----
svm4	svm4_vol002	2HourSnapshot.2016-07-11_1030
		

- Snapshot copy names cannot be longer than 255 characters.



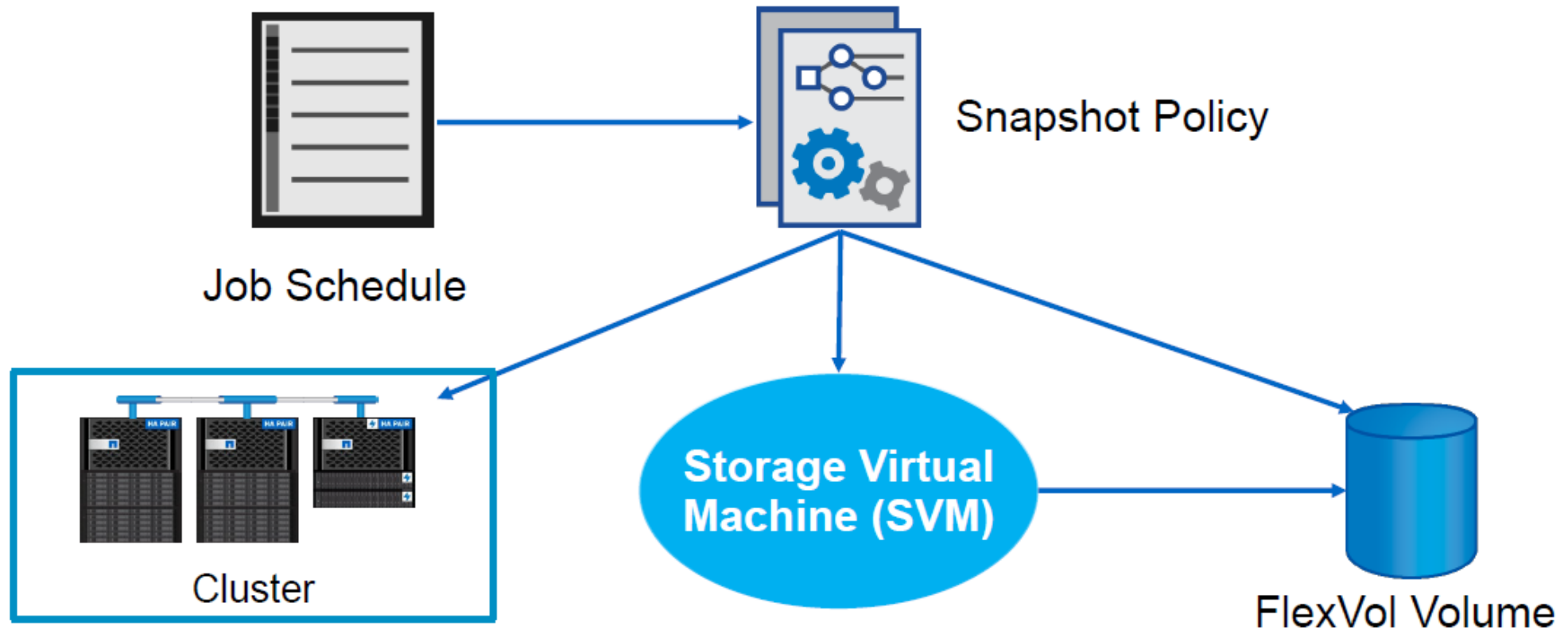
The Snapshot Copy Reserve

- The Snapshot reserve is a storage space set aside inside a volume.
 - Often depicted as a partition
 - Actually a soft quota
- The reserve holds blocks that are no longer in the active file system but are still referenced by Snapshot copies.
- The reserve is not used for file system writes.
- The reserve can be increased or decreased.



The Snapshot Policy

Automatically manage Snapshot copy schedules and retention.



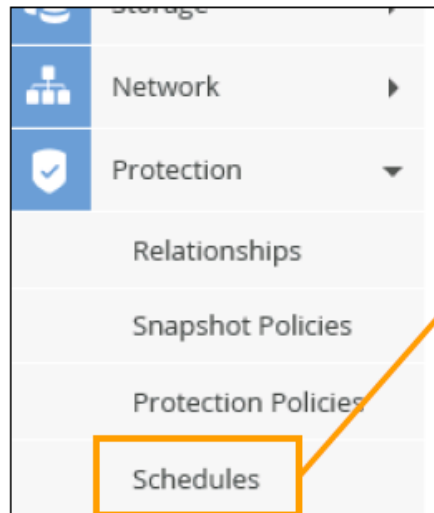
Typical Workflow

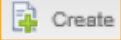


1. Create a job schedule, or use the default.
2. Create a Snapshot policy, and then specify the job schedule.
3. Assign the Snapshot policy to a FlexVol volume, or inherit a Snapshot policy from the SVM.



Create a Job Schedule



Schedules		
 Create  Edit  Delete  Refresh		
Name	Type	
5min	Time based	
8hour	Time based	
Auto Balance Aggregate Scheduler	Interval based	
daily	Time based	

Create Schedule

Schedule Name: 4hrs

☒ Basic

☐ Interval

☐ Advanced

Recurring Days

☒ Daily

☐ Select Days

☐ Sunday

☐ Monday

☐ Tuesday

☐ Wednesday

☐ Thursday

☐ Friday

☐ Saturday

Recurring Schedule

☐ Run Every minutes

☒ Run Every 4 hours

☐ Run Every hours at minutes

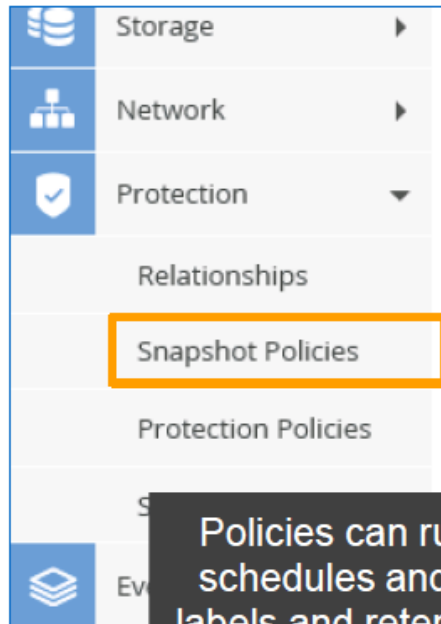
☐ Run At hours minutes

Create

Cancel

```
::> job schedule cron create -name 4hrs -dayofweek all  
-hour 4,8,12,16,20 -minute 0
```

Create a Snapshot Policy



Policies can run on multiple schedules and use different labels and retention counts for each schedule.

Create Snapshot Policy

Name:

Schedules:

Schedule Name	Maximum Snapshot Copies to be Retained	SnapMirror Label
4hrs	5	every_4_hours

OK Cancel

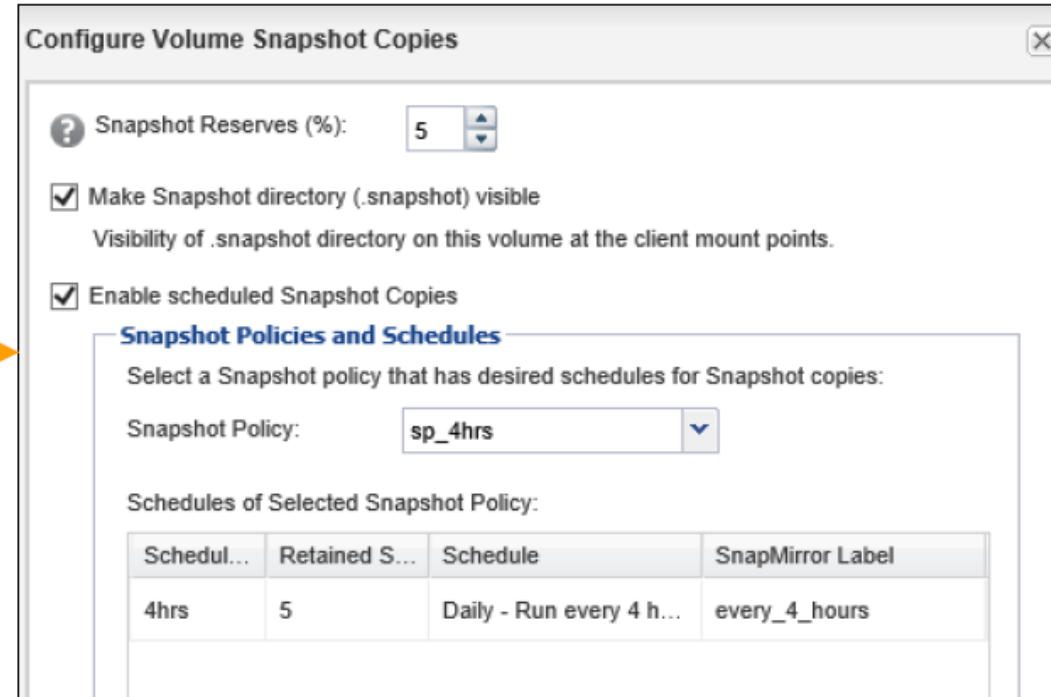
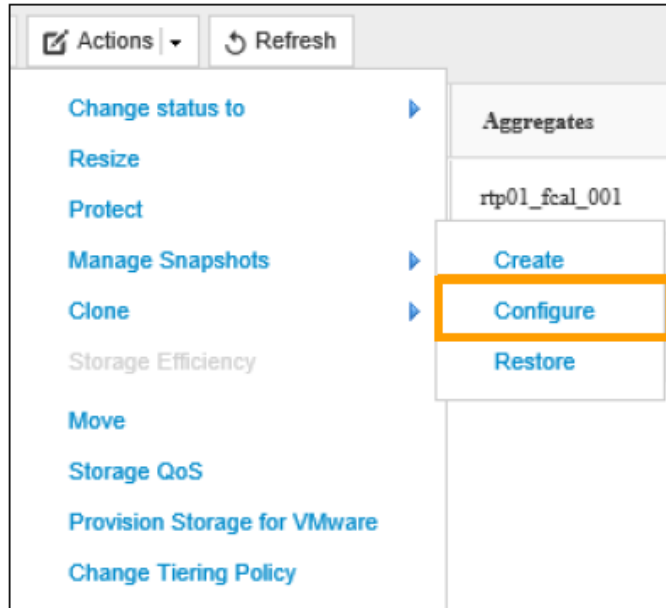
[Tell me more about Snapshot policies](#)

Create Cancel

Snapshot Policies		
Create	Edit	Delete
View as: List Tree Status Refresh		
Policy/Schedule Name	Status	Maximum Snapshots Copies to be Retained
▶ default	enabled	10
▶ default-1weekly	enabled	9
none	disabled	0

```
::> volume snapshot policy create -vserver svm4 -policy sp_4hrs  
schedule1 4hrs -count1 5 -prefix1 every_4_hrs
```

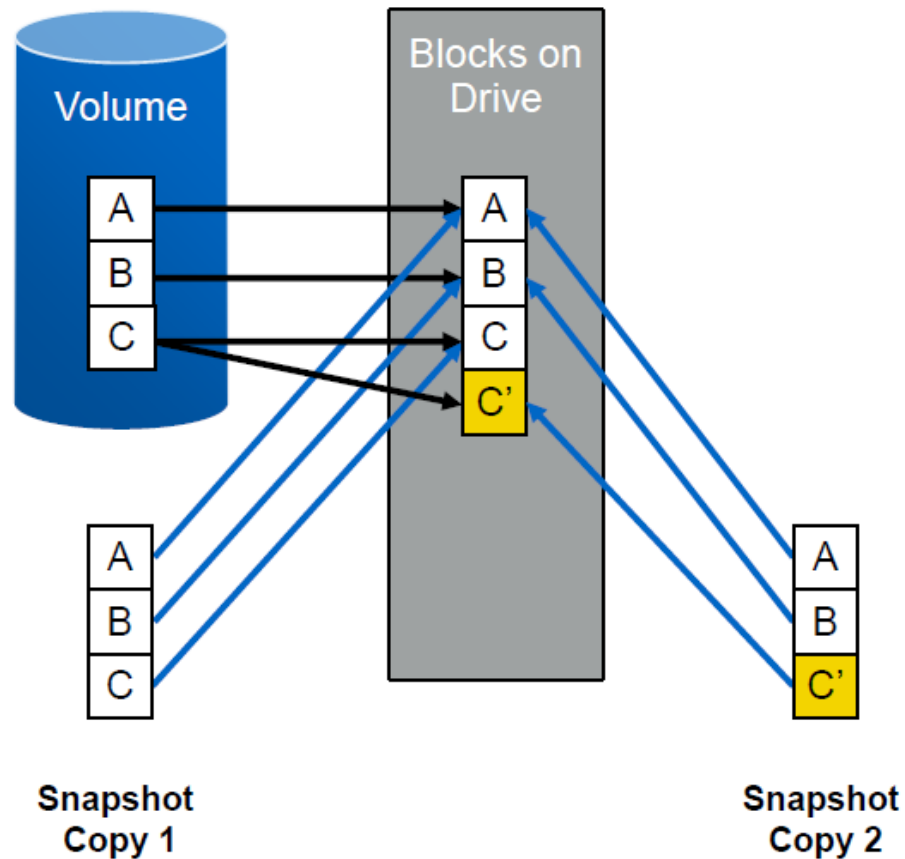
Apply a Snapshot Policy to a Volume



```
::> vol modify -vserver svm4 -volume svm4_vol002 -snapshot-policy sp_4hrs
```

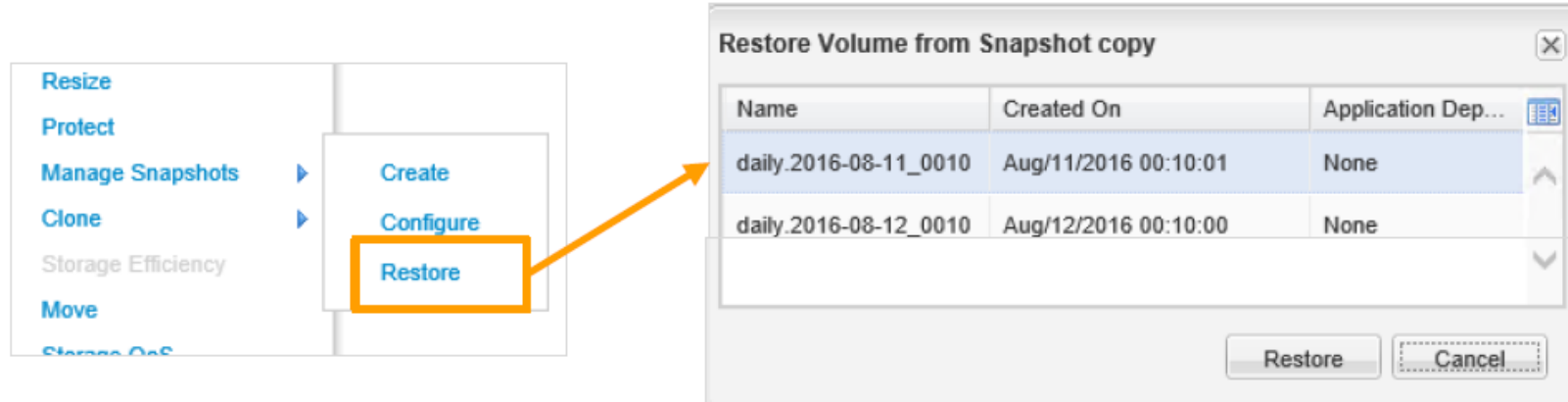
Snapshot Copy Technology

Restore from a Snapshot copy



- To restore a file or LUN from Snapshot copy 1, use SnapRestore data recovery software.
- Snapshot copies that were created after Snapshot copy 1 are deleted.
- Unused blocks on drives are made available as free space.

Reverting and Restoring a File



1. Verify that the volume is online and writable.
2. List the Snapshot copies in the volume.

```
::> volume snapshot show -vserver svm4 -volume svm4_vol_002
```

3. Notify network users about the reversion.
4. Identify the names of the Snapshot copy and the file to restore and initiate the reversion.

```
::> volume snapshot restore-file -vserver svm4 -volume svm4_vol_002 -  
snapshot svm4_vol_002_snap -path /svm4_vol2/myfile.txt
```



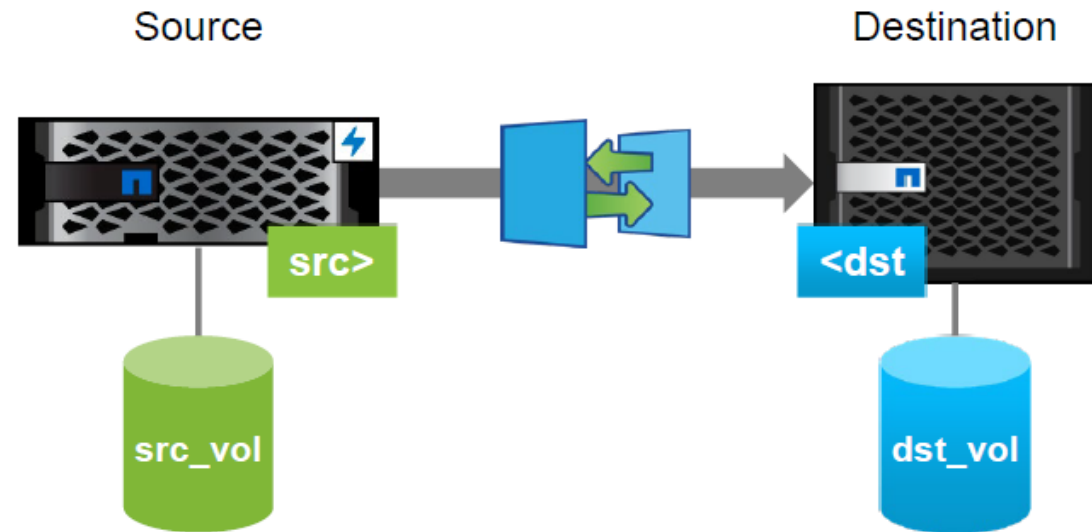
DR – Snap Mirror



Senthilkumar Muthusamy

SnapMirror Technology

SnapMirror technology enables the mirroring of volumes to other local or geographically remote AFF or FAS storage systems or to tape drives.



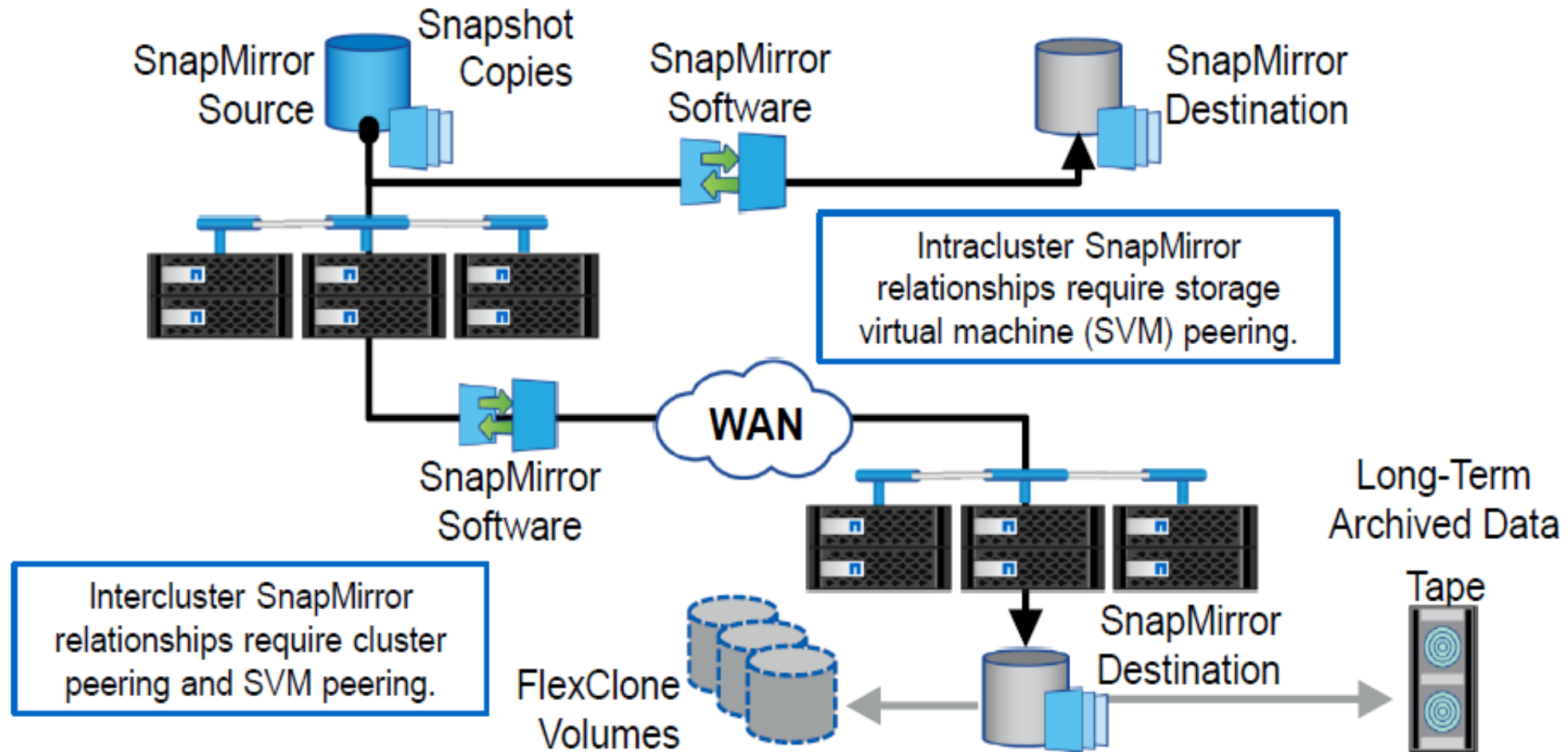
SnapMirror Features and Benefits

- The SnapMirror destination is a replica. Changes to the source are mirrored to the destination.
- Updates to the destination can be made frequently because only new and changed data blocks, rather than entire files, are sent.
- SnapMirror technology uses deduplication and compression and supports multiple paths (Ethernet or FC) to keep latency low and network bandwidth needs to a minimum.
- For backup and recovery, NetApp SnapCenter software manages application and database consistent backups, verification, cloning, and recovery.



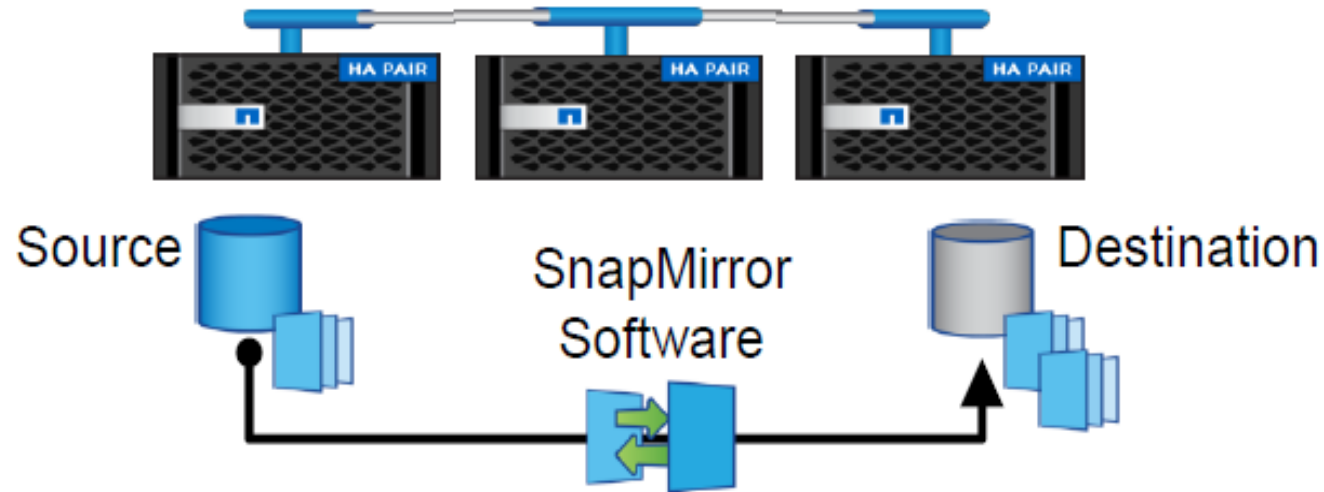
SnapMirror Technology

Provides a remote disaster recovery site



Data Protection Mirror Relationships

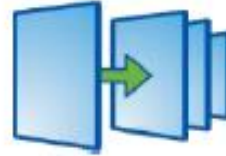
- Mirror relationships between FlexVol volumes
- Mirror relationships for SVM data volumes and configuration



Creating a Mirror Copy for FlexVol Volumes



The SnapMirror
source volume
is online and
writable.

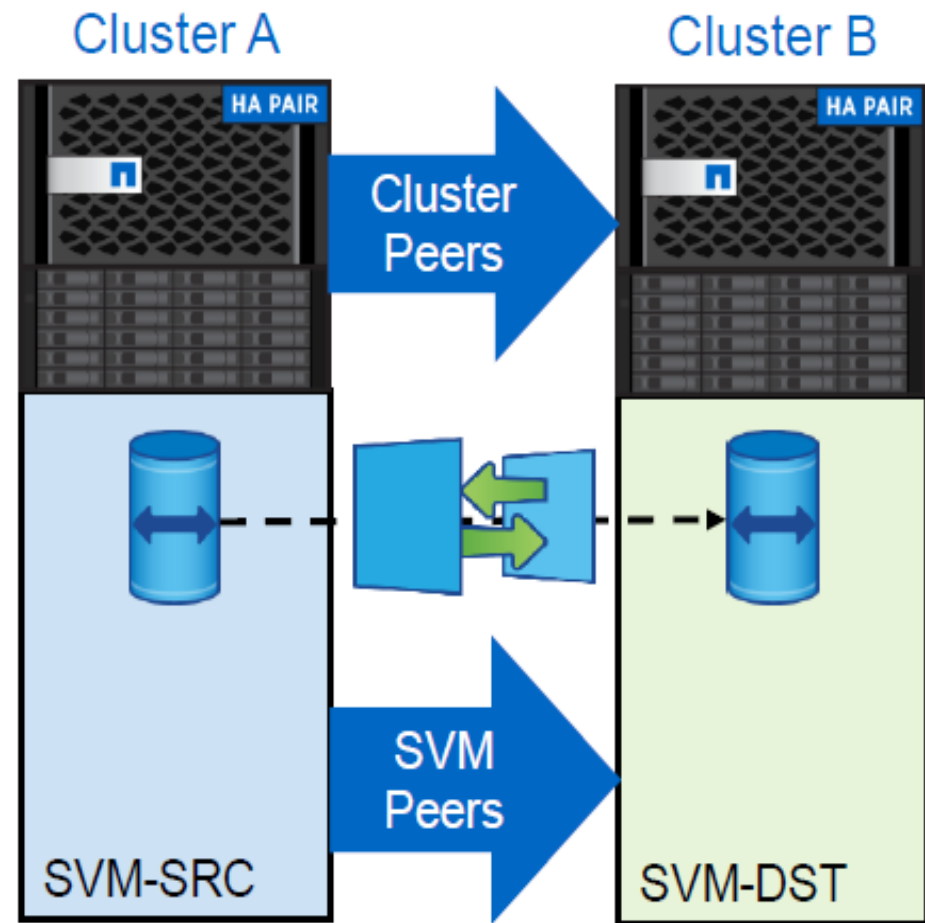


The SnapMirror
destination
volume is online
and read-only.

Steps to Configure SnapMirror Relationships

To create a protection relationship using SnapMirror software, follow these steps:

1. Verify that SnapMirror licenses have been applied on both the source and the destination clusters.
2. Establish cluster and SVM peering.
3. Select the destination cluster and SVM.
4. Create a data protection volume on the destination.
5. Select or create a mirror policy.
6. Select or create a schedule.
7. Create the relationship.
8. Initialize the relationship.





Compliance

SnapLock

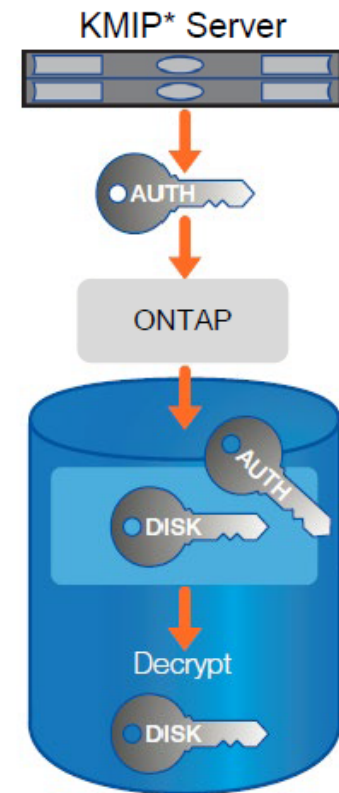


- SnapLock is a high-performance compliance solution for organizations that use WORM (Write Once Read Many) storage to retain files in unmodified form for regulatory and governance purposes.
- Files are locked from modification for an administrator-defined length of time.
 - Files in Snapshots and SnapMirror destinations are also locked until the time limit for all files expires.
- Requires a license
- Can be used in conjunction with storage encryption.
- Recommended Practice: Learn and practice using SnapLock on a simulator before implementing it because some mistakes are not reversible.



What Is NetApp Storage Encryption (NSE)?

- NSE is an ONTAP feature that provides support for self-encrypting drives (SEDs).
- SEDs protect data when it is at rest (when the drive is powered off).
- NSE manages the authorization process with a key management server to grant storage controllers access to the encrypted data on the drives.
- The encryption process is transparent to end users and has a minimal impact on performance.



* KMIP: Key Management Interoperability Protocol

NetApp Volume Encryption (NVE)

- **Software-based, data-at-rest encryption solution:**
 - Encrypts sensitive data, without relying on NSE drives
 - Uses Advanced Encryption Standard (AES)-256 encryption
 - Requires a license
- **Each data volume has a unique encryption key:**
Decide which volumes to encrypt and which to leave unencrypted.
- **Encryption requires zero management:**
Snapshot copies and FlexClone volumes are also encrypted.





Storage Efficiency

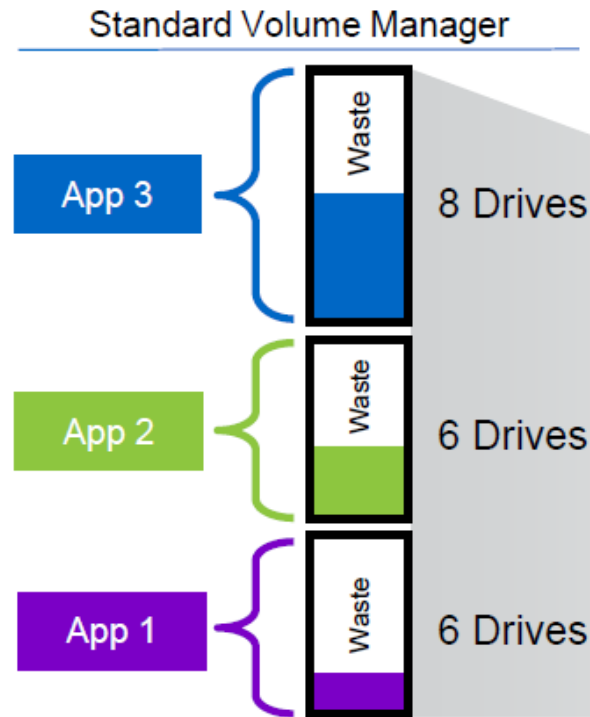
Thick and Thin Provisioning of Volumes

- Thick provisioning of volumes (guarantee = volume):
 - Requires reserved space within the aggregate for volume creation
 - Helps to prevent overcommitment of an aggregate
 - Simplifies storage management
- Thin provisioning of volumes (guarantee = none):
 - Does not require reserved space within the aggregate for volume creation
 - Enables more aggressive allocation
 - Does not prevent overcommitment of an aggregate
 - Requires more complex storage management

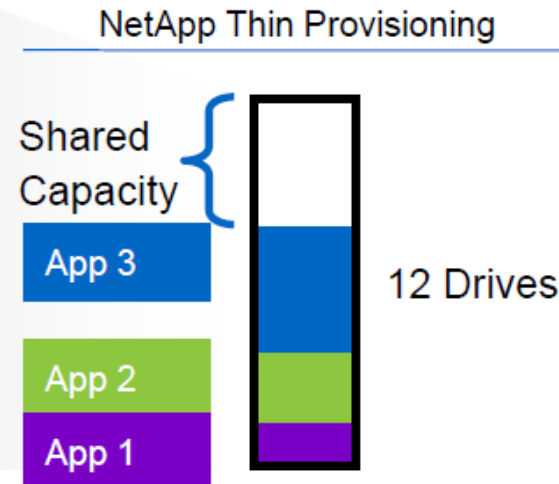


Thin Provisioning

- Typical: Only 40% of provisioned storage is used.



- NetApp: More than 70% is used.
 - Buy 50% less storage.
 - Save 50% power, cooling, and space.



Enable Thin Provisioning

The screenshot shows the NetApp ONTAP web interface. On the left is a navigation menu with options: Dashboard, Applications & Tiers, Storage, Nodes, Aggregates & Disks, SVMs, Volumes, LUNs, Shares, Qtrees, Quotas, and Junction Paths. The main panel is titled 'Volumes' and shows a table of volumes for SVM 'svm4'. The 'Edit' button for the volume 'svm4_NFS' is highlighted with an orange box. A callout bubble with the text 'Thin Provisioned' points to the 'Thin Provisioned' checkbox in the 'Edit Volume' dialog. The 'Edit Volume' dialog has tabs for 'General', 'Storage Efficiency', and 'Advanced'. The 'General' tab is active, showing the volume name 'svm4_NFS_volume', a security style of 'UNIX', and a permissions table. The 'Thin Provisioned' checkbox is checked, and a description explains that space is not allocated in advance. A link 'Tell me more about Thin Provisioning' is also present.

Status	Name
+	svm4_NFS
+	svm4_root

Edit Volume

General | Storage Efficiency | Advanced

Name: svm4_NFS_volume

☐ Encrypted

Security style: UNIX

☐ Configure UNIX permissions (Optional)

	Read	Write	Execute
Owner	☒	☒	☒
Group	☒	☐	☒
Others	☒	☐	☒

☒ Thin Provisioned

When a volume is thin provisioned, space for the volume is not allocated in advance. Instead, as data is written to the volume. The unused aggregate space is available to other thin provisioned LUNs.

[Tell me more about Thin Provisioning](#)

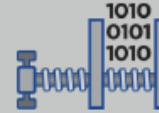
```
::> volume modify -vserver svm4 -volume svm4_vol_002 -guarantee none
```

Volume Efficiency



Deduplication

- Elimination of duplicate data blocks
- Inline or postprocess options
- Inline deduplication for AFF and Flash Pool systems to reduce the number of writes to solid-state drives (SSDs)

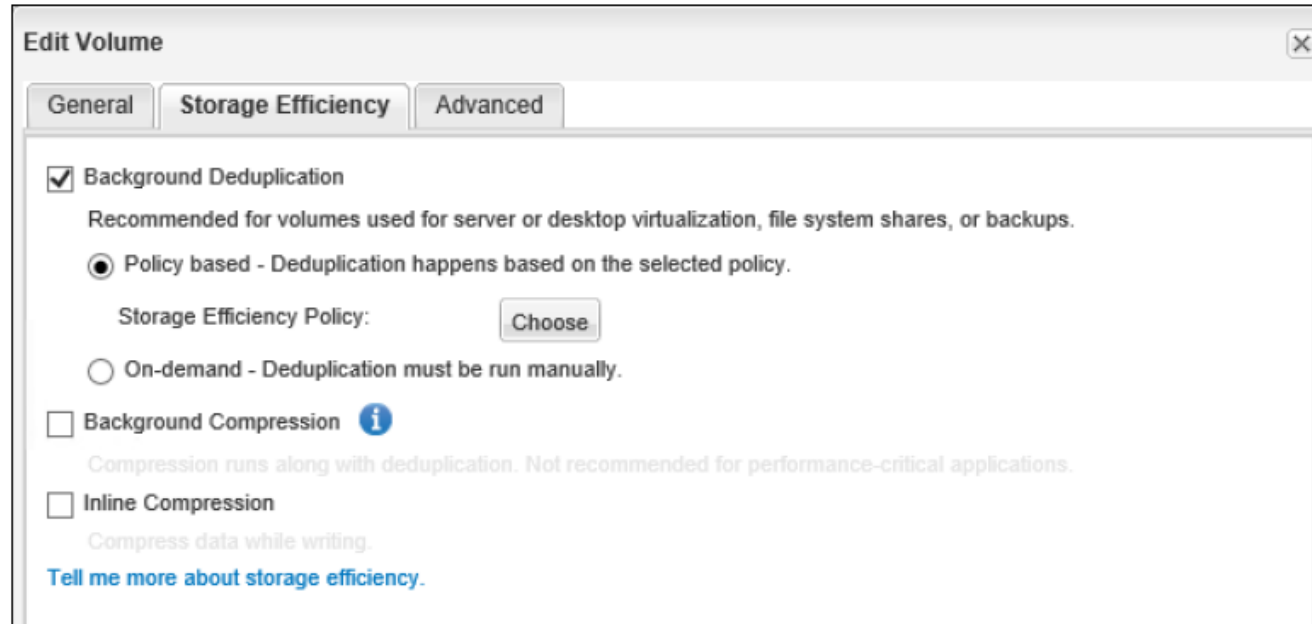


Data compression

- Compression of redundant data blocks
- Inline or postprocess options
- Two compression methods:
 - **Secondary:** 32KB compression groups
 - **Adaptive:** 8KB compression groups, which improve read performance



Enable Deduplication



The screenshot shows the 'Edit Volume' dialog box with the 'Storage Efficiency' tab selected. The 'Background Deduplication' checkbox is checked. Below it, there is a description: 'Recommended for volumes used for server or desktop virtualization, file system shares, or backups.' There are two radio button options: 'Policy based - Deduplication happens based on the selected policy.' (which is selected) and 'On-demand - Deduplication must be run manually.' Below the 'Policy based' option is a 'Storage Efficiency Policy:' label followed by a 'Choose' button. Below the 'On-demand' option is an information icon. There are also two unchecked checkboxes: 'Background Compression' (with an information icon) and 'Inline Compression'. Below 'Background Compression' is a description: 'Compression runs along with deduplication. Not recommended for performance-critical applications.' Below 'Inline Compression' is a description: 'Compress data while writing.' At the bottom, there is a link: 'Tell me more about storage efficiency.'

Edit Volume

General Storage Efficiency Advanced

☒ Background Deduplication
Recommended for volumes used for server or desktop virtualization, file system shares, or backups.

☒ Policy based - Deduplication happens based on the selected policy.
Storage Efficiency Policy: Choose

☐ On-demand - Deduplication must be run manually.

☐ Background Compression ⓘ
Compression runs along with deduplication. Not recommended for performance-critical applications.

☐ Inline Compression
Compress data while writing.

[Tell me more about storage efficiency.](#)

```
::> volume efficiency on -vserver svm4 -volume svm4_vol_002
```

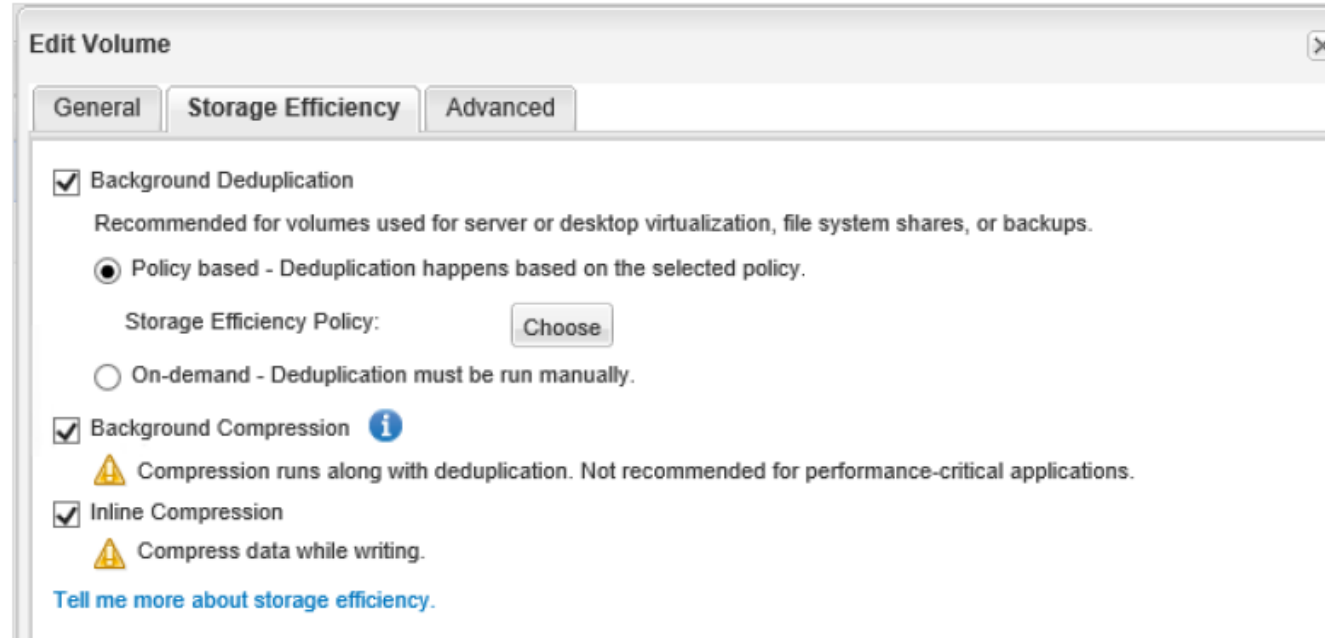


Characteristics of Data Compression

- **Inline compression:**
 - Parallelism increases.
 - Path length decreases.
 - On FAS systems, latency increases.
- **Postprocess compression:**
 - Uncompressed data is compressed during idle time.
 - Only previously uncompressed blocks are compressed.
 - Compression occurs before deduplication.
 - ONTAP software can detect incompressible data before wasting cycles.



Configuring Data Compression

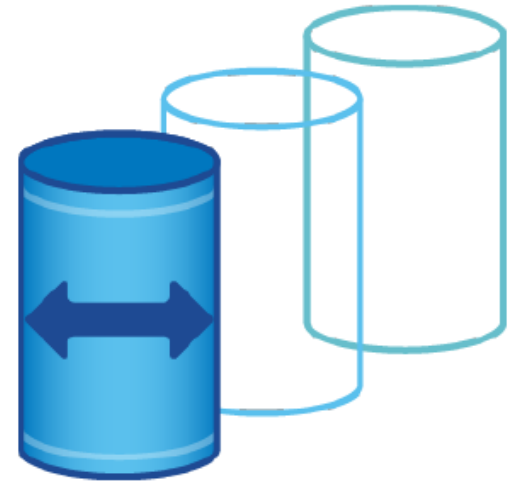


```
::> volume efficiency modify -vserver svm4 -volume svm4_vol002 -  
compression true -inline-compression true
```



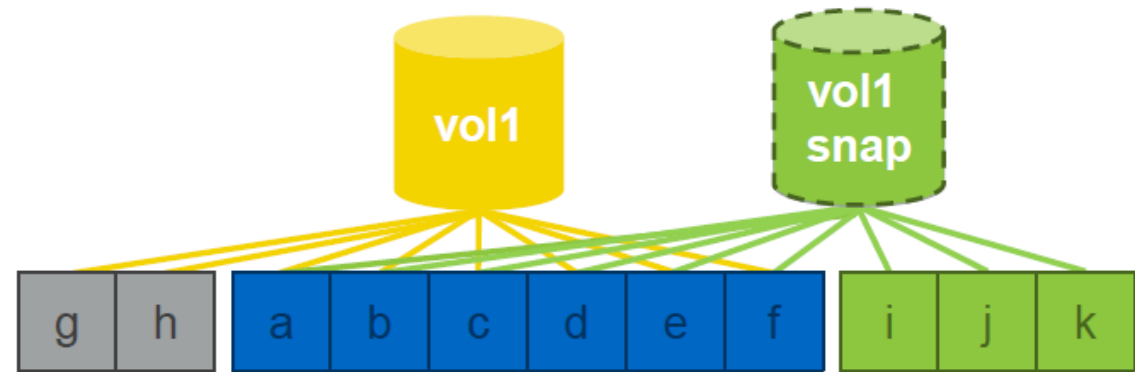
FlexClone

- FlexClone uses blocks pointers to enable you to create multiple, instant dataset clones (files, LUNs, or entire volumes) with no storage overhead.
- FlexClone provides dramatic improvement for application testing and development environments:
 - Create an instantaneous replica of a file or LUN (such as an entire database).
 - Provision thousands of virtual machines (VMs) in seconds by cloning *golden images*.
- Clones can be split from the source but then make copies of all source blocks and consume an equal amount of storage space.



How Cloning Works

- Make a Snapshot copy of the volume.
- Create a clone by adding another pointer to the blocks in the Snapshot (blocks A–H):
 - Modifications to the original volume are separate from modifications to the cloned volume (blocks G and H).
 - Modifications to the clone are separate from the original volume (blocks I – K)
These blocks are the only drive space consumed by the clone.



Clone a Volume

The screenshot shows the NetApp ONTAP web interface. On the left is a navigation sidebar with links to Dashboard, Applications & Tiers, Storage, Nodes, Aggregates & Disks, SVMs, and Volumes. The main content area is titled 'Volumes' and shows a table of volumes for the selected SVM 'svm3'. The table has columns for Status, Name, Thin Provisioning, Available Space, and Total Space. Three volumes are listed: 'exp_svm3_NFS_volume', 'smb3_share_volume', and 'svm3_root'. A context menu is open over the 'exp_svm3_NFS_volume' row, showing options like Change status to, Resize, Protect, Manage Snapshots, Clone, Storage Efficiency, Move, Storage QoS, Provision Storage for VMware, and Change Tiering Policy. The 'Clone' option is selected, and a submenu is open showing 'Volume' and 'File' options, with 'Volume' highlighted.

Status	Name	Thin P...	Avai...	Tota...
+	exp_svm3_NFS_volume	No	971.9 MB	1 GB
+	smb3_share_volume			
+	svm3_root			

Create FlexClone Volume

Name:

☐ Thin Provisioning

Allocate space for the volume as it's used. Otherwise, the system reserves space for the entire volume.

FlexClone parent Snapshot copy

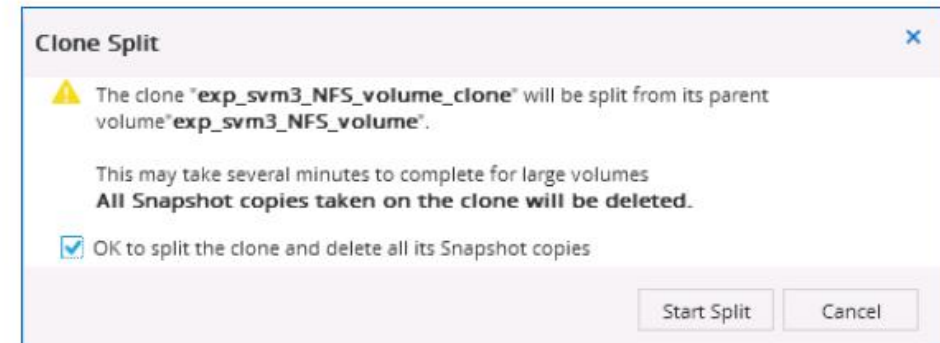
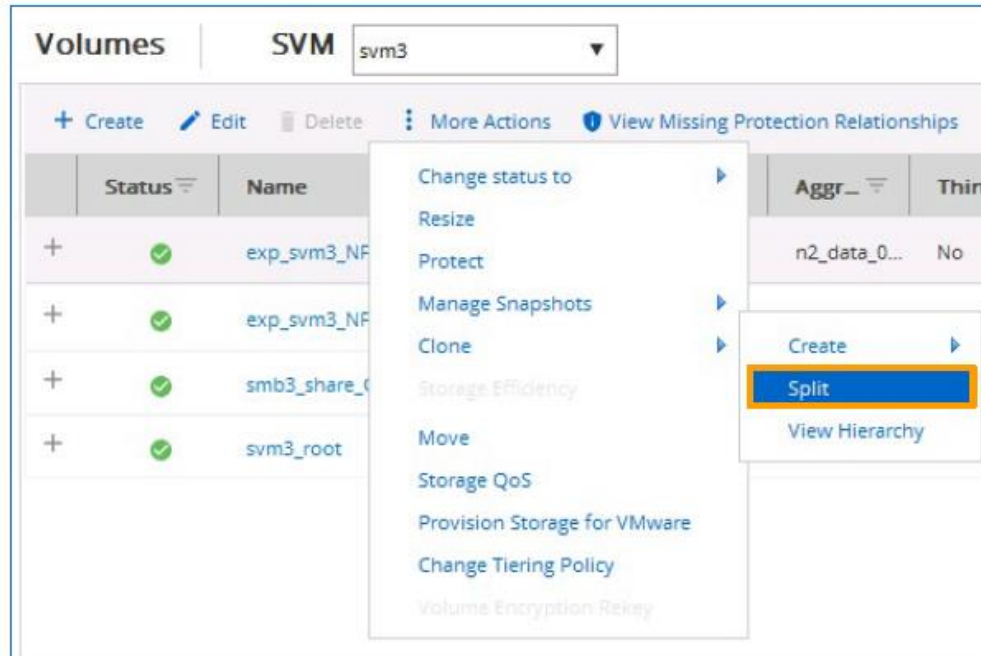
☒ Create new Snapshot copy now

☐ Use an existing Snapshot copy

```
cluster2::> volume clone create -vserver svm3 -flexclone  
exp_svm3_NFS_volume_clone -parent-volume exp_svm3_NFS_volume
```



Split a Cloned Volume



```
::> volume clone split start -vserver svm5 -flexclone svm5_vol_002_clone
```





Cluster Maintenance



Senthilkumar Muthusamy

Alerts

Tools for monitoring the system:

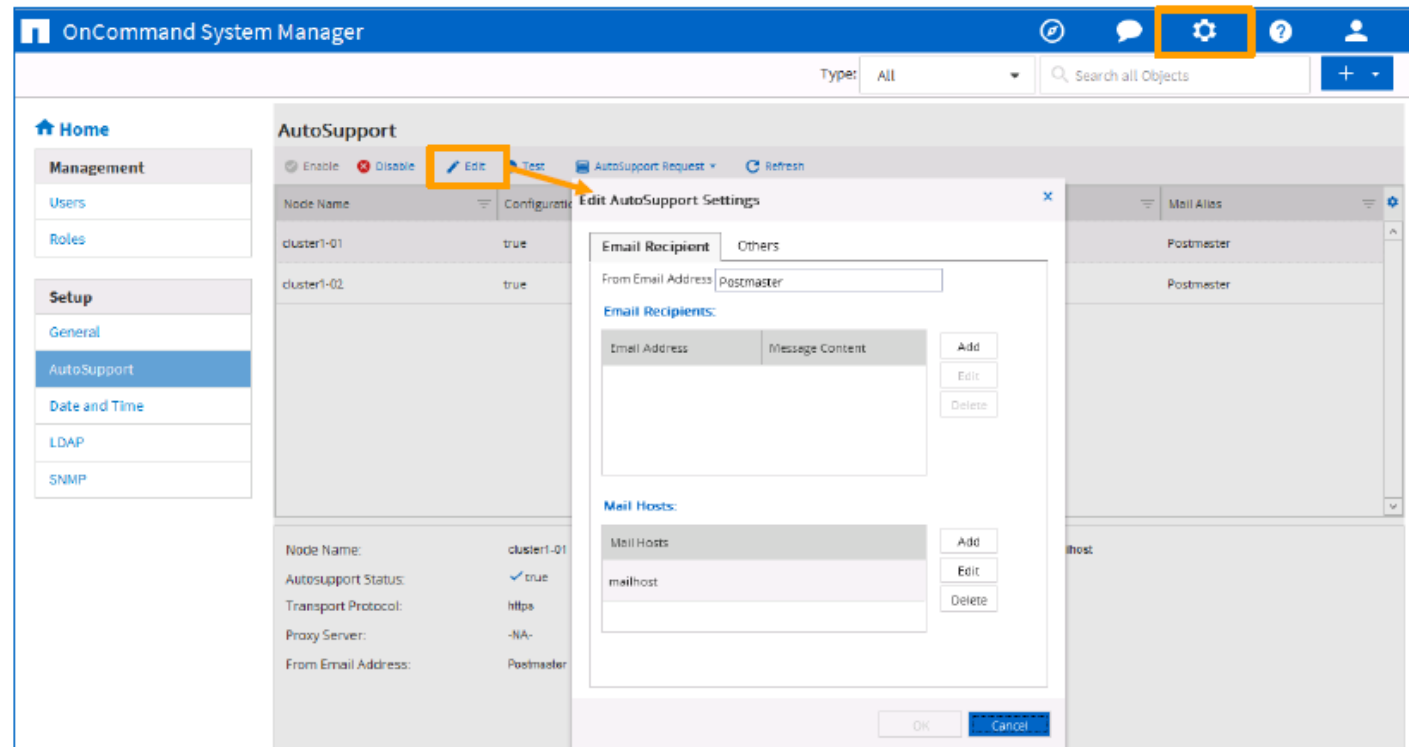
- AutoSupport
- Event management system (EMS)
- OnCommand System Manager
- OnCommand Unified Manager

The screenshot shows the 'System Alerts' window in the OnCommand Unified Manager interface. The window has a title bar 'System Alerts' and a toolbar with buttons for 'Acknowledge' (green checkmark), 'Suppress' (blue download arrow), 'Delete' (red X), and 'Refresh' (green circular arrow). Below the toolbar is a table with columns: 'SubSystem (No. of Alerts)', 'Alert ID', 'Node', and 'Severity'. The table shows one alert under the 'environment' subsystem: 'CriticalPSUFruOffAlert' with node 'conntel-01' and severity 'critical'. On the left, a sidebar menu is visible with options: 'Events & Jobs' (expanded), 'Events', 'System Alerts' (highlighted with an orange border), 'Jobs', and 'Configuration'.

SubSystem (No. of Alerts)	Alert ID	Node	Severity
environment (1 Alerts)	CriticalPSUFruOffAlert	conntel-01	critical

AutoSupport

- Is an integrated monitoring and reporting technology
- Checks the health of NetApp systems
- Should be enabled on each node of a cluster



EMS

- EMS does the following:
 - Writes events to the event log
 - Sends and routes notifications of events
 - Collects events throughout the cluster
 - Can view events of all nodes from any node

```
rtp-nau::> event log show
```

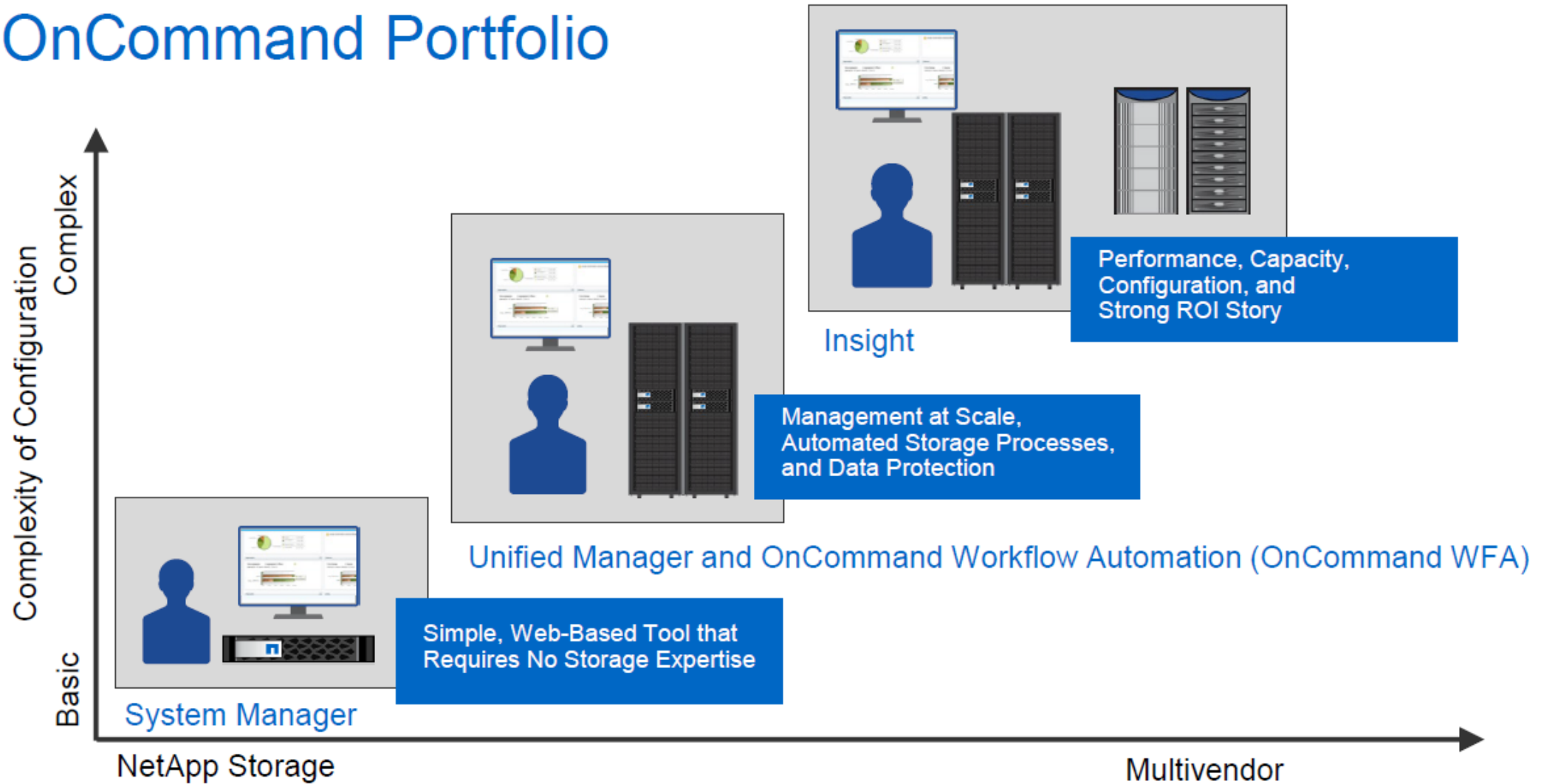
- Each event contains the following:
 - Message name
 - Severity level
 - Description
 - Corrective action, if applicable



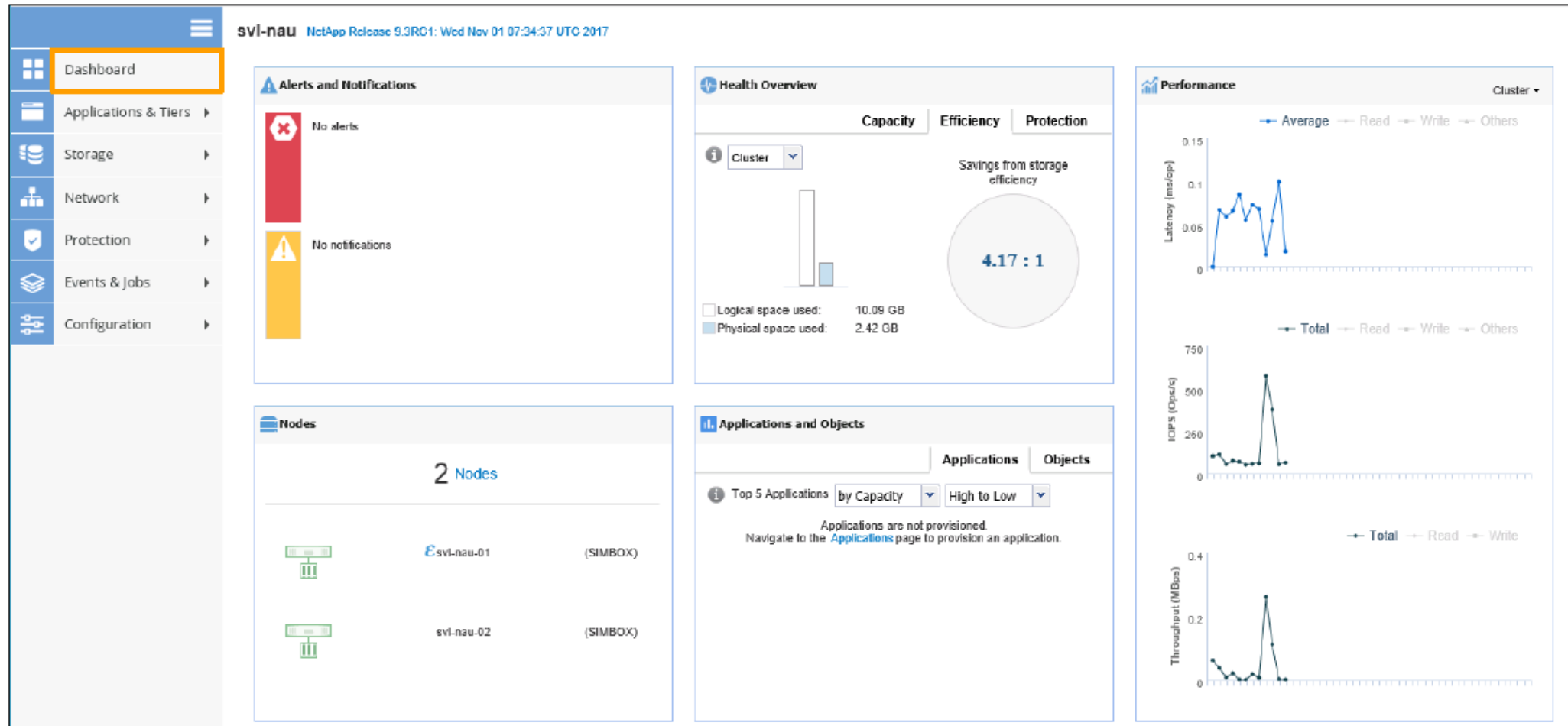
The screenshot shows the 'Events & Jobs' menu in the AWS IAM console. The 'Events' option is highlighted with an orange box.

[illegible]

OnCommand Portfolio



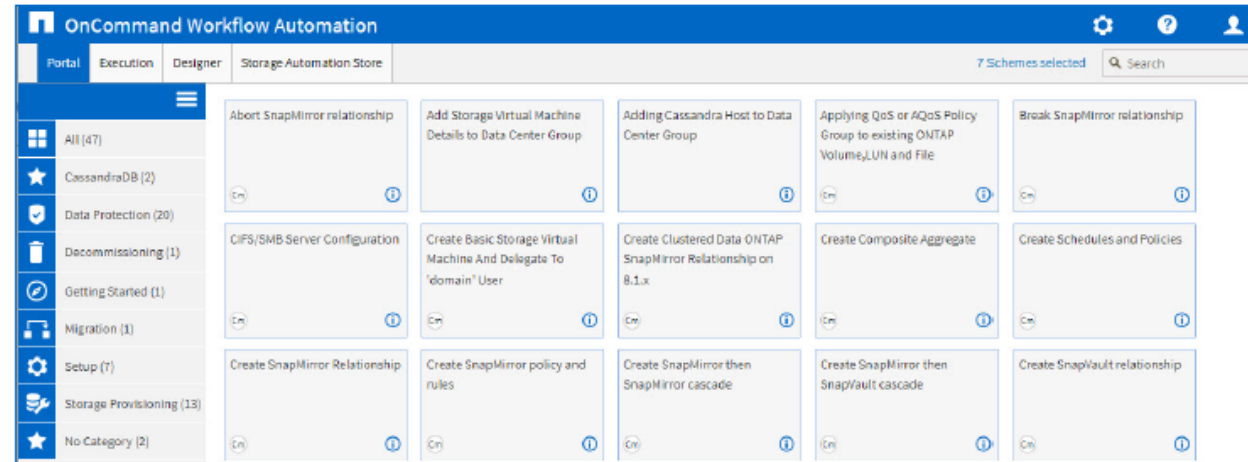
System Manager Dashboard



OnCommand WFA

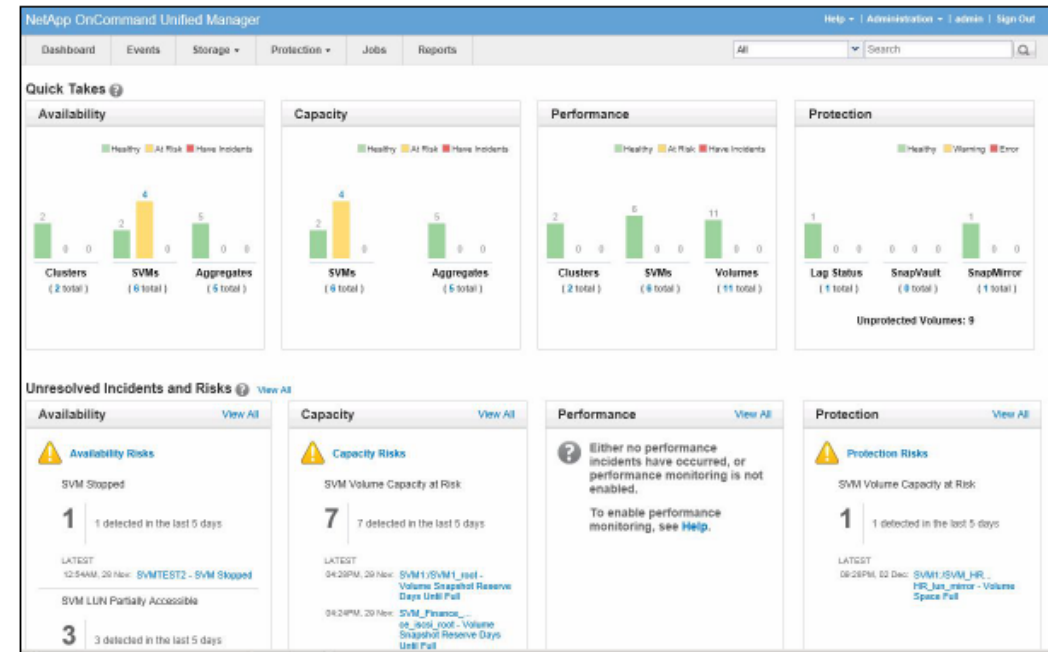
What is OnCommand WFA?

- **Highly flexible automation framework**
Enables automation of simple to complex storage processes
- **Operations portal**
 - One click to perform common tasks, with more than 45 built-in workflows
 - Authentication and authorization
- **Point of integration**
 - Initiate third-party actions
 - Drive OnCommand WFA from web services



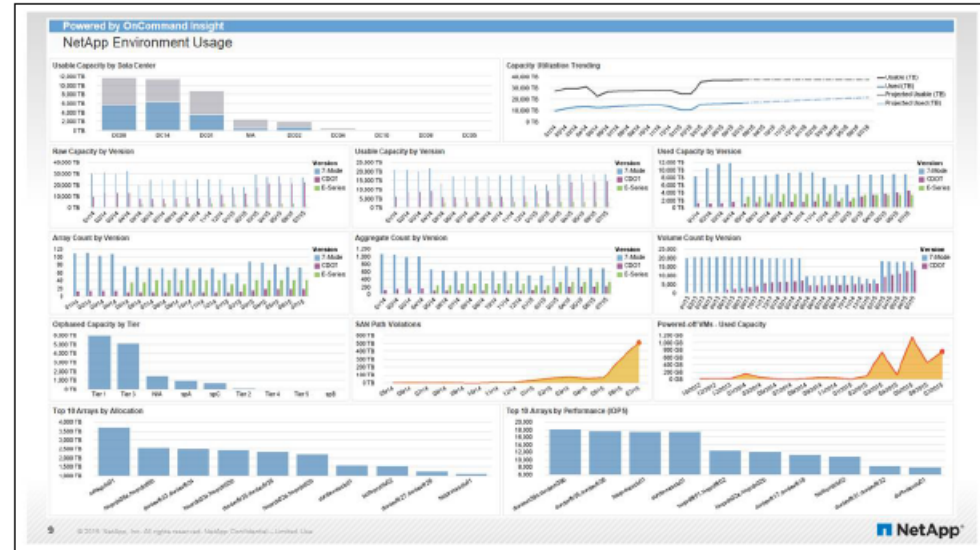
Unified Manager

- NetApp-centric application for managing multiple NetApp storage systems.
- Works with OnCommand System Manager on each storage system.
- Supports plug-in modules to extend functionality.



OnCommand Insight

- Provides proactive monitoring of your entire storage infrastructure: NetApp and competitor storage, SAN fabric switches, virtualization servers, and VM hosts.
- Provides reports on:
 - Inventory
 - Capacity
 - Performance
 - Showback and chargeback
- Is highly customizable through APIs and scripting



Performance Considerations

- Workloads
- I/O operation types:
 - Random
 - Sequential
- Quality of service (QoS)



Workloads

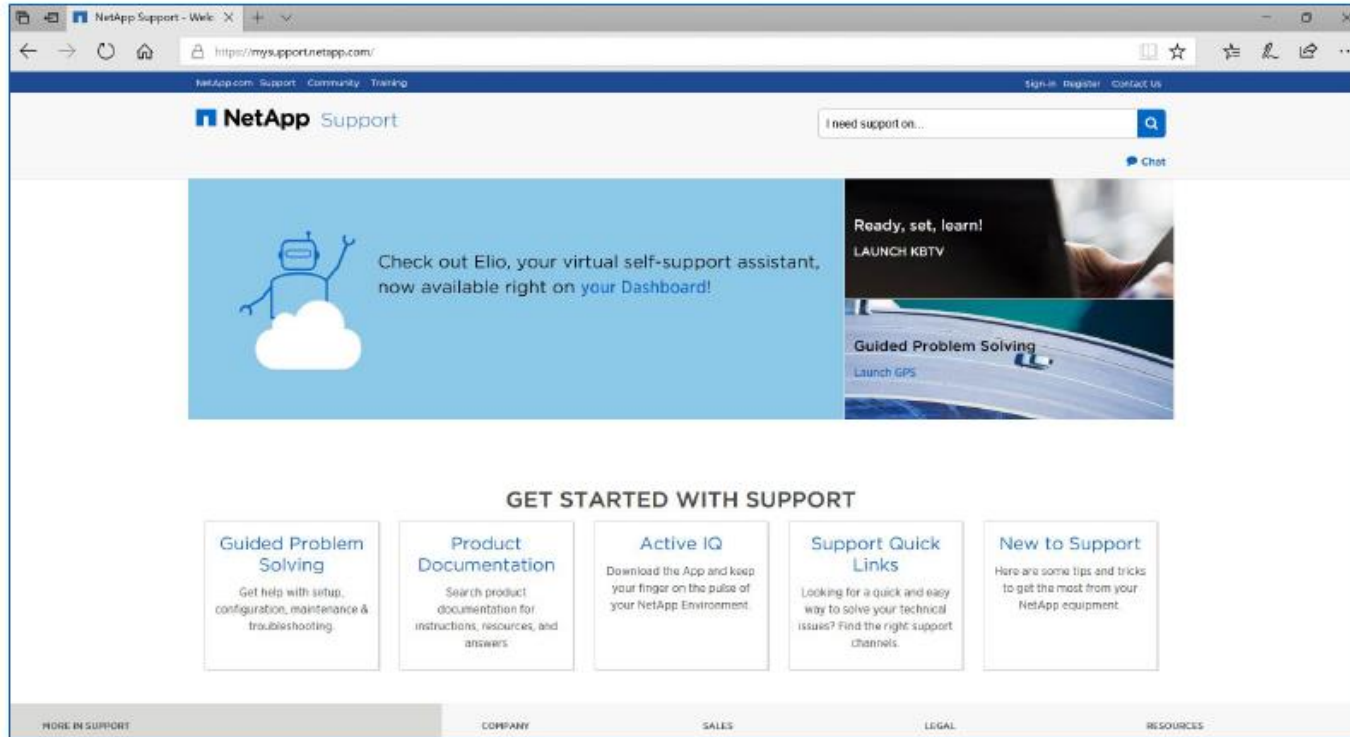
System Logs

- Log messages can be sent to the following:
 - The console
 - The message log
- You can access the message log by using the following:
 - The `debug log` command
 - System Manager
 - OneCollect
 - A web browser:
<https://<cluster-mgmt-ip>/spi/<nodename>/etc/log/>



Use the `debug log` command to browse the `messages.log` file.

NetApp Support



- **NetApp Support:**
mysupport.netapp.com
- **Hardware Universe:**
hwu.netapp.com
- **NetApp IMT:**
mysupport.netapp.com/matrix

QUESTIONS ?????

THANK YOU !!!!!

